

TPS System Implementation Guide for Windows 2003/XP

TP08X-420

R420

10/2011

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About This Document

This document provides instructions for implementing nodes running GUS R420 and APP R420 and greater TPS software. The document includes the following instructions:

- Installation of new Windows 2003 Domain Controllers and Windows 2003/XP nodes
- Migration from Windows 2000 to Windows 2003/XP operating system for Domain Controllers and nodes
- Upgrade of existing Windows 2003 Domain Controllers and Windows 2003/XP nodes to newer TPS software

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References

The following list identifies documents referenced by procedures in this publication.

Document Title	Doc ID
<i>Application Module Implementation Guidelines</i>	AM12
<i>Experion/TPS Backup and Restore User's Guide.</i>	BR20
<i>System Management Configuration Guide</i>	EX08
<i>TPN (LCN) System Startup Guide</i>	SW11-600 (CD) SW11-604 (Cartridge) SW11-614 (Zip)
<i>TPS System Administration Guide for Windows 2003/XP</i>	TP06X
<i>TPS System Planning Guide for Windows 2003/XP</i>	TP10X
<i>TPS System Configuration Utility User's Guide</i>	TP20X

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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE -EXTERNAL: Identifies an additional source of information outside of the Honeywell TPS/TPN bookset.
	REFERENCE - INTERNAL: Identifies an additional source of information within the Honeywell TPS/TPN bookset.
CAUTION	Indicates a situation which, if not avoided, may result in equipment or work (data) on the system being damaged or lost, or may result in the inability to properly operate the process.

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Contents

1. Getting started

1.1 Read me first

Purpose of this document

This document provides instructions to implement GUS R420 and APP R420 TPS software. The document includes the following instructions:

- Installation of new Windows 2003 Domain Controllers and Windows 2003/XP nodes.
- Migration from Windows 2000 to Windows 2003 operating system for Domain Controllers.
- Migration from Windows 2000 to Windows 2003 or Windows XP operating system for TPS nodes.
- Upgrade of existing Windows 2003 Domain Controllers and Windows 2003/XP nodes to newer TPS software release.



ATTENTION

Instructions to implement an LCN-connected Experion node (ESVT, ES-T, or ACE-T) are in the *Integrated Experion-TPS Nodes (ES-T, ESVT, and ACE-T) User Guide (EX23)* in the Experion bookset.

Not included in this document



REFERENCE – INTERNAL

For ongoing system administration tasks (such as user rights and security setup), refer to the *TPS System Administration Guide for Windows 2003/XP*.

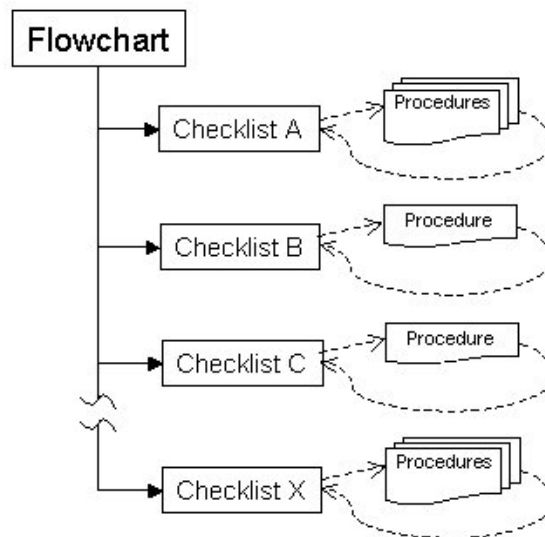
For complete information on backup and recovery of TPS nodes, refer to the *Experion/TPS Backup and Restore User's Guide*.

1. Getting started

1.1. Read me first

How to use this document

This document uses checklists to guide you through related tasks. A flowchart is provided to help you choose the appropriate checklist. Some procedures are included within a checklist; but for more lengthy procedures, the checklist refers to another section within this document. As illustrated in the diagram below, always return to the checklist after performing a procedure in another section.



TIP

Use the appropriate checklist to do a task. If the checklist refers to another procedure, ALWAYS RETURN TO THE CHECKLIST.

Basic assumptions

This document assumes the following:

- You have reviewed the related Software Change Notice (SCN), which provides last-minute changes, special instructions, and workarounds.
- The administrator performing these tasks is familiar with the Honeywell TPS product structure (especially TPS concepts and node types), functionality and security provisions, as outlined in the *TPS System Planning Guide*.
- The administrator performing these tasks is familiar with administering Microsoft Windows 2003/XP and is familiar with installing software and options to Windows 2003/XP.
- The administrator performing these tasks is familiar with TPN (LCN) administration.
- For domain-based installations, the policies and procedures defined here are done on a node-by-node basis within a single Windows 2003 domain. This does not preclude the use of zero administration techniques, trusted domains, or physically separate resource and account domains, but those techniques are not described here.



ATTENTION

Unless stated otherwise in this document, log on to the machine as Administrator to perform all of the installations.

1. Getting started

1.1. Read me first

Definition of implement, upgrade, migrate, and initialize

The following table defines TPS implementation terminology used in this document:

Terminology	Our definition
Implement a new Domain Controller	Implement a new Windows 2003 Domain Controller as a Root Domain Controller.
Implement a new TPS Node	Complete the implementation of a node that is currently at the Honeywell factory-default state. A TPS node at the factory-default state contains: • Windows 2003 server or Windows XP with and the appropriate service pack as defined in the Software Change Notice • TPS Base software
Upgrade a Domain Controller	Migrate an existing TPS Domain to a Microsoft Organization Units (OUs) structure, or implement a new TPS Domain OU structure on an existing Windows 2003 Domain Controller.
Upgrade a TPS Node	Upgrade the existing Honeywell software to a newer version: • TPS Base software • Common component software packages • TPS software packages • TPN (LCN) personality software, if necessary Example: Upgrade GUS R400 to GUS R420.
Migrate a Domain Controller	Change the Microsoft operating system from Windows 2000 Server to Windows 2003 Server – Standard Edition on an existing Domain Controller.
Migrate a TPS Node	Change the Microsoft operating system and upgrade the existing Honeywell software to a newer version: • Windows 2000 Server/Professional to Windows 2003/XP • TPS Base • Common component software packages • TPS Software packages • TPN (LCN) Personality software, if necessary Example: Migrate GUS R360 to GUS R420.

Terminology	Our definition
Initialize a TPS Node	Establishes a TPS node at the Honeywell factory-default state. Performs a clean installation of the Windows 2003 or Windows XP operating system using the Honeywell EXPPlus Initialization DVD, and then you manually install the TPS software from the TPS Software CD. For TPS R400 and greater, the EXPPlus DVD supports TPS node initialization. Honeywell recommends using the EXPPlus DVD to load the Windows 2003/XP operating system on TPS nodes. This technique reformats the disk drive and sets the node to the Honeywell factory-default state minus TPS base. Note: You must backup node data files prior to initialization.

1. Getting started

1.2. Implementation guidelines and recommendations

1.2 Implementation guidelines and recommendations

Domain controller and FTE

Windows 2003 Domain Controllers (including the WINS server) must not be FTE nodes. Either the Domain Controller Root should be singly connected to Tree A and the Domain Controller Peer to Tree B of an FTE network, or the Domain Controller function should be supplied from the other side of a firewall. Refer to FTE documentation for detailed installation instructions.

System Management implications

The Honeywell System Management infrastructure (such as TPS Domain OUs and the System Management Display) makes use of multicast communications and Active Directory. Before upgrading/migrating, make sure you fully understand the implications of using multicast and Active Directory within the process control domain.



ATTENTION

The following table provides basic guidelines when implementing GUS R400 and greater and APP R400 and greater releases of TPS nodes.

Technology used	Considerations/actions
Windows 2000/2003 Active Directory	Windows 2000/2003 Active Directory is required to support GUS R312 and greater and APP R210 and greater. REFERENCE-INTERNAL: The <i>TPS System Implementation Guide</i> (this guide) provides instructions for establishing a Windows 2003 Active Directory Domain.
Dynamic DNS (DDNS)	Windows 2000/2003 requires implementation of a Dynamic DNS server. DDNS must run on a Windows 2000/2003 server machine. The <i>TPS System Implementation Guide</i> provides instructions for establishing a basic Windows 2003 DDNS. ATTENTION: Consult Microsoft documentation and the <i>TPS System Planning Guide</i> for further information.
Multicast Communications	Ensure all ports connected to TPS nodes on network switches are enabled to support multicast communications. REFERENCE-EXTERNAL: See switch vendor documentation for further details.

Site-specific requirements

If installing a new TPS Domain Controller or new TPS nodes, there are site-specific requirements to be considered prior to the new installation.

The following table lists site-specific requirements and refers to TPS documentation for guidelines and recommendations:

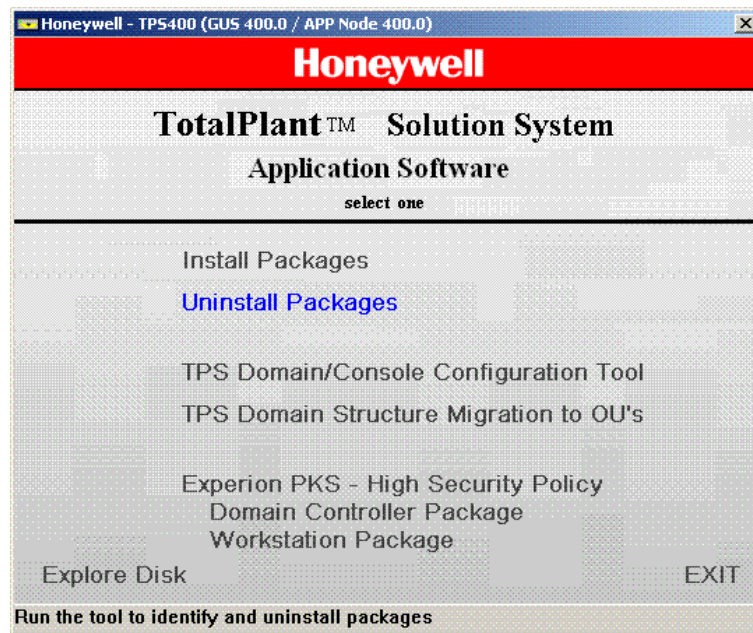
Requirement	Documentation
Determine TPS Domain architecture: • Approach for Implementing Process Control Domain • Implications of Establishing Process Control Domain as Child of IT Domain • Dynamic DNS (domain name service) Server Considerations • DNS Naming Guidelines and Conventions	<i>TPS System Planning Guide:</i> Section: "Windows 2003 Domain Guidelines" Section: "Active Directory Guidelines" Section: "Dynamic DNS Role in Process Control Domain"
Determine Security Requirements	<i>TPS System Administration Guide</i> , "TPS Security"
Determine TPS software packages to be implemented on each node	<i>This guide:</i> For the usage (optional, mandatory, and not allowed) on each node type, refer to section 4.11 "TPS Software Package Usage by Node Type".
Determine Honeywell Common Components requirements for each node: • Fault Tolerant Ethernet Driver • Redirection Manager • System Management	<i>FTE Installation and Service Guide</i> <i>Redirection Manager User's Guide</i> <i>System Management Configuration Guide</i> <i>This guide:</i> For descriptions of the common components and their related documentation, refer to section 1.3, "Summary of common component software". For usage (optional, mandatory, and not allowed) on each node, refer to section 4.11, "TPS Software package usage by node type".
Determine requirements for third-party software	<i>TPS System Planning Guide</i> , "Third Party Component Planning"
Determine the backup methodology	<i>Experion/TPS Backup and Recovery User Guide</i>

1. Getting started

1.2. Implementation guidelines and recommendations

TPS software un-installation guidelines

With GUS R350 and APP R240 and greater, a Honeywell uninstaller is packaged in the TPS Software CD and is accessible through the TPS browser as shown in the following figure:

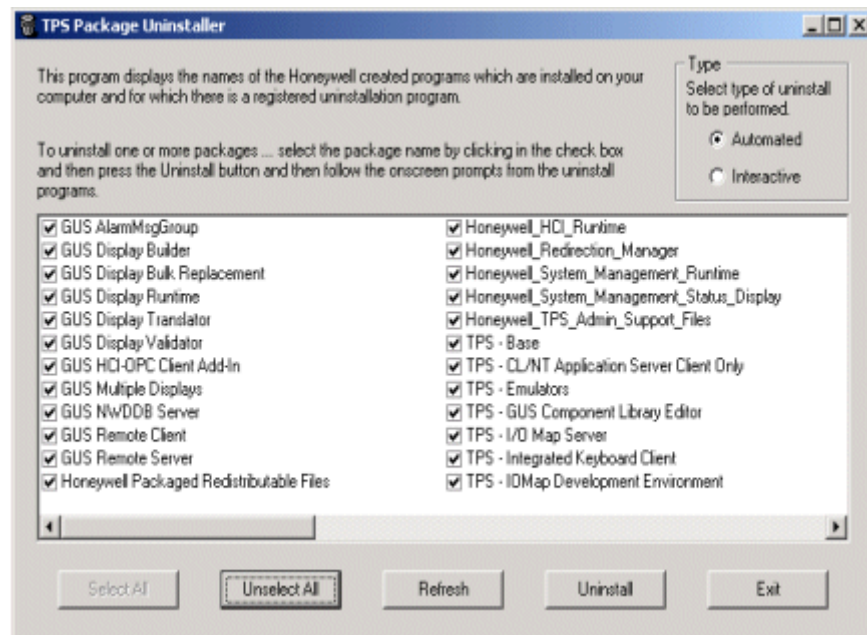


The uninstaller uses the standard uninstall information as recorded in the system registry. The path is \i386\packages\uninstaller.

Using the Honeywell uninstaller to remove the previous version of the TPS software is more efficient than using the existing Add/Remove programs method.

To identify the Honeywell packages, the uninstaller looks for packages with installation paths in the Honeywell directory and for packages that contain a provider entry that indicates the packages are from Honeywell.

The identified packages are displayed in a window as shown in the following figure:



There are two selectable types of uninstall:

- **Automated:** Uninstalls selected packages one by one without any confirmations/questions.
- **Interactive:** For every selected package, the uninstaller asks for a confirmation before uninstallation, and reports any errors during uninstallation.

If there are packages that cannot be uninstalled, then remove them manually through Add/Remove Programs.



REFERENCE - INTERNAL

For the procedure to uninstall existing TPS software, refer to section 4.16, "Un-installation of GUS or APP software".

1. Getting started

1.2. Implementation guidelines and recommendations

TPS node initialization guidelines

Initialization of a TPS node reformats the hard drive, loads the Windows 2003 or Windows XP operating system, and restores the node to the Honeywell factory-default (minus TPS Base).



ATTENTION

Use the proper Microsoft Operating System CD and the applicable Honeywell **EXPPlus Initialization DVD** to install the operating system and hardware drivers to Honeywell factory-default, and then use the **TPS Software CD** to manually load TPS software to Honeywell factory-default.

The initialization procedure in this guide provides instructions for both operations.

Refer to the TPS R4xx software change notice for the applicable **EXPPlus Initialization DVD** release for your TPS release.

Reasons to initialize a TPS node are listed below:

- For a clean installation of Windows 2003 or Windows XP operating system, Honeywell recommends that you perform the Honeywell initialization procedure.
- You replaced a TPS node with a new node that does not have the appropriate operating system.
- You replaced a node's hard drive and want to restore the Honeywell factory-default.
- To restore a corrupted hard drive and to restore the Honeywell factory-default.

Before you initialize a node, you should review the initialization procedure and, for existing nodes, backup the node's data files.



REFERENCE – INTERNAL

For the initialization procedure, refer to section 4.17, "Initializing TPS node using EXPPlus DVD".



REFERENCE – EXTERNAL

For details to backup up a TPS node, refer to the *Experion/TPS Backup and Restore User's Guide*.

After you initialize a node and the factory-default TPS software is installed, you can do either of the following to complete the node implementation:

- Complete the "New TPS Node Implementation checklist" in this guide, or
- Do Network Install of TPS software through Add/Remove Programs.

Note: You must have completed the Administrative Installation procedures.



REFERENCE – INTERNAL

For the Administrative Installation procedures to support Network Installs, refer to the *TPS System Administration Guide*.

1. Getting started

1.2. Implementation guidelines and recommendations

Windows 2003 migration guidelines

Summary of TPS System migration tasks

The following table summarizes the tasks to migrate a Windows 2000-based TPS System to Windows 2003/XP:

Task	Description
1	Upgrade the TPN (LCN), if necessary.
	ATTENTION Before making any kind of TPS migration, you should know whether the TPS release is compatible with the TPN (LCN) release. Consult the version compatibility charts in the front of the TPS Software Change Notice (SCN) for your system. The charts specify what TPN (LCN) releases are compatible with a specific TPS (GUS or APP) release.
2	Migrate the Windows 2000 Domain Controller to a Windows 2003 Domain Controller.
3	If not already installed, install and configure DNS server on the Active Directory Domain Controller.
	ATTENTION If possible, it is preferable to reinitialize the nodes using the appropriate initialization media for steps 5 and 6.
4	Migrate each TPS Windows 2000 Workstation to Windows XP.
5	Migrate each TPS Windows 2000 Member Server to Windows 2003 Member Server.
6	After you are sure the domain is operating correctly, migrate each Windows 2000 peer Domain Controller to Windows 2003 peer Domain Controller.

Use this best practice for TPS/TPN migration

To avoid software compatibility issues and other basic issues that can arise during a migration, review any SCNs packaged with the software release.

**ATTENTION**

Before making any kind of TPS release migration, verify the TPS release is compatible with the TPN (LCN) release.

Consult the version compatibility chart in the front of the Software Change Notice for your system.

Overview of network-based installations

You may choose to install the TPS software through local or network installations:

- Local: The user installs the software on the local machine from the TPS Software CD. This method requires the user to visit each target node and perform the installs sequentially.
- Network-based: The user initiates an automated software installation from a network server to the target machine. To setup this type of installation, the system administrator must establish a distribution point (an accessible place on the network to host the TPS software for deployment); and perform an Administrative Installation (prepare the TPS software packages to allow them to be installed from the distribution point). This method allows multiple computers to be installed simultaneously over the network.

**REFERENCE - INTERNAL**

For details on the Administrative Installation procedures that are required to prepare for network installation of TPS software, refer to the *TPS System Administration Guide*.

1. Getting started

1.3. Summary of common component software

1.3 Summary of common component software

The following table describes the common component software packages available on the **TPS Software CD**, and lists the document to install and configure each common component.

Common components are software packages that are available on both the TPS Software CD and the Experion Applications DVD. The common components are for use by TPS nodes, Experion nodes, and other nodes as needed.

The availability of specific software packages depends on the license key you enter.

Note: Before you install common component software for the first time, you should review system planning documents.

Table 1 Common Component Software

Software	Description	Document
Fault Tolerant Ethernet Driver	An intermediate driver used for implementing a dual (fault tolerant) Ethernet network.	FE05 – Fault Tolerant Ethernet Planning, Installation and Service Guide "Installation of FTE Node Components"
System Management Display	A user interface that allows a user to view the status of all nodes, and configure/control components, in one or more TPS Organizational Units.	EX08 – System Management Configuration Guide (read this first) Read sections – "Getting Started With System Management" "Installing System Management Tools"
System Management Runtime	A Honeywell infrastructure that is required for all other System Management functions, such as System Management Display	
HCI Runtime	A Honeywell Communications Interface that implements OPC Data Access functionality. HCI also provides several extensions to OPC Data Access.	
System Performance Server	An HCI/OPC Data Access server that provides an HCI/OPC interface to performance and configuration information that is internally maintained on a per-node basis using WMI.	

Software	Description	Document
System Event Server	An HCI/OPC event-managed server that integrates Windows XP system events into the Alarm and Event subsystem. Event examples include HCI Server status changes, Experion or TPS node failures, and Experion or TPS configuration changes.	
TPS Domain/Console Configuration Tool	A utility program that creates a TPS Organizational Unit (OU). A TPS OU contains a group of nodes that have some common attributes, for example: same plant or same area/unit.	EX08 – <i>System Management Configuration Guide</i>
TPS Domain Structure Migration to OUs	A utility program that converts a legacy TPS Domain structure to a TPS OU. This is used when you migrate a system that used a legacy TPS Domain.	
Redirection Manager	A software component that provides an OPC client with failover capability to redundant OPC data servers, such as TPN Server or Experion OPC Server.	EX20 – <i>Redirection Manager User's Guide</i>

1. Getting started

1.4. Documentation summary

1.4 Documentation summary

The following table briefly describes documents that are helpful when installing and configuring software installed from the TPS Software CD.

Table 2 Document Descriptions

Document	Description
Most recent SCN	
<i>The Software Change Notice (SCN) document contains an overview of new functionality along with recommendations, requirements, and special considerations for the latest software release.</i>	
TPS SCN	The TPS software change notice (SCN) document for the TPS Software CD ships with the software and can be obtained from the Honeywell SSOL website. Starting with the TPS 400 release, these previously separate SCNs are combined into one TPS SCN: - GUS SCN - APP SCN - Common Components SCN - File Transfer SCN - GUS ProTrend/ProMenu SCN
Common Component documents	
<i>System Management Configuration Guide (EX08)</i>	Describes the purpose of and uses for System Management as well as defining planning and system requirements. Includes installation and configuration procedures and some troubleshooting information.
<i>System Management Operations Guide (EX07)</i>	Explains the purpose of and uses for System Management. Includes procedures for managing components and monitoring status.
<i>FTE Status Server and Auxiliary Display User's Guide (EX22)</i>	Provides an overview of the FTE Status Server and Auxiliary Display including uses. Includes installation and operation procedures.
<i>FTE Overview and Implementation Guide (FE02)</i>	Provides an overview of Fault Tolerant Ethernet including system planning, requirements and FTE terminology. Also includes network architecture, guidelines, and requirements.
<i>FTE Installation and Service Guide (FE05)</i>	Includes installation and configuration procedures as well as troubleshooting information for Fault Tolerant Ethernet.

Document	Description
TPS System documents	
<i>TPS System Planning Guide for Windows 2003/XP (TP10X)</i>	Provides guidelines for planning and defining a TPS system, including system level planning, network planning, security, and performance considerations. Intended to be used in conjunctions with TPN (LCN) planning guides and other TPS system applications specific planning topics.
<i>TPS System Administration Guide for Windows 2003/XP (TP06X)</i>	Provides procedures for TPS system and TPS node administrative activities.



REFERENCE - INTERNAL

For additional documentation for TPS software packages, see section 4.11, "Documentation for configuring installable TPS software packages".

1. Getting started

1.5. System Management tasks and tools

1.5 System Management tasks and tools

The following table lists the tools and related documentation to perform the various System Management tasks:

Table 3 System Management Tasks

Tool	Task	Documentation
Alias Generator	Configure (create) an alias file when you need to access the name service repository of a remote node that is not within the scope of the multicast group.	<i>System Management Configuration Guide</i> (EX08)
Configuration Utility	Configure your local nodes if the System Management Display is not available for the TPS system.	<i>System Management Configuration Guide</i> (EX08)
Event Filtering Snap-In	Configure (specify) an event you are interested in viewing – to receive notification when a system or process event occurs.	<i>System Management Configuration Guide</i> (EX08)
FTE Status Server and Auxiliary Display	Monitor status of FTE nodes.	<i>Fault Tolerant Ethernet Status Server and Auxiliary Display User's Guide</i> (EX22)
HCI Configuration	Configure an HCI server from an installed base component. Re-configure an existing or previously configured HCI server. Remove the configuration of an HCI server.	<i>System Management Configuration Guide</i> (EX08)
HCI Name Service Alias Generator	Configure (create) a file of alias names, which is used by the name service provider on that node to publish the alias names of servers outside the Multicast Scope.	<i>System Management Configuration Guide</i> (EX08)
HCI Name Service display	Access the HCI Name Service display when you need to examine a list of available published HCI/OPC alias names. Remove alias names from a computer that is no longer available.	<i>System Management Configuration Guide</i> (EX08)

Tool	Task	Documentation
Node Administration Snap-In	Manage, monitor, administer, and troubleshoot Honeywell nodes. • Monitor node and component status. • Create TPS Domain OU and Console OU. • Manage Organizational Units and computers. • Configure HCI/OPC Components. • Administer HCI Name Service. • Troubleshoot System Management.	<i>System Management Configuration Guide (EX08)</i>
System Management Display	Configure and monitor managed nodes and their HCI managed components.	<i>System Management Configuration Guide (EX08)</i>
System Management Display	Configure the TPN Server as an HCI component.	<i>Configuring the TPN Server in the TPN Server User's Guide, TP11X</i>
System Management Display	Start or stop the TPN server.	<i>Monitoring TPN Server Status in the TPN Server User's Guide, TP11X</i>
System Management Display	Use the TPN Server's Auxiliary Status Display.	<i>Auxiliary Status Display - TPN Server in the TPN Server User's Guide, TP11X</i>
System Management Display	Manage component operations.	<i>HCI Managed Component Operations in the System Management Operations Guide, EX07</i>

1. Getting started

1.6. Configuration data sheets

1.6 Configuration data sheets

Definition of TPS node types

This table summarizes the functionality that defines each TPS node type. The procedures in this guide state whether a procedure applies to a particular node type or not; consequently, you must be aware of your node type to perform the procedures properly.

TPS R4xx node type	Connected to Ethernet network?	Member of domain?	This is shown in Native Window status displays	Comments
Network GUS	YES	NO	GUS	Supports GUS and Native Window functionality. Does not support HCI. Although it is not a requirement, it may be a member of a Windows 2003 domain. Does not require the Honeywell High Security model and Honeywell-specific settings on the Domain Controller, so users are free to configure the Domain Controller as necessary to comply with site-specific policies.
Standalone GUS	NO	NO	GUS	Supports GUS and Native Window functionality. Ethernet network and Domain Controller are not present.
AM Replacement (AMR)	NO	NO	AM	Supports standard TPN (LCN) AM functionality only.
US Replacement (USR)	NO	NO	UNVL or OPR	Supports standard TPN (LCN) US (Native Window) functionality only.

TPS R4xx node type	Connected to Ethernet network?	Member of domain?	This is shown in Native Window status displays	Comments
TPS GUS	YES	YES	GUS	Supports GUS, Native Window, and HCI functionality. Must be a member of a Windows 2003 domain. Requires Honeywell High Security model and Honeywell-specific settings on the Domain Controller.
APP Member Server	YES	YES	APP	Supports APP functionality. Must be a member of a Windows 2003 domain. Requires Honeywell High Security model and Honeywell-specific settings on the Domain Controller.

Domain Controller configuration data sheet

Prior to implementation of a Domain Controller for TPS nodes, fill in this blank configuration data sheet with your site-specific data. Procedures in this document refer to this data sheet when site-specific data must be entered into the system.

Data description	Your system data
Node Options	
Hardware Platform Type:	
Ethernet Identification	
Computer Name: The Computer Name should be unique and conform to DNS naming standards (underscores are not permitted).	
Domain Controller Role (root or child):	

1. Getting started

1.6. Configuration data sheets

Data description	Your system data
Windows 2003 Domain Name: The name of the Windows 2003 domain that is to contain the TPS nodes. If the DC is the first DC in a new forest, append “.local” to domain name.	
 TIP DNS Naming Conventions The domain and computer names can be any combination of A-Z letters, 0-9 numerals, and the hyphen character. Underscores are not allowed.	
IP Address: The IP address should conform to site-specific standards. If the network is isolated, use the private 192.168.1.X subnet.	
Subnet Mask: The subnet mask is relative to your network configuration. Typically, the subnet mask is assigned to 255.255.255.0.	
Default Gateway: The default gateway should conform to the site-specific network topology. If the network is isolated, do not configure the default gateway.	
DNS Server Addresses: If you do not have a DNS Server that supports Dynamic DNS, you must establish one in your own domain. The Windows 2003 Domain Controller installation and migration procedures can automatically create a new Dynamic DNS service.	
WINS Server Addresses: Typically WINS is not configured in Windows 2003 environments. If you support legacy Windows NT 4.0 domain architectures, you may need to configure WINS servers.	
Windows 2003 domain administrator name/password	

TPS Node configuration data sheet

Prior to the implementation of a TPS Node, fill in this blank configuration data sheet with your site-specific data. Procedures in this document refer to this data sheet when site-specific data must be entered into the system.



TIP

If you plan to install more than one TPS node, make multiple copies of these pages.

Data description	Your system data
TPS Node Identification	
Honeywell Node Type:	
Windows Domain Name: For nodes that are to be members of a Windows Domain, enter the name of the Domain that is to contain the node.	
TPS Domain OU Name: For nodes that are to be members of a TPS Domain OU, enter the name of the TPS Domain OU that is to contain the node.	
TPS Software Release:	
TPN (LCN) Identification	
TPN (LCN) Software Release:	
LCNP Address: Valid identifier: Integer from 0 to 96	
LCNP Board Name:	
NG Name: If you do not have an NG (Network Gateway), then enter a space.	
Ethernet Identification	
Computer Name: The Computer Name should be unique and conform to DNS naming standards (underscores are not permitted).	
Domain Name: For nodes that are to be members of a Windows Domain, enter the Windows 2003 domain that is to contain the node.	

1. Getting started

1.6. Configuration data sheets

Data description	Your system data
IP Address: The IP address should conform to site-specific standards. If the network is isolated, use the private 192.168.1.X subnet.	
Subnet Mask: The subnet mask is relative to your network configuration. Typically, the subnet mask is assigned to 255.255.255.0.	
Default Gateway: The default gateway should conform to the site-specific network topology. If the network is isolated, do not configure the default gateway.	
DNS Server Addresses: Windows 2003 requires domain members to use DNS servers supporting dynamic DNS (addresses should be the same as the primary and backup Dynamic DNS Servers configured on the Domain Controller for this node's domain).	
WINS Server Addresses: Typically WINS is not configured. To support legacy Windows NT 4.0 domain architectures, you may need to configure WINS servers.	
Administrator username and password to join Windows 2003 domain: (or an administrator can add the node to the domain using Active Directory tools):	
Node Options:	
Windows Operating System Type: (Windows 2003 or Windows XP)	
Honeywell Hardware Platform Type: (WS470-based Honeywell workstation, WS1600SC-based Honeywell server, Other)	
Multi-Monitor Orientation: (Side-by-Side, Over-Under, or Two-by-Two for Honeywell Icon Series console)	
Multi-Monitor Card: (None or Matrox) - Matrox G450 Multi-Monitor Series (MMS). Both 2 and 4 monitor options. - Matrox QID. Supports 4 monitor option only.	
Integrated Keyboard (IKB): (Yes or No)	
Icon Series Console Operator Entry Panel (OEP): (Yes or No)	
Touch Screen: (None, MicroTouch, CarrollTouch or SAW)	
Iomega Zip Drive: (Yes or No)	

1.7 Windows domain implementation overview

Creating a Windows domain

A domain is the building block of a Windows network. It can exist in itself or be a part of a tree or forest.

A domain with its associated name is created when a server computer is promoted to a Domain Controller. When a server is promoted as a Domain Controller, you must decide whether it must be a root Domain Controller (create a root domain) or a child Domain Controller (create a child domain in the specified domain).

CAUTION If a domain is required for your TPS system, create a separate Domain Controller for control systems. This control system domain should not be in the same Active Directory tree or forest as the enterprise domain.

Tasks to implement a new domain

The following table summarizes the tasks required to implement a new domain.

High level task	Reference
Install Windows 2003 Server. Configure the following: - Administrator password - Date/Time - Computer Name - TCP/IP settings	Refer to section 2.2 "New Domain Controller implementation checklist".
Use DCPROMO to promote your server as a Root Domain Controller.	Refer to section 2.2 "New Domain Controller implementation checklist," subsection "Begin Promotion of Windows 2003 Server to Root DC".
Install Active Directory and DNS for root Domain Controller.	Refer to section 3.3, "Installing Active Directory and DNS for root Domain Controller".
Install additional software and servers on Domain Controller that are appropriate for your domain. It varies according to the requirements. Note: A TPS system does not require any additional software or servers on the Domain Controller.	

1. Getting started

1.7. Windows domain implementation overview

High level task	Reference
Install the Honeywell High Security Domain Controller package.	Refer to section 3.5, "Installing Domain Controller package on DC for Honeywell High Security Policy".
Install the Honeywell TPS Domain/Console (OU) Configuration Tool.	Refer to section 3.10, "To install 'TPS domain/console configuration tool' on DC".
Create Users and Groups for TPS.	Refer to section 3.6, "Configuring domain users and groups for TPS".
Create the OUs that are to contain the TPS nodes.	Refer to section 3.10, "Performing 'TPS domain' OU procedures.
Assign the "TPS Domain" attribute to an OU. Assign the "Console" attribute to an OU if needed.	Refer to section 3.10, "Performing 'TPS domain' OU procedures.
Customize Group Policies if needed.	Refer to section 3.13, "Implementing Group Policy".
Implement the TPS nodes	Refer to section 2, "TPS implementation checklists". Refer to section 4.3, "To add a computer to a domain". Refer to section 4.19, "Performing LinkDomainGroups".
Implement a Peer Domain Controller.	Refer to section, 3.14, "Implementing a Peer Domain Controller".



REFERENCE – INTERNAL

For a general description of user and computer accounts, OUs, and security groups, refer to the [TPS System Planning Guide](#).

For information on changing the TPS user names and groups, see [TPS Security](#), in the [TPS System Administration Guide](#).

2. TPS implementation checklists

2.1 Selecting the correct checklist

The following flowchart shows how to select the correct checklist to guide you through the implementation tasks. Follow the path from the top to the bottom to locate the checklist that best fits your situation.

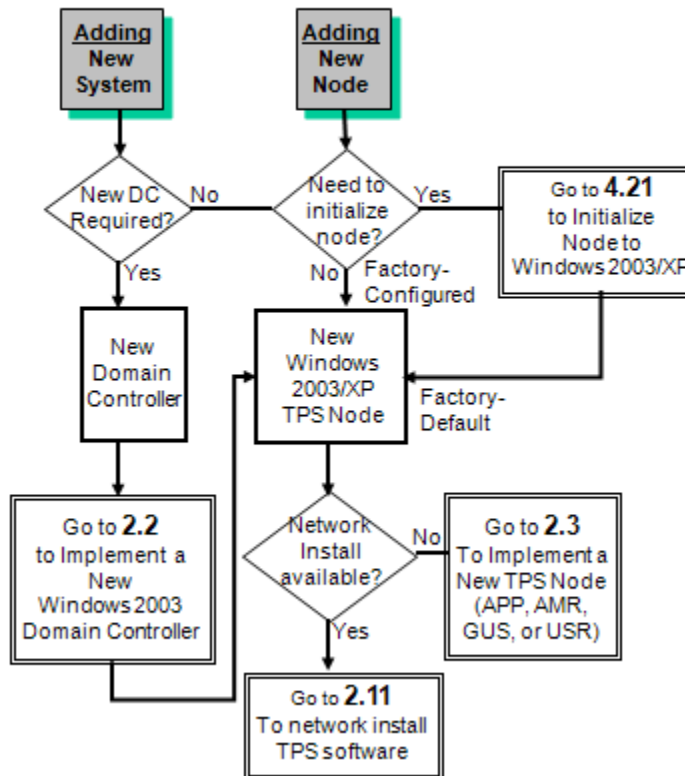
Use this flowchart to add New systems and nodes:

1) Pick one gray box:



2) Follow the path that best fits your situation.

3) Go to the specified sections and do the steps in the checklists.



2. TPS implementation checklists

2.1. Selecting the correct checklist

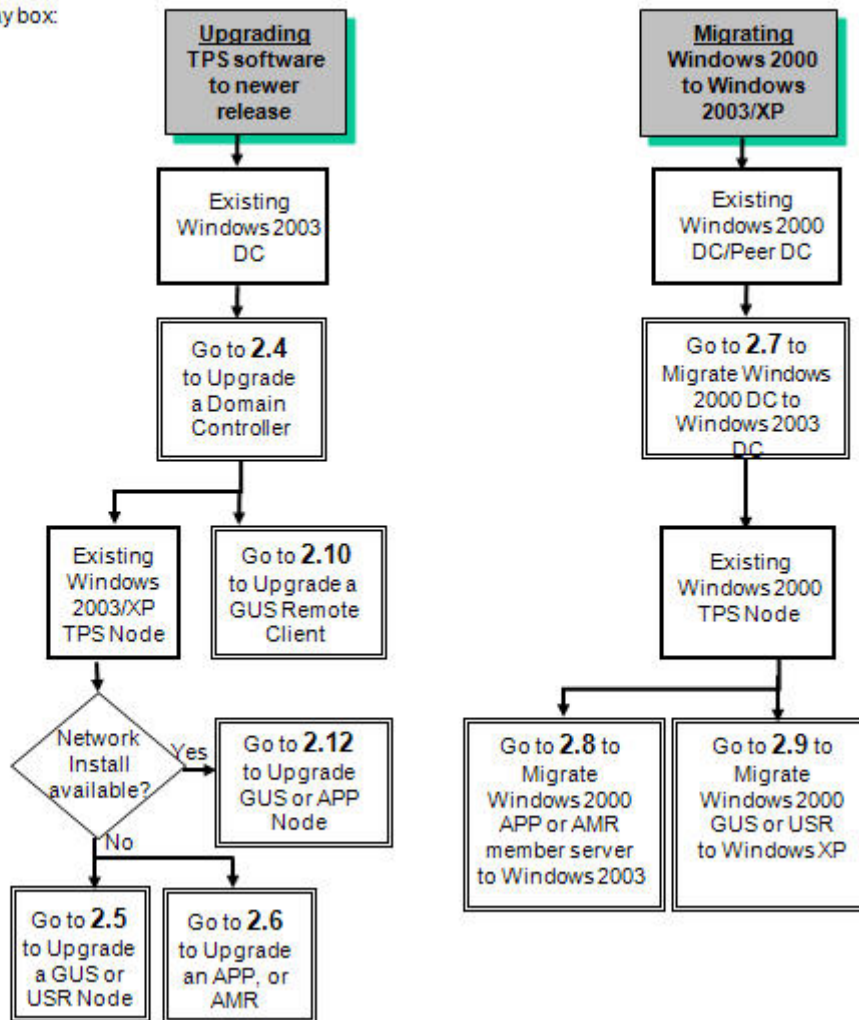
Use this flowchart for Upgrades and Migrations:

1) Pick one gray box:



2) Follow the path that best fits your situation.

3) Go to the specified sections and do the steps in the checklists.



2.2 New Domain Controller implementation checklist

Purpose of this checklist

This checklist guides you through the steps required to implement a new TPS Domain Controller as a root Domain Controller. The flowchart in section 2.1, "Selecting the correct checklist" illustrates when you would choose to use this checklist.



ATTENTION

The instructions in this checklist are based upon a single domain model with one Windows 2003 Domain Controller controlling all the members of the domain.



TIP

With TPS R420, Domain Controllers are supported on Windows 2008.



REFERENCE - INTERNAL

Refer to the *TPS System Planning Guide* for additional Honeywell guidelines on Windows 2003 domain architectures.

Guidelines for using the checklist

Follow the steps in the checklist. Some steps refer to another procedure in this document, and then require that you return to the checklist. You may want to make copies of this checklist if you are working on more than one node. Place a checkmark in the right column once you have completed a step.

Before you begin

To perform this implementation, you need the following items:

- A retail copy of the Windows 2003 Server – Standard Edition media, if Windows 2003 is not already installed
- TPS Software CD
- *Software Change Notice*
- *TPS System Administration Guide*

2. TPS implementation checklists

2.2. New Domain Controller implementation checklist

- *System Management Configuration Guide*
- A completed Domain Controller Configuration Data Sheet for the machine to be implemented. The Domain Controller Configuration Data Sheet is in section 1.6.

New Domain Controller implementation checklist



TIP

If you leave the checklist to go to a procedure in another section, ALWAYS RETURN TO THE CHECKLIST.

Step	Action	✓
Install Windows 2003 Server, if necessary		
(If the machine already has Windows 2003 Server installed, skip steps 1 and 2.)		
1	Insert the Windows 2003 Server media and boot the machine to begin the installation process.	
2	At each Windows 2003 Server prompt, answer the questions as appropriate. When Windows 2003 Server is installed, continue to the next step.	
Install Microsoft Service Pack		
3	If not already installed, install the latest Microsoft Windows 2003 Service Pack that is available from Honeywell for TPS systems, as defined in the TPS R4xx Software Change Notice.	
Configure local Administrator Password, if necessary		
(The Administrator password normally gets entered during installation of Windows 2003 Server. When a factory-configured node is powered on for the first time, user is prompted to enter an administrator password.)		
4	Log on to the machine as Administrator.	
5	Press <Ctrl> <Alt> <Delete> and change the password, if necessary.	

CAUTION

Record and store the domain administrator password in a secure place. If you forget the password, you have to reinstall Windows 2003 Server to recover.

Step	Action	✓
Set Date and Time, if necessary		
(This is normally done as part of the Windows 2003 installation. The time on your system may already be correct and you may not need to do this step.)		
6	Display the Control Panel: Start > Settings > Control Panel . Set the Time Zone, Date, and Time as needed.	
Change Computer Name, if necessary		
7	Display the System Properties and change the name of the node, if necessary: a) Right-click the My Computer icon on Start menu and select Properties. b) Select the Computer Name tab and click Change. c) Change the computer name of the server. d) Reboot the machine.	
Configure TCP/IP Settings		
8	Log on to the machine as Administrator.	
9	Configure TCP /IP Settings: IP Address, Subnet Mask, Default Gateway, DNS Servers, and WINS Servers. For detailed instructions for configuring the TCP/IP settings, see Section 3.2, "Configuring TCP/IP settings".	
	TIP For the actual data that needs to be entered, refer to your <i>Domain Controller Configuration Data Sheet</i> that was completed prior to beginning this procedure.	
Connect Network Cable		
10	Physically connect the network (Ethernet) cable, if not already connected.	
Begin Promotion of Windows 2003 Server to Root DC		
11	To begin the promotion of the standalone Windows 2003 server machine to a root or peer Domain Controller, run the Microsoft application dcpromo.exe: Start > Run , type dcpromo , and click OK . RESULT: The dcpromo application initiates the Active Directory Installation Wizard.	

2. TPS implementation checklists

2.2. New Domain Controller implementation checklist

Step	Action	✓
<i>Install Active Directory and DNS</i>		
12	At the Active Directory Installation Wizard, enter the appropriate configuration to install the Active Directory for a Root Domain Controller and install DNS, if necessary. For detailed instructions, refer to section 3.3, "Installing Active Directory and DNS for root Domain Controller".	
<i>Add a Reverse Lookup Zone to DNS</i>		
13	Using the DNS MMC, add a Reverse Lookup Zone: Start > Administrative Tools > DNS. For detailed instructions to add a Reverse Lookup Zone, refer to section 3.4, "Adding a Reverse Lookup Zone To DNS".	
<i>Install Domain Controller Package</i>		
14	Insert the TPS Software CD and click Domain Controller Package (under the Experion PKS – High Security Policy menu item) to establish a set of default Honeywell global groups (such as Operators, Supervisors, Engineers, and View) and security policies. For detailed instructions to install TPS security, see Section 3.5, "Installing Domain Controller package on DC for Honeywell High Security Policy".	
<i>Install TPS Domain/Console Configuration Tool</i>		
15	Using the TPS Software CD, install the TPS Domain/Console Configuration Tool. For detailed instructions see Section 3.10, "To install 'TPS domain/console configuration tool' on DC".	
<i>Copy and Edit HOSTS and LMHOSTS Files</i>		
16	Use Notepad to copy and edit the HOSTS file. Modify the HOSTS file to include the GUS nodes, APP nodes, and Domain Controllers. For detailed instructions, see Section 3.7, "Editing the HOSTS and LMHOSTS files", "Using Notepad to edit the HOSTS file".	

Step	Action	✓
17	Use Notepad to copy and edit the LMHOSTS file. Modify the LMHOSTS file to include the GUS nodes, APP nodes, and Domain Controllers. For detailed instructions to edit the LMHOSTS file, see Section 3.7, "Editing the HOSTS and LMHOSTS files", subsection "Using Notepad to edit the LMHOSTS file".	
Create users and groups for TPS		
18	Create users and groups. Customize the TPS users and groups, as necessary. See section 3.6, "Configuring domain users and groups for TPS".	
 REFERENCE – INTERNAL For instructions on customizing TPS users and groups, see "TPS Security", in the <i>TPS System Administration Guide</i>.		
Perform 'TPS domain' OU procedures		
19	Create the OUs that are to contain the TPS nodes. See section 3.10, "To create a new OU". Or If applicable, see section 3.10, "To migrate existing legacy TPS domain structures to Active Directory OUs".	
20	Assign the "TPS Domain" attribute to the appropriate OU. See section 3.10, "To assign the 'TPS domain' attribute to an OU".	
21	Assign the "Console" attribute to the appropriate OU. See section 3.10, subsection "To assign the 'console' attribute to an OU within a 'TPS Domain' OU".	
Implement Group Policy		
22	Customize the group policies as needed for your site. Refer to section 3.13, subsection, "Implementing Group Policy".	
Create and Edit the COPY_UTILITY.BAT File		
The COPY_UTILITY batch file can be used to distribute the HOSTS, LMHOSTS, and Operator.bat files to all members of the Domain Controller's Windows 2003 domain.		

2. TPS implementation checklists

2.2. New Domain Controller implementation checklist

Step	Action	✓
23	To ensure all nodes receive the necessary files, create the COPY_UTILITY.BAT file and modify it to include all GUS, APP, and peer Domain Controller nodes. For detailed instructions, see Section 3.9, "Editing the COPY_UTILITY.BAT file". Implement TPS Nodes 24 Implement the GUS and APP nodes, add them to the domain, and move them into the appropriate OUs, as necessary. For detailed instructions to implement the TPS nodes, refer to section 2.1, "Selecting the correct checklist". Perform Backup Procedures 25 Backup the Domain Controller and TPS nodes, as necessary.	
	REFERENCE – INTERNAL For descriptions of data to be backed up, refer to the <i>Backup and Recovery Guidelines</i> section of the <i>TPS System Administration Guide</i>.	
Implement TPS Peer Domain Controller, if necessary 26 After the Windows 2003 domain is operating as desired, implement a peer Domain Controller, if necessary. For details, refer to section 3.14, "Implementing a Peer Domain Controller". Conclusion 27 The implementation of the new Windows 2003 TPS Domain Controller is complete.		

2.3 New TPS node implementation checklist

Purpose of this checklist

This checklist guides you through the steps required to implement a new TPS Node. For example, you would use this checklist to implement any of the following node types as a new node: TPS GUS, Network GUS, Standalone GUS, US Replacement, APP, and AM Replacement. A specialized APP node used to integrate TPS and Experion systems, conforms to the APP node implementation procedures. The flowchart in section 2.1, "Selecting the correct checklist" describes when you would choose to use this checklist.

Guidelines for using the checklist

Follow the steps in the checklist. Some steps require you to refer to another procedure in this document, and then return to the checklist. Some steps refer to other documents that provide additional background information. Most steps identify the node types to which the step applies; if you are working on a different node type, skip to the next step. You may want to make copies of this checklist if you are working on more than one node. Place a checkmark in the right column once you have completed a step.

Before you begin

To perform this implementation, you need the following items:

- TPS Software CD
- GUS-TPN Software CD, to implement a GUS or US Replacement node
- Application Module CD, to implement APP, or AM Replacement
- Utilities and Load Modules Software CD
- *Software Change Notices (SCNs)*
- *TPS System Administration Guide for Windows 2003/XP*
- *System Management Configuration Guide*
- a completed TPS Node Data Sheet for the node to be implemented (see Section 1.6, "Configuration data sheets")

2. TPS implementation checklists

2.3. New TPS node implementation checklist

Prerequisites



ATTENTION

TPS GUS and APP nodes require a Windows 2003 Domain Controller. Before you start this procedure, if you plan to implement TPS GUS and APP nodes, verify that the Domain Controller was migrated to Windows 2003 or that the Windows 2003 Domain Controller exists and is available on the network.

New TPS Node implementation checklist

Prerequisites

The following prerequisite software must have been installed before using this checklist:

- Experion PKS – High Security Policy – Workstation Package
- TPS Base (GUS Base or APP Base)

A Honeywell factory-configured node meets the prerequisites.

If you performed a node initialization with the EXPPlus DVD, then you must manually install the prerequisite software from the TPS Software CD. The node initialization procedure in section 4.17 includes the steps to install the TPS Base software and the Workstation security package. Make sure that you perform these manual steps before using this checklist.



TIP

If you leave the checklist to go to a procedure in another section, ALWAYS RETURN TO THE CHECKLIST.

Step	Action	✓
Configure Local Administrator Password, if necessary		
(The administrator password is normally entered during installation. When a factory-configured node is powered on for the first time, the user is prompted to enter an administrator password.)		
1	Log on to the machine as Administrator.	
2	Press <Ctrl> <Alt> <Delete> and change the default password, if necessary.	
	CAUTION: Record and store the domain administrator password in a secure place. If you forget the password, you have to reinstall Windows 2003/XP to recover.	
	ATTENTION: Do not rename the local administrator account.	

Step	Action	✓
Set Date and Time, if necessary		
(This is normally done as part of the Windows 2003/XP installation. The time on your system may already be correct and you may not need to do this procedure.)		
3	Display the Control Panel: Start > Settings > Control Panel . Set the Time Zone, Date, and Time as needed.	
Change Computer Name		
4	Display the System Properties and change the name of the node, if necessary: a) Right-click the My Computer icon from Start menu and select Properties. b) Select the Computer Name tab. c) Change the computer name of the machine. d) Reboot the machine.	
Configure TCP/IP Settings		
5	Log on to the machine as Administrator. Configure TCP/IP Settings: IP Address, Subnet Mask, Default Gateway, DNS Servers, and WINS Servers. For detailed instructions to configure the TCP/IP settings, see Section 4.2, "Configuring TCP/IP settings".	
	TIP: For the actual data that needs to be entered, refer to your <i>TPS Node Configuration Data Sheet</i> that was completed prior to beginning this procedure.	
Connect Network Cable		
6	Physically connect the network (Ethernet) cable, if not already connected.	

2. TPS implementation checklists

2.3. New TPS node implementation checklist

Step	Action	✓
Edit HOSTS and LMHOSTS Files, if necessary		
7	If not already done, add the local node name and IP address to the HOSTS and LMHOSTS files. If the node will be in a domain, then edit the HOSTS and LMHOSTS files on the Domain Controller and run COPY_UTILITY on Domain Controller to distribute the files to the other nodes. For detailed instructions to edit the files, see section 3.7, "Using Notepad to edit the HOSTS file". For detailed instructions for distributing the files, see section 4.9, "Distributing HOSTS, LMHOSTS and Operator.bat files". If the node will not be in a domain, then edit the HOSTS and LMHOSTS files and use standard site procedures to place them in the correct locations. For detailed instructions to edit the files, see Section 3.7, "Using Notepad to edit the LMHOSTS file".	
Join Windows 2003 Domain		
8	This step is required only for TPS GUS, Network GUS, and APP nodes. Join the Windows 2003 domain.	
 TIP Domain configuration can be found by right-clicking the My Computer icon on the desktop and selecting Properties . Select the Computer Name tab and click Change .		
For detailed instructions to join a computer to a domain, see section 4.18, "To add a computer to a domain".		
Configure Local User Accounts		
9	Configure the local user account (TPS Local Server account) and the logon script, if needed. For detailed instructions to configure the local accounts, see Section 4.3, "Configuring local user accounts for TPS".	

Step	Action	✓
<i>Install Microsoft .NET Framework</i>		
10	This step is required only for GUS and APP nodes (not USR and AMR nodes). Using the TPS Software CD, install the .NET Framework Redistributables package, and then restart the machine. For detailed instructions, refer to section 4.20, "Installing Microsoft .NET Framework".	
	TIP A Honeywell factory-configured node or a node that you initialized with the EXPPlus DVD/TPS CD already has .NET installed. If needed, refer to Microsoft documentation for the steps to determine the version(s) of .NET that are installed on the node.	
<i>Install System Management Software Package</i>		
<i>This step is required only for TPS GUS nodes and APP nodes.</i>		
11	Using the TPS Software CD, install the Honeywell System Management Display (under the Install Packages menu item). For the Honeywell System Management Display software package usage (optional, mandatory, or not allowed) on each node type, see Section 4.11 TPS Software package usage by node type. For detailed instructions to install the System Management Display package, refer to section 4.13, <i>Install System Management Display Software Package</i>.	
<i>Install Redirection Manager (optional)</i>		
12	Using the TPS Software CD, install the Redirection Manager, if needed on the node, and as authorized by your license. For detailed instructions to install Redirection Manager, refer the <i>Redirection Manager User's Guide</i>. For the Redirection Manager usage (optional or not allowed) on each node type, see Section 4.13, TPS Software package usage by node type.	
13	If it was not done in the previous step, restart the machine.	

2. TPS implementation checklists

2.3. New TPS node implementation checklist

Step	Action	✓
Add TPS Global Groups to TPS Local Groups		
14	At the TPS node, log on as a user with Administrator privileges in the domain (for example, domain Administrator).	
15	This step applies only to TPS GUS, Network GUS, and APP nodes.	
	ATTENTION Execute the following script after you have made the node a member of a domain that supports the TPS global groups – that is, the domain controller has the Honeywell domain controller high security package installed. Navigate to the Program Files\Honeywell\WkStaSecurity directory and execute the LinkDomainGroups.vbs script. RESULT: This script file identifies standard global groups existing on a Domain Controller (such as Engineers, Supervisors, and Operators), and adds them to the equivalent local groups.	
Install TPS Software Packages		
These steps apply only to GUS nodes and APP nodes (not USR and AMR nodes).		
16	Using the TPS Software CD , install the desired TPS software packages authorized by your license. For detailed instructions to install TPS software packages, refer to section 4.13 <i>Install TPS Software Packages</i> , subsection <i>Procedure to Install TPS Software Packages</i> . TPS software packages requiring HCI Runtime to be installed prior to installing the package itself, are indicated with (H) in section 4.13 <i>TPS Software Package Usage by Node Type</i> .	
Add LCNP Board to Node		
17	Using the Honeywell TPS Add Board utility, remove and re-add the LCNP board to the node and give the board a name. For detailed instructions to remove/re-add the LCNP board, see Section 4.4, "Using AddBoard utility to add LCNP board to node".	

Step	Action	✓
<i>Install TPN (LCN) Personality Software on Local Drive</i>		
18	This step applies only to GUS and APP nodes. Depending on the node type being implemented, use the appropriate TPS CD to install either the GUS or APP personality software onto the node's local drive, and then restart the machine For detailed instructions to install the GUS or APP personality, see Section 4.5 <i>Install TPN Personality Software on Local Drive</i>.	
<i>Configure LCNP-Related Settings</i>		
19	Using the TPS Configuration Utility, configure the LCNP-related settings, as applicable. Note: If you are doing this step as part of a migration, you may have already recorded the Configuration Utility settings that existed prior to the migration. For detailed instructions to configure LCNP-related settings, refer to the following topics in the section 4.6 Configure LCNP (Board0) Settings: ___ Edit LCNP Settings (all nodes) ___ Edit Native Window Settings (GUS and US nodes) ___ Edit Keyboard Settings (GUS and US nodes) ___ Configure Printer Settings (GUS and US nodes)	
<i>Connect TPN (LCN) Cables</i>		
20	Connect the transceiver and TPN (LCN) cables, if not connected already. For detailed instructions to connect the TPN cables, see Section 4.7 "Connecting the transceiver and TPN cables".	
<i>Configure Signon Manager</i>		
21	Perform the Signon Manager configuration (required for all TPS nodes). For detailed instructions to configure Signon Manager, see Section 4.8 "Configuring Signon Manager".	

2. TPS implementation checklists

2.3. New TPS node implementation checklist

Step	Action	✓
<i>Move the node to the TPS Domain OU</i>		
22	This step applies only to TPS GUS nodes and APP nodes. At the Domain Controller with domain administration privileges, access Active Directory Users and Computers, and then do the following: a) If not already present, create a 'TPS Domain' OU. See section 3.10, "To assign the 'TPS domain' attribute to an OU". Or If applicable, migrate existing legacy 'TPS Domain' structures to OUs. See section 3.10, "To migrate existing legacy TPS domain structures to Active Directory OUs". b) Move the new computer (TPS node) to the 'TPS Domain' OU. See section 3.10, subsection "To move a TPS GUS or APP node to a 'TPS domain' OU".	
<i>TPN (LCN) Procedure – Load GUS as US, if necessary</i>		
23	This applies to systems where an existing GUS or US is not available to access the TPN History Module (HM) and NCF. Load a GUS as a US temporarily, to perform the TPN procedures. For detailed instructions to load a GUS as a US, see Section 5.1, <i>Load GUS as US from Emulated Disks</i>.	
<i>TPN (LCN) Procedure – Initialize History Module, if New TPN System</i>		
24	This applies only to new TPN Systems.  ATTENTION This step needs to be done only once per TPN. Use Native Window to Initialize the TPN History Module, if necessary. For detailed instructions to initialize the TPN History Module, see Section 5.2 <i>Use Native Window to Initialize the TPN History Module</i>.	

Step	Action	✓
TPN (LCN) Procedure – Copy Load Modules to History Module		
25	This step applies only to GUS nodes.  ATTENTION This step needs to be done only once per TPN. If it has not been done already as part of a previous GUS node implementation, use the Native Window to copy the contents of the &CUS subdirectory from CD to the TPN System History module. For detailed instructions to update the GUS files, refer to section 5.3 <i>Use Native Window to Copy GUS Load Modules to HM</i>. REFERENCE EXTERNAL: Refer to the TPS SCN for the versions of GUS TPN load modules that apply to your TPN release.	
26	This step applies only to APP nodes. ATTENTION: This procedure needs to be done only once per TPN. If you have not already done so as part of a previous APP node implementation, from a Native Window, copy the contents of the &CUS subdirectory from CD to the TPN System History module: For detailed instructions to update the APP files, refer to section 5.4 <i>Use Native Window to Copy APP Load Modules to HM</i>. REFERENCE EXTERNAL: Refer to the TPS SCN for the versions of APP TPN load modules that apply to your TPN release.	
TPN (LCN) Procedure – Configure Nodes in NCF		
27	Use a Native Window to update an existing NCF with the node configuration. For detailed instructions to configure a GUS node in the NCF, refer to section 5.5 <i>Use Native Window to Update an Existing NCF for GUS</i>. For detailed instructions to configure an APP node in the NCF, refer to section 5.6 <i>Use Native Window to Update an Existing NCF for APP</i>.	
TPN (LCN) Procedure – Configure Display Search Paths in Area Database		
28	This step applies only to GUS nodes. ATTENTION: This procedure needs to be done only once per Area. Use a Native Window to verify that the GUS display search path(s) are already configured (System Menu > Organizational Summary > Schematic Titles). If the search paths are not present, then configure the GUS display search paths in the appropriate Area Database. For detailed instructions to configure the Area Database, refer to section 5.7 <i>Using Native Window to update display search path in Area Database</i>.	

2. TPS implementation checklists

2.3. New TPS node implementation checklist

Step	Action	✓
Load LCNP Board With TPN (LCN) Personality		
29	Load the node's TPN Personality. For detailed instructions to load the GUS TPN Personality, refer to section 5.8 <i>Load Native Window (GUS or US) Personality</i>. For detailed instructions to load the APP TPN Personality, see Section 5.8 <i>Load the APP/AM Personality when a Checkpoint Does Not Exist</i>.	
30	This step applies only to APP nodes and AM Replacement nodes. From a GUS Native Window or US System Status display, perform the procedure to save the APP/AM checkpoint to the TPN History Module.	
Configure Additional TPS Functionality, as required		
31	Configure remote access to Native Window and LCNP Status, if needed. For detailed instructions to configure remote access functionality, refer section 6.1 <i>Remote Access Procedures</i>.	
32	Configure GUS local or network printer, if needed. For detailed instructions to configure GUS printer functionality, see Section 6.2 <i>Local or Network Printer Configuration</i>.	
33	Configure Windows 2003 offline files and folders for replication. ATTENTION: Files such as GUS display files and SafeView configuration files can be stored on a Domain Controller and distributed to network nodes through the Windows 2000/2003 "offline folders" capability (Windows NT 4.0 replication is no longer supported in Windows 2000/2003). For detailed instructions to configure offline files and folders functionality, see Section 6.3 <i>Configuring Windows 2003 offline files and folders for replication</i>.	
34	Configure GUS startup to ensure the desired results for operations personnel (Operator.bat logon script and SafeView workspace configuration). For detailed instructions to configure GUS startup, see Section 6.4 <i>Startup of a GUS Node for Operations</i>.	

Step	Action	✓
35	Set the node's Dr. Watson default crash dump paths to C:\TEMP. For detailed instructions to set the default paths, see Section 6.5 <i>Setting Dr. Watson Crash Dump Paths</i>.	
Configure Installable TPS Software Packages		
36	Perform the configuration necessary to fully implement the TPS Software Packages installed on the node. For a list of documents providing configuration instructions on each installable TPS Software package, see Section 4.11. "TPS Software Package Usage by Node Type," subsection "Documentation for Configuring Installable TPS Software Packages".	
Configure Honeywell System Management Display		
37	Add the Honeywell Node Administration Snap-in to the Microsoft Management Console (MMC). For detailed instructions, refer to section "Adding the Node Administration Snap-In to the MMC" in the <i>System Management Configuration Guide</i>. RESULT: The <i>default</i> configuration of your Honeywell System Management Display is complete.	
38	Configure the Node Administration Snap-in properties as necessary. For detailed instructions, refer to "Node Administration Properties" in the <i>System Management Configuration Guide</i>. RESULT: An MMC console (.msc) file is configured and saved with a name of your choice (the default is Console1). End users can view a new System Management Display by opening its .msc file through the Honeywell Node Administration MMC Snap-in.	
Perform Backup Procedures		
39	Backup the node, as necessary. For descriptions of data to be backed up, refer to "Node-Specific Application and Configuration Data Backup Guidelines" in the <i>TPS System Administration Guide</i>.	
Conclusion		
40	The implementation of the new TPS node is complete.	

2.4 Domain controller upgrade checklist

Purpose of this checklist

This checklist guides you through the steps required to upgrade an existing Windows 2003 Domain Controller in preparation for upgrading TPS nodes to TPS GUS R400 and greater and APP R400 and greater. The main purpose of the steps is to implement the TPS domain structures as organization units in Active Directory Users and Computers.

The flowchart in section 2.1, "Selecting the correct checklist" illustrates when you would choose to use this checklist.

Guidelines for using the checklist

Follow the steps in the checklist. You may want to make copies of this checklist if you are working on more than one node. Place a checkmark in the right column once you have completed a step.

Before you begin

To perform this implementation, you need the following items:

- Honeywell TPS Software CD
- *Software Change Notice*
- *TPS System Administration Guide*
- *System Management Configuration Guide*

Domain Controller upgrade checklist



TIP

If you leave the checklist to go to a procedure in another section, ALWAYS RETURN TO THE CHECKLIST.

Step	Action	✓
Install TPS Domain/Console Configuration Tool		
1	Using the TPS Software CD, install the TPS Domain/Console Configuration Tool (if not already installed). ATTENTION: The TPS Domain tab does may not appear in the OU Properties dialog box until after the node is restarted. For detailed instructions to install the TPS Domain/Console Configuration Tool, refer to the 3.6, "Configuring domain users and groups for TPS".	
Retain the existing OU for Windows 2000-based nodes		
2	Retain the existing OU to support Windows 2000-based TPS nodes that you do not migrate / upgrade.	
Create New TPS Domain OU and Move New TPS Nodes Under OU		
3	If your TPS system does not have an existing TPS Domain OU for TPS 4xx configured, use Active Directory Users and Computers to create a new TPS Domain OU for Windows 2003/XP-based nodes. See section 3.10, "Performing 'TPS domain' OU procedures". Or If applicable, see section 3.10, "To migrate existing legacy TPS domain structures to Active Directory OUs".	
4	Using Active Directory Users and Computers, move Windows 2003/XP – based TPS GUS and APP nodes (computers) under the new TPS Domain OU. See section 3.10, subsection "To move a TPS GUS or APP node to a 'TPS domain' OU".	

2. TPS implementation checklists

2.4. Domain controller upgrade checklist

Step	Action	✓
Install Domain Controller Package		
5	Insert the TPS Software CD and click Domain Controller Package . When the installation completes, click OK . Results: Establishes a set of default Honeywell global groups (such as Operators, Supervisors, Engineers, and View Only Users) and installs a set of security policies.	
6	Ensure that all members of the pre-R400 TPS Administrators group and TPS Domain Servers group are members of the R4xx DCS Administrators group and DCS Domain Servers group respectively.	
Conclusion		
7	The Windows 2003 Domain Controller upgrade is complete. Continue to the checklists for implementing GUS and APP nodes, as necessary. To choose the correct implementation checklist, refer to the subsection 2.1 <i>Selecting the correct checklist</i> .	

2.5 GUS node upgrade checklist

Purpose of this checklist

This checklist guides you through the steps required to upgrade a GUS or US Replacement node to a newer TPS software release. The node must have already been migrated to Windows XP. For example, you would use this checklist to upgrade GUS R400 to GUS R410.

The flowchart in section 2.1 *How to Select the Correct Checklist* illustrates when you would choose to use this checklist.

Guidelines for using the checklist

Follow the steps in the checklist. Some steps require you to refer to another procedure in this document, and then return to the checklist. Some steps refer to other documents that provide additional background information.

Most steps identify the node types to which the step applies; if you are working on a different node type, skip to the next step. Place a checkmark in the right column once you have completed a step.



TIP

You may want to make copies of this checklist if you are upgrading more than one GUS node.

Before you begin

To perform this implementation, you need the following items:

- TPS Software CD
- Utilities and Load Modules Software CD
- GUS-TPN Software CD
- *Software Change Notices (SCNs)*
- *TPS System Administration Guide*
- *System Management Configuration Guide*
- a completed TPS Node Data Sheet for the node to be implemented (see Section 1.6, “Configuration data sheets”)

2. TPS implementation checklists

2.5. GUS node upgrade checklist

Prerequisites



ATTENTION

If Upgrading a TPS GUS Node

- Before starting the upgrade, a TPS GUS node requires a Windows 2003 Domain Controller. The Domain Controller must have a TPS Domain configured as an OU in Active Directory (to support Honeywell System Management):
 - Verify that the Windows 2000 Domain Controller has been migrated to Windows 2003 Domain Controller or that the Windows 2003 Domain Controller exists and is available on the network.
 - Verify that the Windows 2003 Domain Controller has a TPS Domain OU, and if not, perform the “*Domain Controller Upgrade Checklist*” in section 2.4.
- Before starting the upgrade of a TPS GUS node, it must be a member of a Windows 2003 domain.
 - Verify that the node is a member of a Windows 2003 domain.



TIP

You can find Domain configuration by right-clicking the **My Computer** icon on the Start menu and selecting **Properties**. Select the **Computer Name** tab.

GUS node upgrade checklist



TIP

If you leave the checklist to go to a procedure in another section, **ALWAYS RETURN TO THE CHECKLIST.**

Step	Action	✓
Backup Node Data		
1	Backup the node data files, as necessary.	
Remove Node from Legacy TPS Domain, if applicable		
2	This step applies only if upgrading a GUS that is a member of a legacy (nonOU-based) TPS Domain. Remove the GUS node from the legacy TPS domain configuration and replicate the change to the remaining nodes. TIP: Removing the node from the legacy TPS Domain stops the remaining nodes from polling for the HCI Component status of the GUS being upgraded.	
TPN (LCN) Procedure – Copy Load Modules to History Module		
3	This step needs to be done ONLY ONCE for each TPN, not for each node. This step applies only to GUS nodes (not US Replacement nodes). For the GUS software to operate properly, certain TPN system files are required in the &CUS directory on the TPN System History Module. If not already done as part of a previous upgrade of the TPN copy the new GUS load modules from CD to the TPN System History module. REFERENCE EXTERNAL: Refer to the TPS SCN for the versions of load modules that apply to your TPN release. For detailed instructions to install GUS Load Modules, refer to section 5.3 <i>Use Native Window to Copy GUS Load Modules to HM.</i>	
Record TPS Devices, Services, and Applications		
4	Log on to the machine as the Administrator.	
5	Open the Configuration Utility: Start > Programs > Honeywell TPS > Configuration Utility.	

2. TPS implementation checklists

2.5. GUS node upgrade checklist

Step	Action	✓
6	Select Configure. Record the configuration settings for the following menu items, if present: • Board 0 (or custom board name) • LCNI18N • Signon Manager • File Transfer – HM Node numbers: • Devices/Services (Record which boxes are checked): ___ TPS Signon Manager Service ___ Device Driver (tdcaint.sys) ___ Services (Tdcaserver) ___ Time Synchronization Service Enable ___ Synchronization Interval ___ LCN File Service ___ GUS Remote LXS ___ Carbon Copy ___ TPN Communication Service ___ Integrated Keyboard Driver Deactivate TPS Devices and Services	
7	Uncheck all of the checkboxes on the Devices/Services page, and then click OK. For detailed instructions to deactivate TPS Devices/Services, refer to section 4.10 "Installing TPS software packages" and the subsection "Deactivate devices/services".	
8	Restart the machine if not done in the previous step. (A reboot is necessary to make sure all services are not running.)	

Step

Action

9

If not done in step 7, hold down the **Shift** key as you log on to the machine as an Administrator.

Hold down the **Shift** key as you press the **Enter** key. Continue holding down the **Shift** key until the hourglass disappears and the cursor becomes the normal cursor.

(This step locks out AutoRun items, keeping certain TPS software from starting automatically.)

10

The next step provides two optional methods to uninstall TPS software. Regardless of the method you choose, the following lists indicate the packages that you need to uninstall and the packages that you do NOT need to uninstall:

ATTENTION: Do not uninstall the TPN (LCN) personality if it is already at the correct release. See the **ATTENTION** in the next Step.

Packages to uninstall: GUS AlarmMsgGroup GUS Display Builder GUS Display Runtime (not compatible with Remote Client) GUS MultipleDisplays GUS Display Translator GUS Display Validator GUS Remote Client GUS Remote Server SafeView TPS – Remote IKB Client (not compatible with Remote Client) TPS – Base (allowed if Remote Server) TPS – GUS Component Library Editor TPS – Native Window TPS – Touch Screen TPS Reusable Application Components (RAC)	DO NOT uninstall the following Packages: Adobe Acrobat Reader Control Drawing to GUS Picture Honeywell Uniformance Desktop Microsoft ODBC Total Plant Information Server TPS Builder Base TPS Builder HG/Data Hiway TPS Builder IOMap TPS Builder LCN/UCN
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2. TPS implementation checklists

2.5. GUS node upgrade checklist

Step	Action	✓
11	Uninstall the TPS Software packages using one of the methods shown below. ATTENTION: Please do not uninstall the TPN (LCN) personality if it is already at the correct release. If there is not a personality listed on the uninstaller and you wish to uninstall a, it may be necessary to manually remove the current personality if the personality is being upgraded. If using rmvtps.exe, notice that all of the packages are checked to be deleted by default. Uncheck the TPN Personality if the personality on the local drive is already at the desired TPN release and you do not want to uninstall it. Uninstalling the TPN Personality is not required for this step. Method 1: Use Microsoft Add/Remove Programs to remove all TPS, GUS, and SafeView, packages. You do not need to uninstall TPS Builder for this step. Method 2: Use the Honeywell Uninstaller to remove the existing TPS Software packages (this is the Uninstaller on the TPS Software CD.) NOTE: The Uninstaller may not uninstall all of the packages that need to be uninstalled. After running the utility, refer to the previous list of packages to be uninstalled and use Microsoft Add/Remove Programs to uninstall any remaining packages. Answer Yes to all if prompted to delete files and ignore all other error messages with the following exception: If an error is received when uninstalling TPS Base that indicates that the uninstall failed, open Add/Remove Programs and manually uninstall it.	
Restart		
12	Restart the machine. (This reboot deletes all .exe and .dll files that were still in use during package removal.) After you reboot, if all of the directories and files under \Honeywell\TPS were not removed, manually delete them. NOTE: If you chose to NOT update your TPN personality, Honeywell\TPS\Emulators\Personalities\US remains. If you receive error messages that some files are read-only, select OK.	

Step	Action	✓
Service Pack and Hot Fixes		
13	If not already installed, install the Windows 2003/XP Service Packs/Hot Fixes available from Honeywell for TPS Systems, as defined in the TPS R4xx Software Change Notice Note: A node that is at the Honeywell factory-default state contains appropriate service packs and hot fixes. Restart the machine.	
Install System Management Display Software Package		
This step is required only for TPS GUS nodes.		
14	Using the TPS Software CD, install the System Management Display (under the Licensed Packages menu item). For the System Management Display usage (optional, mandatory, or not allowed) on each node type, see Section 4.11 "TPS Software package usage by node type". For detailed instructions to install the System Management Display package, see Section 4.13 "Installing System Management display software package".	
Install Redirection Manager (optional)		
15	Using the TPS Software CD, install the Redirection Manager, if needed on the node, and as authorized by your license. If you receive a prompt asking if the TPN Server password should be changed, click NO. REFERENCE INTERNAL: For detailed instructions to install Redirection Manager, refer to the <i>Redirection Manager User's Guide</i>. For the Redirection Manager usage (optional or not allowed) on each node type, see Section 4.11 "TPS Software package usage by node type".	
16	If not done in the previous step, restart the machine.	
Install Workstation Security Package		
17	Using the TPS Software CD, install Workstation Package.	
18	Restart the machine and log on as Administrator.	

2. TPS implementation checklists

2.5. GUS node upgrade checklist

Step	Action	✓
Install TPS Software Packages		
19	Using the TPS Software CD, install the desired TPS software packages authorized by your license. For the list of TPS software packages usable (optional, not allowed, and mandatory) on each TPS node, see Section 4.11 <i>TPS Software Packages by Node Type</i>. TPS software packages requiring HCI Runtime to be installed prior to installing the package itself are indicated with an (H) in section 4.11. For detailed instructions to install TPS Software Packages, see Section 4.10 <i>Install TPS Software Packages</i>, subsection <i>Procedure to Install TPS Software Packages</i>.	
Add TPS Global Groups to TPS Local Groups		
20	ATTENTION: Execute the following script after you have made the node a member of a domain that supports the TPS global groups – that is, the domain controller has the Honeywell domain controller high security package installed. At the TPS node, log on as a user with Administrator privileges in the domain (for example, domain Administrator). Navigate to the Program Files\Honeywell\WkStaSecurity directory and execute the LinkDomainGroups.vbs command script. RESULT: This script file identifies standard global groups existing on a Domain Controller (such as Engineers, Supervisors, and Operators), and adds them to the equivalent TPS local groups.	
Reactivate Devices, Services and Applications		
21	On the TPS Configuration utility Devices/Services page, make sure all of the items that were previously selected, as well as those items that are currently selected, are all selected. Refer to a previous step where you recorded the node's configuration settings. For detailed instructions to activate Devices/Services, see Section 4.10 <i>Install TPS Software Packages</i>, subsection <i>Activate Devices/Services</i>.	
22	Make sure that all of the other configuration pages reflect the configuration settings you recorded earlier.	
23	Click OK and close the Configuration Utility.	
24	Restart the machine and log on as Administrator.	

Step	Action	✓
Configure Alarm Summary Display, if Necessary		
25	This step applies only to GUS nodes. If you intend to use GUS Displays for the Group and/or Alarm Summary, then do the following: • Select Configure > GUS Alarm/Message/Group Display. • Check the appropriate options.	
26	Click OK and close the Configuration Utility.	
Install GUS Personality Software on Local Drive		
27	This step applies only to GUS nodes (not US Replacement nodes). If you have not already done so as part of a previous upgrade of the TPN, use the GUS-TPN Software CD to install the GUS Personality onto the machine's local drive.  REFERENCE EXTERNAL: Refer to the TPS SCN for the version of GUS (Universal) Personality that applies to your TPN release. For detailed instructions to install the GUS Personality software, see Section 4.5 <i>Install TPN Personality Software on Local Drive</i>.	
Load the TPN (LCN) Personality		
28	Display the Native Window and load the node's LCNP board with its TPN Personality. RESULT: When the loading completes, a TPS GUS node appears with a node type of GUS, and a US Replacement node appears with a node type of US. For detailed instructions to load the GUS Personality, see Section 5.8 <i>Load LCNP Board with TPN Personality</i>, subsection <i>Load Native Window (GUS or US) Personality</i>.  ATTENTION: If the initial load fails, you may need to use the Add Board utility to remove and re-add the LCNP board configuration. If you change the Add Board configuration, you must cycle power on the node.	
29	Restart the machine and log on as Administrator.	

2. TPS implementation checklists

2.5. GUS node upgrade checklist

Step	Action	✓
Migrate GUS Displays, if Necessary		
These steps apply only to GUS nodes being upgraded from a release prior to R210. It is NOT necessary to migrate or revalidate displays that work on GUS 210 or greater.		
30	If upgrading from a GUS release prior to GUS 210, then install the appropriate GUS Display Migrator from the TPS Software CD. For detailed instructions to install the GUS Display Migrator, refer to section 4.12 <i>Migrating GUS Displays</i>.	
31	Migrate (with validation) the GUS displays from releases prior to GUS 210.	
	REFERENCE EXTERNAL: Refer to the <i>Migration Issues</i> section of the TPS SCN for important information.	
Upgrade GUS SafeView Operation, if not already done		
32	Add the following executable to the operator.bat or custom logon script file: C:\Program Files\Honeywell\TPS\Base\ enablesetforeground.exe In addition to modifying the operator.bat file, you must grant the affected users and groups Read and Execute permissions to the enablesetforeground.exe file. For detailed instructions for editing the operator.bat file, see Section 3.8 Editing OPERATOR.BAT file.	
	TIP As of GUS R350, the GUS startup is set to enableSetForeground, by default. This option is the equivalent of including enableSetForeground.exe in a startup .bat file. This new option can be turned off by the Allow SetForeground during Startup checkbox on the GUS Display Runtime Settings in the Configuration Utility. Note: Unchecking this option only prevents enablesetforeground from running on future logins - it does NOT change the registry settings for existing users (That can still be done through the AllowSetForeground utility).	

Step	Action	✓
Configure Honeywell System Management Display, if not done		
33	If not already done, add the Honeywell Node Administration Snap-in to the Microsoft Management Console (MMC). RESULT: The <i>default</i> configuration of your Honeywell System Management Display is complete.  REFERENCE – INTERNAL: For detailed instructions, refer to section “Adding the Node Administration Snap-In to the MMC” in the <i>System Management Configuration Guide</i>.	
34	If not already done, configure the Node Administration Snap-in properties as necessary. RESULT: An MMC console (.msc) file is configured and saved with a name of your choice (the default is Console1). End users can view a new System Management Display by opening its .msc file through the Honeywell Node Administration MMC Snap-in.  REFERENCE – INTERNAL: For detailed instructions, see “Node Administration Properties” in the <i>System Management Configuration Guide</i>.	
Restore Data		
35	Restore the customer-specific data files that were backed up prior to the upgrade, as necessary.	
Conclusion		
36	The GUS node upgrade is complete.	

2.6 APP node upgrade checklist

Purpose of this checklist

This checklist guides you through the steps required to upgrade an APP or AMR node running Windows 2003 Server (member server) to a newer TPS software release.

For example, you would use this checklist to upgrade APP 400 to APP 410 or greater.

The flowchart in section 2.1 *How to Select the Correct Checklist* illustrates when you would choose to use this checklist.

Guidelines for using the checklist

Follow the steps in the checklist. Some steps require you to refer to another procedure in this document, and then return to the checklist. Some steps refer to other documents that provide additional background information.

Most steps identify the node types to which the step applies; if you are working on a different node type, skip to the next step. Place a checkmark in the right column once you have completed a step.



TIP

You may want to make copies of this checklist if you are upgrading more than one APP node.

Before you begin

To perform this implementation, you need the following items:

- TPS Software CD
- Utilities Load Modules CD
- Application Module CD
- *Software Change Notices*
- *TPS System Administration Guide*
- *System Management Configuration Guide*
- a completed TPS Node Data Sheet for the node to be implemented (see Section 1.6, "Configuration data sheets")

Prerequisites



ATTENTION

If Upgrading an APP Node

- Before starting the upgrade, an APP node requires a Windows 2003 Domain Controller. The Domain Controller must have an OU with the TPS Domain attribute configured in Active Directory (to support Honeywell System Management):
 - Verify that the Windows 2000 Domain Controller has been migrated to a Windows 2003 Domain Controller, or that the Windows 2003 Domain Controller exists and is available on the network.
 - Verify that the Windows 2003 Domain Controller has a TPS Domain OU, and if not, perform the “*Domain Controller Upgrade Checklist*” in section 2.4.
- Before starting the upgrade of an APP node, it must be a member of a Windows 2003 domain.
 - Verify that the APP node is a member of a Windows 2003 domain.



TIP

You can find domain configuration by right-clicking the **My Computer** icon on the Start menu and selecting **Properties**. Select the Computer Name.

2. TPS implementation checklists

2.6. APP node upgrade checklist

APP/AMR node upgrade checklist



TIP

If you leave the checklist to go to a procedure in another section, **ALWAYS RETURN TO THE CHECKLIST.**



TIP – Startup Errors

Newly installed services (such as CL Server and TPN Server) may not start up properly if the accounts they have been assigned to run under do not use the Honeywell default passwords, but use site-specific passwords.

If startup errors occur, use the Honeywell **Windows Services and DCOM Servers Log on tool** to update the password information for the services registered to use the account.



REFERENCE - INTERNAL

For instructions on using the **Windows Services and DCOM Servers Log on tool**, refer to the *TPS System Administration Guide*.

Step	Action	✓
Backup Node Data		
1	Backup the node data files, as necessary.	
Record TPS Devices, Services, and Applications		
2	Log on to the machine as administrator.	
3	Open the TPS Configuration Utility and select Configure .	

Step	Action	✓
4	Record the configuration settings for the following menu items, if present: • Board 0 (or custom board name) • LCNI18N • Signon Manager • File Transfer – HM Node numbers: • Devices / Services (Record which boxes are checked): ☐ TPS Signon Manager Service ☐ Device Driver (tdcaint.sys) ☐ Services (Tdcaserver) ☐ Time Synchronization Service Enable ☐ Synchronization Interval ☐ LCN File Service ☐ File Transfer ☐ GUS Remote LXS ☐ TPN Communication Service ☐ Carbon Copy Deactivate TPS Devices and Services	
5	Uncheck all of the checkboxes on the Devices/Services page, and then click OK. For detailed instructions to deactivate TPS Devices/Services, refer to section 4.10 Installing TPS software packages and the subsection Deactivate Devices/Services.	
6	Restart the machine if not done in the previous step. (A reboot is necessary to make sure all services are not running.)	
7	If not already done in a previous step, hold down the Shift key as you log on to the machine as Administrator. Hold down the Shift key as you press the Enter key. Continue holding down the Shift key until the hourglass disappears and the cursor becomes the normal cursor. (This step locks out AutoRun items, keeping certain TPS software from starting automatically.)	

2. TPS implementation checklists

2.6. APP node upgrade checklist

Step	Action	✓
Uninstall Honeywell System Management Components, if any		
These steps apply only if upgrading from APP R210 to a newer release.		
8	From the Microsoft Services window, stop the Honeywell System Management services: a) Stop Windows Management Instrumentation service. This stops the following four (of five) Honeywell System Management services: – sm-System Event Provider – sm-Component Admin (CAS) – sm-FTE (heartbeat) Provider – sm-Name Service Provider b) Stop the sm-Remote Configuration service.	
9	Using Microsoft Add/Remove Programs or the Honeywell Uninstaller, uninstall all Honeywell System Management components that are on the node, such as the following: a) Honeywell System Management Status Display b) Honeywell System Management Runtime c) Honeywell TPS Admin Support Files	
	ATTENTION: If you are upgrading the Fault Tolerant Ethernet (FTE) driver on your node, refer to the <i>TPS SCN</i> for specific upgrade instructions then return to this procedure.	
Uninstall TPS Software		
10	Uninstall the TPS Software packages using one of the following methods: Method 1: Use Microsoft Add/Remove Programs to remove all TPS software packages. Method 2: Use the Honeywell Uninstaller to remove the existing TPS software packages.	
Restart		
11	Restart the machine. (This reboot deletes all .exe and .dll files that were still in use during package removal.)	

Step	Action	✓
Service Pack and Hot Fixes		
12	If not already installed, install the Windows 2003/XP Service Pack/Hot Fixes available from Honeywell for TPS Systems, as defined in the TPS R4xx Software Change Notice. Note: A node that is at the Honeywell factory-default state contains appropriate service packs and hot fixes. Restart the machine.	
Install System Management Software Package		
<i>This step is required only for APP nodes.</i>		
13	Using the TPS Software CD, install the System Management Display (under the Licensed Packages menu item). For the System Management Display usage (optional or not allowed) on each node type, see Section 4.11 TPS Software package usage by node type. For detailed instructions to install the System Management Display package, see Section 4.13 <i>Install System Management Software Package</i>.	
Install Redirection Manager (optional)		
14	Using the TPS Software CD, install the Redirection Manager, if needed on the node, and as authorized by your license. For the Redirection Manager usage (optional or not allowed) on each node type, see Section 4.11 TPS Software package usage by node type.  REFERENCE INTERNAL: For detailed instructions to install Redirection Manager, see the <i>Redirection Manager User's Guide</i>.	
Install TPS Software Packages		
15	Using the TPS Software CD, install the desired TPS software packages authorized by your license. For the list of TPS software packages (optional and mandatory) on each TPS node, see Section 4.11 TPS Software package usage by node type. TPS software packages requiring HCI Runtime to be installed prior to installing the package itself are indicated with an (H) in section 4.11. For detailed instructions to install TPS Software Packages, see Section 4.10 <i>Install TPS Software Packages, Procedure to Install TPS Software Packages</i>	
Install Workstation Security Package		
16	Using the TPS Software CD , install Workstation Package .	

2. TPS implementation checklists

2.6. APP node upgrade checklist

Step	Action	✓
17	Restart the machine. Note: This step is mandatory for a smooth upgrade.	
Add TPS Global Groups to TPS Local Groups		
18	 ATTENTION: Execute the following script <u>after</u> you have made the node a member of a domain that supports the TPS global groups – that is, the domain controller has the Honeywell domain controller high security package installed. At the TPS node, log on as a user with Administrator privileges in the domain (for example, domain Administrator). Navigate to the Program Files\Honeywell\WkStaSecurity directory and execute the LinkDomainGroups.vbs command script. RESULT: This script file identifies standard global groups existing on a Domain Controller (such as Engineers, Supervisors, and Operators), and adds them to the equivalent TPS local groups.	
Reactivate Devices/Services		
19	Open the TPS Configuration Utility and select Configure .	
20	Refer to a previous step where you recorded the node's configuration settings and make sure all of the items that were previously selected, as well as those items that are currently selected, are all selected. For detailed instructions to activate Devices/Services, see Section 4.10 <i>Install TPS Software Packages</i> , subsection <i>Activate Devices/Services</i> .	
21	Click OK and close the Configuration Utility.	
22	Restart the machine and log on as Administrator .	
Install TPN (LCN) Personality Software on Local Drive, If Necessary		
23	If you have not already done so as part of a previous upgrade of the TPN, use the Application Module CD to install the APP Personality onto the machine's local drive.  REFERENCE EXTERNAL: See the TPS SCN for the version of APP (amw) Personality that applies to your TPN release. For detailed instructions to install the APP Personality software, see Section 4.5 <i>Install TPN Personality Software on Local Drive</i> .	

Step	Action	✓
Reload the LCNP Board with the TPN (LCN) Personality, if Necessary		
24	From a GUS, display the Native Window: Programs > Honeywell TPS > Native Window . Or From a US, display the System Status display .	
25	From the System Status Display, select the node and verify that it is in the PWR_ON state. If it is not, shut it down.	
26	Once the node transitions to PWR_ON or FAIL, reload the TPN Personality into the node using the AUTOLOAD NET target. RESULT: When the loading completes, the node appears with a node type of APP or an AM Replacement node appears with a node type of AM .	
Configure Honeywell System Management Display, if not done		
27	If not already done, add the Honeywell Node Administration Snap-in to the Microsoft Management Console (MMC). RESULT: The <i>default</i> configuration of your Honeywell System Management Display is complete.	
	 REFERENCE – INTERNAL: For detailed instructions, see Section “Adding the Node Administration Snap-In to the MMC” in the <i>System Management Configuration Guide</i> .	
28	If not already done, configure the Node Administration Snap-in properties as necessary.	
	 REFERENCE – INTERNAL: For detailed instructions, see the “Node Administration Properties” section in the <i>System Management Configuration Guide</i> . RESULT: An MMC console (.msc) file is configured and saved with a name of your choice (the default is Console1). End users can view a new System Management Display by opening its .msc file through the Honeywell Node Administration MMC Snap-in.	
Restore Data		
29	Restore the customer-specific data files that were backed up prior to the upgrade, as necessary.	
Conclusion		
30	The APP node upgrade is complete.	

2. TPS implementation checklists

2.6. APP node upgrade checklist



TIP – Startup Errors

Newly installed services (such as CL Server and TPN Server) may not start up properly if the accounts they have been assigned to run under do not use the Honeywell default passwords, but use site-specific passwords.

If startup errors occur, use the Honeywell **Windows Services and DCOM Servers Log on tool** to update the password information for the services registered to use the account.



REFERENCE - INTERNAL

For instructions on using the **Windows Services and DCOM Servers Log on tool**, refer to the *TPS System Administration Guide*.

2.7 Migrate Windows 2000 Domain Controller to Windows 2003 Domain Controller checklist

Purpose of this Checklist

This checklist guides you through the steps required to migrate the operating system of

- Windows 2000 Domain Controller to a Windows 2003 **root or child** Domain Controller, or
- Windows 2000 peer Domain Controller to a Windows 2003 **peer** Domain Controller.

The checklist includes the most common settings for the migration. The flowchart in section 2.1 *How to Select the Correct Checklist* describes when you would choose to use this checklist.



ATTENTION

APP on Domain Controller is not supported in this release.

Guidelines for using the checklist

Follow the steps in the checklist. Some steps require you to refer to another procedure in this document, and then return to the checklist. Some steps refer to other documents that provide additional background information. Place a checkmark in the right column once you have completed a step.

Before you begin

To perform this migration, you need the following items:

- Retail copy of the Windows 2003 Server – Standard Edition media
- TPS Software CD
- *TPS System Implementation Guide*
- A completed Domain Controller Configuration Data Sheet for the machine to be implemented (see section 1.6)

2. TPS implementation checklists

2.7. Migrate Windows 2000 Domain Controller to Windows 2003 Domain Controller checklist

Prerequisites



ATTENTION

- Verify that the role you intend to establish for the Domain Controller is an appropriate choice for your site.
 - Administrators must be very familiar with the implications of establishing a process control domain as a child of the IT domain architecture. In addition, the IT and Process Control departments should spend time planning the best possible approach, as the current version of Windows 2003 does not offer tools to modify the domain architecture after it has been established.
 - Honeywell recommends establishing the Domain Controller of a process control domain as a root Domain Controller.
-



REFERENCE – INTERNAL

See the *TPS System Planning Guide* for additional Honeywell guidelines on Windows 2003 domain architectures.

Migrate Windows 2000 Domain Controller to Windows 2003 Domain Controller checklist



TIP

If you leave the checklist to go to a procedure in another section, ALWAYS RETURN TO THE CHECKLIST.

Step	Action	✓
Backup Node Data		
1	Backup the node data files, as necessary.	
Set System BIOS Boot Order		
2	Enter the BIOS menu and ensure that the boot order is configured such that the hard drive boots prior to the CD and floppy drives. For detailed instructions to set the boot order, see Section 4.14, "Set System BIOS Boot Order".	

Step	Action	✓
Migrate Operating System		
3	Insert the Windows 2003 Server – Standard Edition media and select Upgrade. Note: An option to upgrade may not be available if both of the following conditions are true: • Windows 2000 Server or Windows NT Server 4.0 was installed from a retail version of the Windows 2000 CD-ROM. • You try to upgrade to Windows Server 2003 by using an Original Equipment Manufacturer (OEM) CD-ROM.	
4	At each prompt, answer the questions as appropriate. While the Operating System is migrated to Windows 2003, the Active Directory Installation Wizard is launched	
5	This step applies only to the migration of a peer Domain Controller. While migrating a peer Domain Controller, select the option Additional Domain Controller for an existing domain.	
	 REFERENCE – EXTERNAL For detailed instructions, refer to Microsoft documentation.	
6	This step applies only to the migration of a peer Domain Controller. Perform the steps necessary to install Active Directory.	
	 REFERENCE – EXTERNAL For detailed instructions, refer to Microsoft documentation.	

2. TPS implementation checklists

2.7. Migrate Windows 2000 Domain Controller to Windows 2003 Domain Controller checklist

Step	Action	✓
7	This step applies only to the migration of a Domain Controller. If migrating a DC, do the following to establish the domain role: <u>To establish a root domain role:</u> At the Active Directory Installation Wizard, select Domain in a new forest and complete the configuration as necessary to configure the Windows 2003 server as a root Domain Controller and install DNS, if this machine is to be a DNS Server. For detailed instructions on installing Active Directory and DNS for a root Domain Controller, see Section 3.3, "Install Active Directory and DNS for Root Domain Controller". <u>To establish a child domain role:</u> At the Active Directory Installation Wizard, select Child domain in an existing domain tree and complete the configuration as necessary to configure the Windows 2003 server as a child Domain Controller. For detailed instructions on installing Active Directory for a child Domain Controller, see Section 3.11, "Create a New Child Domain in an Existing Domain Tree". <i>Install DNS On Child Domain Controller, if necessary</i> 8 If the child Domain Controller is to be a DNS Server, then perform the steps to install DNS. For detailed instructions on installing DNS for a child Domain Controller, see Section 3.12, "Install DNS on Child Domain Controller". <i>Install Microsoft Windows 2003 Service Pack and Hot fixes</i> 9 If not already installed, install the Microsoft Windows 2003 Service Packs and Hotfixes that is available from Honeywell, as defined in the TPS R4xx Software Change Notice.  TIP Service Pack installation can take 10-20 minutes and require at least one reboot.	

Step	Action	✓
<i>Proceed to Upgrade</i>		
10	Follow instructions in section 2.4, "Domain Controller Upgrade Checklist" subsection "Domain Control Upgrade Checklist".	
<i>Conclusion</i>		
11	The Windows 2000 Domain Controller to Windows 2003 Domain Controller migration is complete.	

2.8 Migrate APP member server checklist

Purpose of this checklist

This checklist guides you through the steps required to migrate the operating system of a Windows 2000 member server APP node to a Windows 2003 member server APP node, and then upgrade the node to a newer TPS software release. For example, you would use this checklist to migrate APP R2xx to APP R400 and greater.

The flowchart in section 2.1 *How to Select the Correct Checklist* illustrates when you would choose to use this checklist.

Guidelines for using the checklist

Follow the steps in this checklist to perform the migration. Some steps require you to jump out to another procedure, and then return to this checklist. Some steps refer to other documents that provide additional background information.

Place a checkmark in the right column once you have completed the step.



TIP

You may want to make copies of this checklist if you are migrating more than one APP node.

Before you begin

To perform this implementation, you need the following items:

- EXPPlus Initialization DVD
- TPS Software CD
- Application Module CD
- Utilities and Load Modules Software CD
- *Software Change Notices*
- *TPS System Administration Guide*
- *System Management Configuration Guide*
- A completed TPS Node Data Sheet for the node to be implemented (see Section 1.6, “Configuration data sheets”)



If Upgrading an APP Node

- Before starting the upgrade, an APP node requires a Windows 2003 Domain Controller. The Domain Controller must have an OU in Active Directory configured with the TPS Domain attribute (to support Honeywell System Management):
 - Verify that the Windows 2000 Domain Controller has been migrated to Windows 2003 Domain Controller or that the Windows 2003 Domain Controller exists and is available on the network.
 - Verify that the Windows 2003 Domain Controller has a TPS Domain OU, and if not, perform the *“Domain Controller Upgrade Checklist”* in section 2.4.
- Before starting the upgrade of an APP node, it must be a member of a Windows 2003 domain.
 - Verify that the node is a member of a Windows 2003 domain.

If you leave the checklist to go to a procedure in another section, ALWAYS RETURN TO THE CHECKLIST.

Step	Action
Backup Node Data	
1	Backup the node data files, as necessary. Note: The standard data files/directories from R3xx should be saved prior to a migration. These would include the GUS displays, HWIAC (for checkpoints and such) and the C:\Symantec for EBR configuration.
 REFERENCE – INTERNAL See “Backup and Recovery Guidelines” in the <i>TPS System Administration Guide</i>.	

2. TPS implementation checklists

2.8. Migrate APP member server checklist

Step	Action	✓
Record TPS Devices, Services, and Applications		
2	Log on to the machine as Administrator.	
3	Open the TPS Configuration Utility and select Configure .	
4	Record the configuration settings for the following menu items, if present: • Board 0 (or custom board name): • LCNI18N: • Signon Manager: • File Transfer – HM Node numbers: • Devices/Services (Record which boxes are checked): ☐ TPS Signon Manager Service ☐ Device Driver (tdcaint.sys) ☐ Services (Tdcaserver) ☐ Time Synchronization Service Enable ☐ Synchronization Interval ☐ LCN File Service ☐ GUS Remote LXS ☐ TPN Communication Service ☐ Carbon Copy	
Set System BIOS Boot Order		
5	Enter the BIOS menu and ensure that the boot order is configured such that the hard drive boots prior to the CD-ROM and floppy drives. For detailed instructions to set the boot order, see Section 4.14, "Set System BIOS Boot Order".	
Migrate the Operating System		
Note: The EXPPlus Initialization DVD does not provide an option to upgrade to Windows Server 2003. Instead, it wipes out Windows 2000 Server installation and installs Windows Server 2003 afresh.		
6	Follow instructions from Section 4.17 "Initializing TPS node using EXPPlus DVD" to perform a clean install of Windows Server 2003 on the node.	
Implement the APP Node		
7	After the initialization completes, follow the instructions from section 2.3, "New TPS node implementation checklist" to implement the APP node.	

2 TPS implementation checklists
2.8 Migrate APP member server checklist

Step	Action	✓
<i>Restore Node Data</i>		
8	Restore the node-specific data files that were backed up prior to the migration/upgrade, as necessary.	
<i>Conclusion</i>		
9	The Windows 2000 APP Member Server to Windows 2003 APP Member Server migration and upgrade is complete.	

2.9 Migrate GUS checklist

Purpose of this checklist

This checklist guides you through the steps required to migrate the operating system of a GUS node from Windows 2000 to Windows XP, and then upgrade the node to a newer GUS software release. For example, you would use this checklist to migrate GUS R3xx to GUS R400 and greater.

The flowchart in section 2.1 describes when you would choose to use this checklist.

Guidelines for using the checklist

Follow the steps in the checklist. Some steps require you to refer to another procedure in this document, and then return to the checklist. Some steps refer to other documents that provide additional background information.

In most cases, a statement at the beginning of the step defines what TPS nodes are affected. If the step does not apply to the node type you are implementing, then skip to the next step. Place a checkmark in the right column once you have completed a step.



TIP: You may want to make copies of this checklist if you are migrating more than one GUS node.

Before you begin

To perform this implementation, you need the following items:

- EXPPlus Initialization DVD
- TPS Software CD
- GUS-TPN Software CD
- Utilities and Load Modules Software CD
- Software Change Notices
- TPS System Administration Guide
- System Management Configuration Guide
- A completed TPS Node Data Sheet for the node to be implemented (see Section 1.6, “Configuration data sheets”)

Prerequisites



ATTENTION

If Migrating a TPS GUS Node

- Before starting the migration, a TPS GUS node requires a Windows 2003 Domain Controller. The Domain Controller must have an OU in Active Directory configured with the TPS Domain attribute (to support Honeywell System Management):
 - Verify that the Windows 2000 Domain Controller has been migrated to Windows 2003 Domain Controller or that the Windows 2003 Domain Controller exists and is available on the network.
 - Verify that the Windows 2003 Domain Controller has a TPS Domain OU, and if not, perform the “Domain Controller Upgrade Checklist” in section 2.4.
 - Before starting the migration of a TPS GUS node, it must be a member of a Windows 2003 domain.
 - Verify that the node is a member of a Windows 2003 domain.
-

Migrate GUS or USR checklist



TIP

If you leave the checklist to go to a procedure in another section, ALWAYS RETURN TO THE CHECKLIST.

Step	Action	✓
Backup Node Data		
1	Backup the node data files, as necessary.	
 REFERENCE – INTERNAL		
See the “Backup and Recovery Guidelines” in the <i>TPS System Administration Guide</i> .		
Record TPS Devices, Services, and Applications		
2	Log on to the machine as administrator.	
3	Open the Configuration Utility and select Configure .	

2. TPS implementation checklists

2.9. Migrate GUS checklist

Step	Action	✓
4	Record the configuration settings for the following menu items, if present: • Board 0 (or custom board name): • LCNI18N: • Signon Manager: • File Transfer – HM Node numbers and other information: • Devices/Services (Record which boxes are checked): ___ TPS Signon Manager Service ___ Device Driver (tdcaint.sys) ___ Services (Tdcaserver) ___ Time Synchronization Service Enable ___ Synchronization Interval ___ LCN File Service ___ GUS Remote LXS ___ Carbon Copy ___ TPN Communication Service ___ Integrated Keyboard Driver	
5	Record the configuration of all other settings appearing in the node's Configure menu.	
Migrate the Operating System		
Note: The EXPPlus Initialization DVD does not provide an option to upgrade to Windows XP. Instead, it wipes out the Windows 2000 installation and installs Windows XP afresh.		
6	Follow instructions from <i>Section 4.17 "Initializing TPS node using EXPPlus DVD"</i> to install Windows 2003 afresh on the node.	
Implement the GUS Node		
7	After the initialization completes, follow the instructions from section 2.3, "New TPS node implementation checklist" to implement the GUS node.	
Restore Node Data		
8	Restore the node-specific data files that were backed up prior to the migration/upgrade, as necessary.	
Conclusion		
9	The Windows 2000 Professional to Windows XP Professional GUS node migration and upgrade is complete.	

2.10 Upgrading a GUS remote client

Uninstall TPS software



ATTENTION

- Some of the packages listed below are optional and may not need to be uninstalled on every node.
- It is not necessary to uninstall any other third party or other packages to perform this upgrade.
- Answer 'Yes to all' when prompted to delete files and ignore all other error messages. Ignore messages suggesting that you reboot after any uninstall, because the instructions ask you to reboot after you complete all of the installations.
- For reference, when installing the new software, note which packages you removed for each specific node.
- There is a list of packages NOT to uninstall, following this **ATTENTION**.



ATTENTION

DO NOT uninstall any of the following packages, as they are not part of this upgrade and uninstalling them unnecessarily complicates your upgrade process.

Packages not required to uninstall:

- Adobe Acrobat Reader
- Control Drawing to GUS Picture
- Honeywell Uniformance Desktop
- Microsoft ODBC
- Total Plant Information Server
- TPS Builder Base
- TPS Builder HG/Data Hiway
- TPS Builder IOMap
- TPS Builder LCN/UCN

2. TPS implementation checklists

2.10. Upgrading a GUS remote client

Step	Action
1	Log off the client node.
2	Log back on with the "Shift" key held until the hourglass disappears.
3	Uninstall the TPS Software using Control Panel >Add/Remove Programs , or use the Honeywell Uninstaller. TPS – System Packages to Uninstall: GUS AlarmMsgGroup GUS Display Builder GUS Display Runtime (not compatible with Remote Client) GUS MultipleDisplays GUS Display Translator GUS Display Validator GUS Remote Client GUS Remote Server SafeView TPS – Remote IKB Client (not compatible with Remote Client) TPS – Base (allowed if Remote Server) TPS - GUS Component Library Editor TPS - Native Window TPS - Touch Screen TPS Reusable Application Components (RAC)
4	After performing the last uninstall, shutdown and reboot the node.

Install new GUS software

Step	Action
1	Insert the TPS Software CD in the GUS node's CD drive.
2	Select the TPS System item in the TPS Installation Browser to launch the TPS Installer.
3	Select Install Packages .
4	When the <i>Welcome</i> dialog displays, read the details and if you agree, click Next .
5	To accept the software license agreement terms, click Next .
6	The User Information dialog is displayed. Enter the Name , Company , System ID and Serial Number information from your GUS Software License sheet, and then click Next . RESULT: When the license information has been validated, the <i>Package Selection</i> dialog displays with the list of available licensed packages.
7	Select ONLY packages from the following list, and only if you intend to use that function on this remote node. Hold down the CTRL key while clicking to selectively pick multiple packages. TPS – System Packages to Install: GUS Display Builder GUS Remote Displays Client GUS Multiple Displays Native Window Client SafeView Touch Screen

2. TPS implementation checklists

2.11. Checklist for network install of TPS software on a new Windows XP/2003 TPS node

2.11 Checklist for network install of TPS software on a new Windows XP/2003 TPS node

Purpose of this checklist

This checklist guides you through the steps required to perform a network install of TPS software on a new Windows XP/2003 TPS node. The flowchart in section 2.1 *How to Select the Correct Checklist*, describes when to use this checklist.

Guidelines for using the checklist

Follow the steps in the checklist. Some steps require you to refer to another procedure in this document, and then return to the checklist. Some steps refer to other documents that provide additional background information.

If a step does not apply to the node type you are implementing, then skip to the next step. Place a checkmark in the right column after you have completed a step.



TIP

You may want to make copies of this checklist if you are installing more than one TPS node.

Prerequisites

The Administrative Installation procedures must have been performed to prepare a distribution point that supports network installs of TPS software.



REFERENCE – INTERNAL

For Administrative Installation setup procedures, refer to the *TPS System Administration Guide*.

Before you begin

Before you begin the TPS software installation, verify the following:

- The client (target) TPS node is running.
- All non-TPS software is installed.
- The client (target) TPS node has access to the Distribution Point.

2.11 Checklist for network install of TPS software on a new Windows XP/2003 TPS node

- Windows XP/2003 and required Service Packs and Hot fixes are installed, as defined in the TPS R4xx Software Change Notice.
- McAfee is installed.

Checklist for network install of TPS software on a new Windows XP/2003 TPS node**TIP**

If you leave the checklist to go to a procedure in another section, ALWAYS RETURN TO THE CHECKLIST.

Step	Action	✓
<i>Configure Local Administrator Password, if necessary</i>		
(The administrator password is normally entered during installation of Windows 2000 Server. When Honeywell ships a system as a factory-configured node, password is configured when user first starts up the node.)		
1	Log on to the machine as the Administrator.	
2	Press <Ctrl> <Alt> <Delete> and change the default password, if necessary.	
CAUTION		
Record and store the local Administrator password in a secure place. If you forget the password, you have to reinstall Windows XP/2003 to recover.		
 ATTENTION		
Do not rename the local Administrator account.		
<i>Set Date and Time, if necessary</i>		
(This is normally done as part of the Windows XP/2003 installation. The time on your system may already be correct and you may not need to do this procedure.)		
3	Display the Control Panel: Start > Settings > Control Panel.	
	Set the Time Zone, Date, and Time as needed.	

2. TPS implementation checklists

2.11. Checklist for network install of TPS software on a new Windows XP/2003 TPS node

Step	Action	✓
Change Computer Name		
4	If it is necessary to change the name of the node: a) Right-click the My Computer icon and select Properties. b) Select the Network Identification tab and click Properties. c) Change the computer name of the machine. d) Reboot the machine.	
Configure TCP/IP Settings		
5	Log on to the machine as the Administrator.	
6	Configure TCP/IP Settings: IP Address, Subnet Mask, Default Gateway, DNS Servers, and WINS Servers. For detailed instructions to configure the TCP/IP settings, see Section 4.2, "Configure TCP/IP Settings".	
	TIP For the actual data that needs to be entered, refer to your TPS Node Configuration Data Sheet that was completed prior to beginning this procedure.	
Connect Network Cable		
7	Physically connect the network (Ethernet) cable, if not already connected.	
Edit HOSTS and LMHOSTS Files, if necessary		
8	If not already done, add the local node name and IP address to the HOSTS and LMHOSTS files. <u>If the node is in a domain</u> , then edit the HOSTS and LMHOSTS files on the Domain Controller and run COPY_UTILITY to distribute the files to the other nodes. <u>If the node is not in a domain</u> , then edit the HOSTS and LMHOSTS files and use standard site procedures to place them in the correct locations. For detailed instructions to edit the files, see Section 3.7 Edit HOSTS and LMHOSTS Files.	

2.11 Checklist for network install of TPS software on a new Windows XP/2003 TPS node

Step	Action	✓
Join Domain		
9	This step is required only for TPS GUS, Network GUS, and APP nodes. Join the Windows 2003 domain.	
	TIP Right-click My Computer and select Properties . Select the Computer Name tab and click Change .	
Configure Local User Accounts		
10	Configure the local user accounts (TPS Local Server account) and the logon script, if needed. For detailed instructions to configure the local accounts, see Section 4.3, "Configuring local user accounts".	
11	Restart the machine.	
Network install of TPS software on a new Windows XP/2003 TPS node		
12	Use Add/Remove Programs to install GUS or APP TPS software on the new Windows XP/2003 TPS node. For detailed instructions to install the GUS or APP TPS Software on a new TPS node, see Section 4.15, "Network install of TPS software".	

2. TPS implementation checklists

2.12. Checklist for network install to upgrade GUS or APP node

2.12 Checklist for network install to upgrade GUS or APP node

Purpose of this Checklist

This checklist guides you through the steps required to upgrade a GUS or APP node to a newer TPS software release using a network-based installation. The node must have already been migrated to the appropriate operating system (Windows XP or Windows 2003 Server).

For example, you would use this checklist to upgrade GUS R340, GUS R350, and GUS R360 to GUS R400 or greater.

The flowchart in section 2.1 *How to Select the Correct Checklist* illustrates when you would choose to use this checklist.

Guidelines for using the checklist

Follow the steps in the checklist. Some steps require you to refer to another procedure in this document, and then return to the checklist. Some steps refer to other documents that provide additional background information.

Most steps identify the node types to which the step applies. If you are working on a different node type, skip to the next step. Place a checkmark in the right column once you have completed a step.



TIP

You may want to make copies of this checklist if you are upgrading more than one GUS or APP node.

Before you begin

To perform GUS or APP node upgrade, you need the following items:

- *Software Change Notices* (SCNs)
- A completed TPS Node Data Sheet for the node to be implemented (see section 1.5)

Prerequisites



ATTENTION

If Upgrading a TPS GUS or APP Node

- Before starting the upgrade, a TPS GUS or APP node requires a Windows 2003 Domain Controller. The Domain Controller must have an OU in Active Directory configured with the TPS Domain attribute (to support Honeywell System Management):
 - Verify that the Windows 2000 Domain Controller has been migrated to Windows 2003 Domain Controller or that the Windows 2003 Domain Controller exists and is available on the network.
 - Verify that the Windows 2003 Domain Controller has a TPS Domain OU, and if not, perform the procedures in the “Domain Controller Upgrade Checklist” in section 2.4.
- Before starting the upgrade of a TPS GUS or APP node, it must be a member of a Windows 2003 domain.
 - Verify that the node is a member of a Windows 2003 domain.



TIP

To find the Domain configuration, right-click the **My Computer** icon on the desktop and select **Properties**. Select the **Computer Name** tab.

2.12. Checklist for network install to upgrade GUS or APP node

Checklist for GUS or APP upgrade using network-based installation



If you leave the checklist to go to a procedure in another section, ALWAYS RETURN TO THE CHECKLIST.

Step	Action
Backup Node Data 1 Backup the node data files, as necessary.	
Un-installation of GUS or APP 2 You must uninstall the older version of the software before you upgrade the GUS or APP node with the new software version. For detailed instructions to uninstall the existing software, see Section 4.16, "Un-installation of GUS or APP".	
 ATTENTION Use the Honeywell Uninstaller to uninstall the existing software. The uninstaller uninstalls the previous versions of the TPS software more efficiently than the existing Add/Remove Programs function.	
Network install of GUS or APP 3 Install the new version of the GUS or APP software in the target machine. For detailed instructions about how to do a network install of the GUS or APP software on a new TPS node, refer to section 4.15, "Network install of TPS software".	

3. Domain controller procedures

3.1 Using the Domain Controller procedures

The procedures in this section are to be used in conjunction with a Domain Controller checklist. In most cases, you should have arrived in this section from a checklist. The checklist guides you to procedures in this section.

The procedures in this section are based upon a single domain model with one Windows 2003 Domain Controller controlling all the members of the domain.

For instructions on using the checklists, refer to section 2, "TPS Implementation Checklists".



ATTENTION

This section does not represent the complete set of procedures to implement, migrate, or upgrade a Domain Controller. To obtain the complete set of procedures, you must refer to one of the following checklists:

- "New Domain Controller Implementation Checklist"
 - "Domain Controller Upgrade Checklist"
 - "Migrate Windows 2000 DC to Windows 2003 DC Checklist"
-



REFERENCE – EXTERNAL

If necessary, consult Microsoft documentation for instructions on implementing Windows 2003 peer Domain Controllers.

3.2 Configuring TCP/IP settings



ATTENTION

This procedure is to be used in conjunction with one of the "TPS Implementation Checklists".

TCP/IP network configuration is required for the Windows 2003 Server machine. The following table provides guidelines for configuring TCP/IP settings.

Step	Action
1	Log on as the Administrator .
2	Select Start > Settings > Control Panel > Network Connections > Local Area Connection .
3	On Local Area Connection Status , select Properties .
4	Double-click the Internet Protocol (TCP/IP) item.
5	Select Use the Following IP Address option.
6	Enter the IP Address , Subnet Mask , Default Gateway , and DNS Servers from your <i>Domain Controller Configuration Data Sheet</i> .



ATTENTION

If you do not have a DNS Server that supports Dynamic DNS, you must establish one in your own domain. The Windows 2003 Domain Controller installation and migration procedures can automatically create a new Dynamic DNS service.

7	Click Advanced .
8	Enter WINS servers, if applicable.
9	Ensure the Enable LMHOSTS Lookup box is selected under the WINS tab.
10	Click OK to close the Advanced TCP/IP Settings dialog box.
11	Click OK to close the Internet Protocol (TCP/IP) Properties dialog box.
12	Click OK to close the Local Area Properties dialog box.
13	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

3.3 Installing Active Directory and DNS for root Domain Controller



ATTENTION

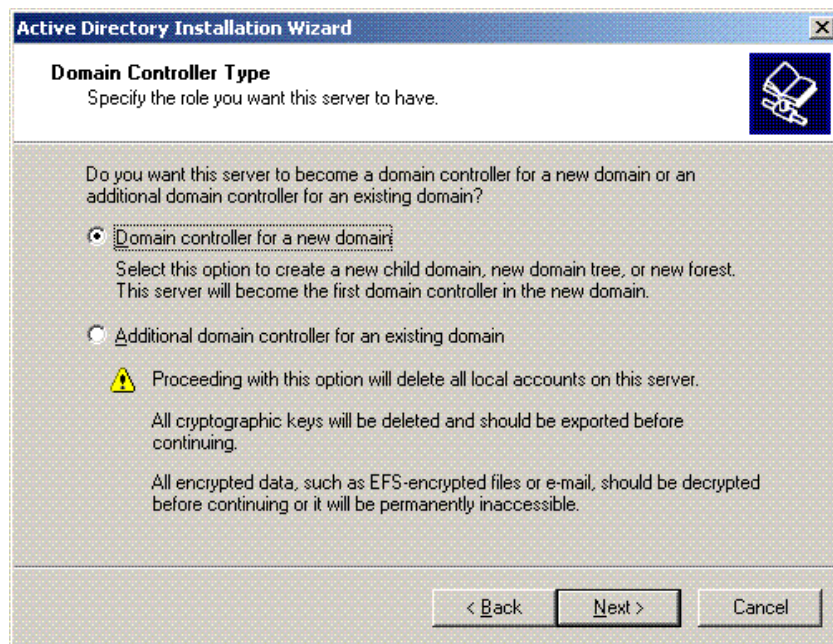
- This procedure is to be used in conjunction with one of the *TPS Implementation Checklists*.
 - Windows 2003 requires a Dynamic DNS server. You must configure a Dynamic DNS server if one is not established already in the enterprise. (DNS services on a Windows NT 4.0 server do not support Dynamic DNS.)
 - The Windows 2003 Dynamic DNS server can be directed to resolve Windows 2003 names itself and redirect to a Windows NT 4.0/2000-based DNS server to resolve other names within the enterprise.
-

Step	Action
1	To initiate the Active Directory Installation Wizard, do one of the following, depending on the type of installation being performed: • If this is an installation of a new Root Domain Controller, then select Start > Run and enter DCPROMO. • If this is a migration of a Windows 2000 Root Domain Controller to Windows 2003 Root Domain Controller, then insert the Microsoft Windows 2003 Server CD and select Upgrade.

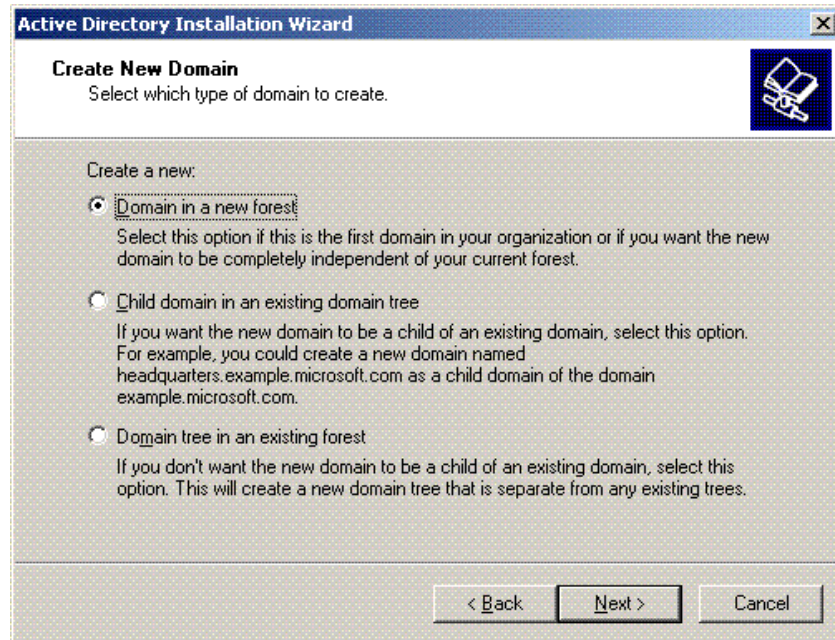
3. Domain controller procedures

3.3. Installing Active Directory and DNS for root Domain Controller

Step	Action
2	At the Active Directory Installation Wizard, click Next on the initial Welcome Screen and the Operating System Compatibility screen. In the Domain Controller Type screen, select Domain controller for a new domain. Click Next to proceed.



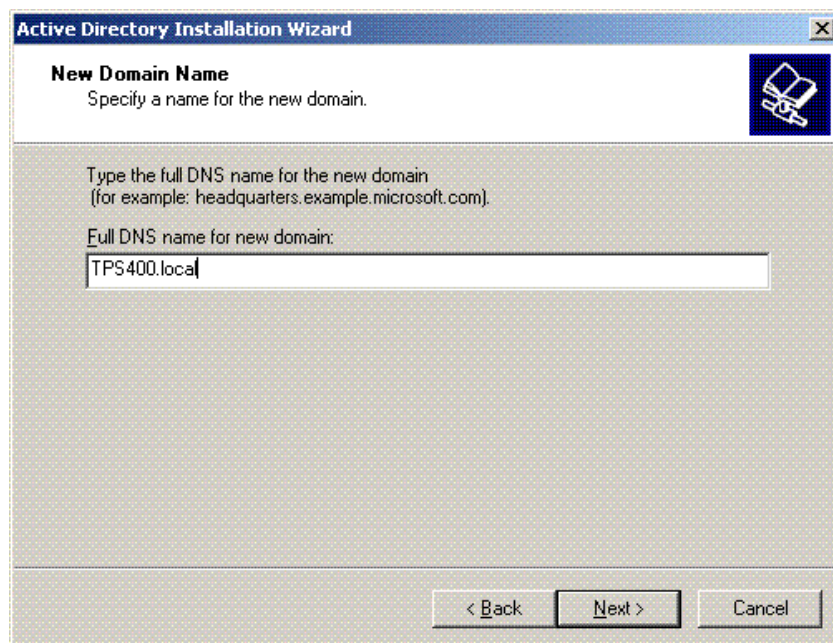
Step	Action
3	Select Domain in a new forest . Click Next to proceed.



3. Domain controller procedures

3.3. Installing Active Directory and DNS for root Domain Controller

Step	Action
4	Enter the full DNS name of this root domain. The DNS name must adhere to DNS conventions. Refer to your Domain Controller Data Sheet for details on the DNS name. Click Next to proceed.



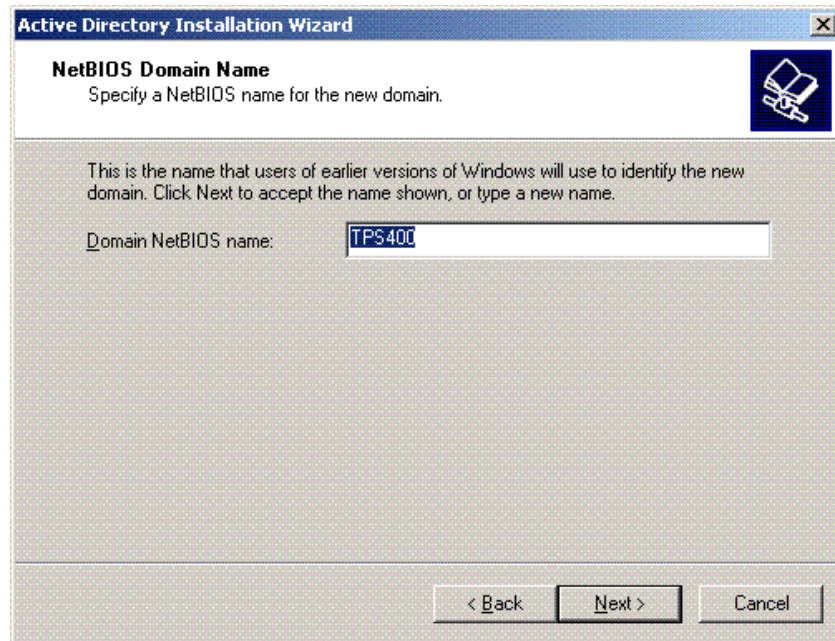
REFERENCE – INTERNAL

Refer to the *TPS System Planning Guide* for additional information regarding DNS naming. In general, a root domain contains a .local suffix.

3 Domain controller procedures

3.3 Installing Active Directory and DNS for root Domain Controller

Step	Action
5	If requested, enter the appropriate NetBIOS name. Typically, you select the default that matches the DNS name. Click Next to proceed.



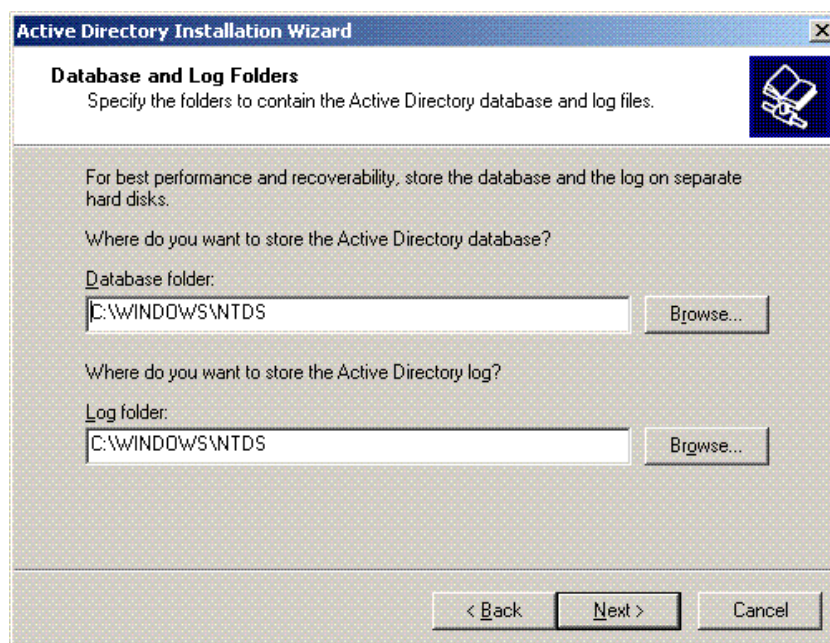
3. Domain controller procedures

3.3. Installing Active Directory and DNS for root Domain Controller

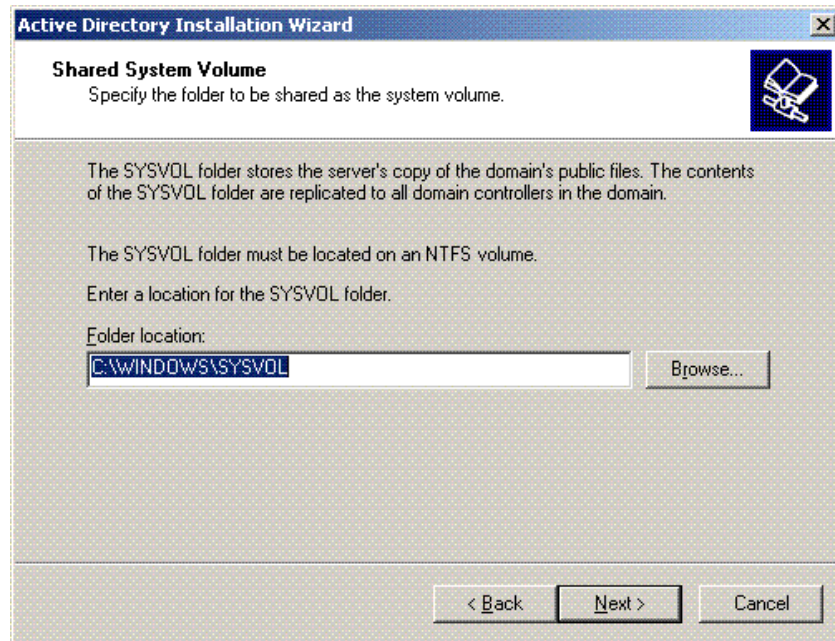
Step	Action
------	--------

- | | |
|---|--|
| 6 | The following dialog box requests directory paths to store the Active Directory database and log files. Typically, the default directories are adequate. If disk space is an issue, an alternate location can be selected. |
|---|--|

Click **Next** to proceed.



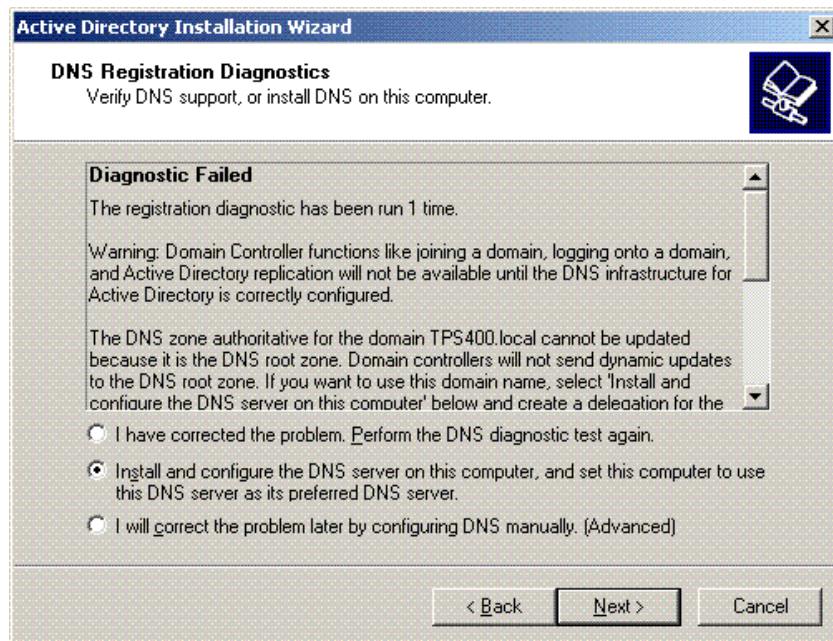
Step	Action
7	Click Browse, and then select the directory location for SYSVOL. Typically, this directory is stored in C:\WINDOWS\SYSVOL. Click Next to proceed.



3. Domain controller procedures

3.3. Installing Active Directory and DNS for root Domain Controller

Step	Action
8	The message illustrated below appears if the wizard cannot contact the DNS server that handles the root domain you specified. If the following error message appeared, select Install and configure DNS server on this computer and click Next to continue.



ATTENTION

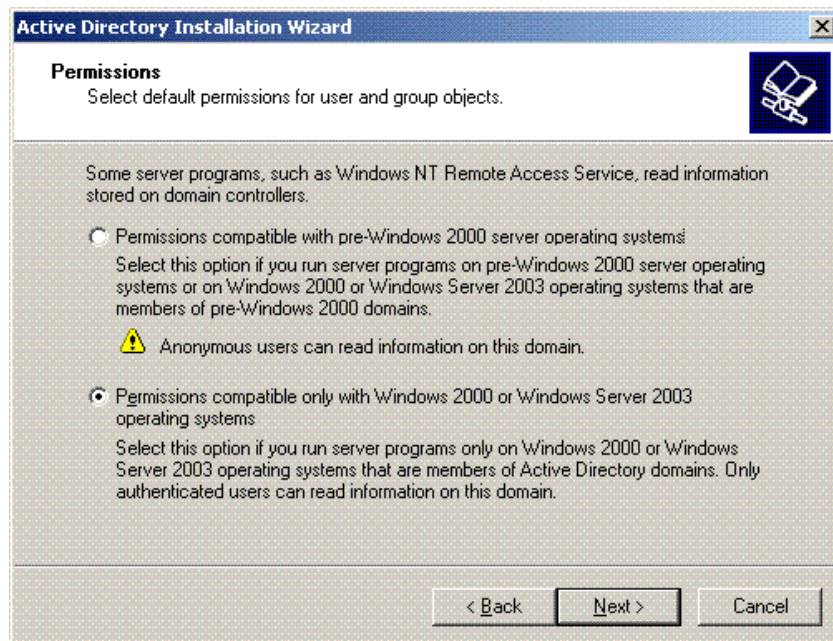
Windows 2003 requires a dynamic DNS server (for example, Dynamic DNS is not supported by NT 4.0 DNS services). As a result, you must configure a Dynamic DNS server if one is not already established in the enterprise. The Windows 2003 Dynamic DNS can subsequently be directed to resolve Windows 2000 names itself, and redirected to an NT 4.0/2000 based DNS to resolve other names within the enterprise.

Step	Action
	ATTENTION A root Domain Controller requires the existence of a Dynamic DNS server. In this step, the Microsoft upgrade procedure offers to configure a Dynamic DNS server on the first Domain Controller in the new domain. Unless a Dynamic DNS server already exists on the network, Honeywell recommends selecting Install and configure the DNS server on this computer to accept the auto configuration. If you answer Install and configure the DNS server on this computer, the Active Directory installation wizard performs the following tasks to install and configure DNS: • Installs the DNS Server service. • Creates a forward lookup zone with the same name as the Active Directory domain you are creating. • Configures the forward lookup zone as an Active Directory integrated zone. • Enables secure dynamic updates on the forward lookup zone. The Active Directory installation wizard does not create a reverse lookup zone. If you want a reverse lookup zone, you need to create one after you complete the Active Directory installation.
	TIP If you select I will correct the problem greater by configuring DNS manually, you are required to manually create the Dynamic DNS server. Dynamic DNS server creation and maintenance is beyond the scope of this procedure and should only be performed by those familiar with the technology.

3. Domain controller procedures

3.3. Installing Active Directory and DNS for root Domain Controller

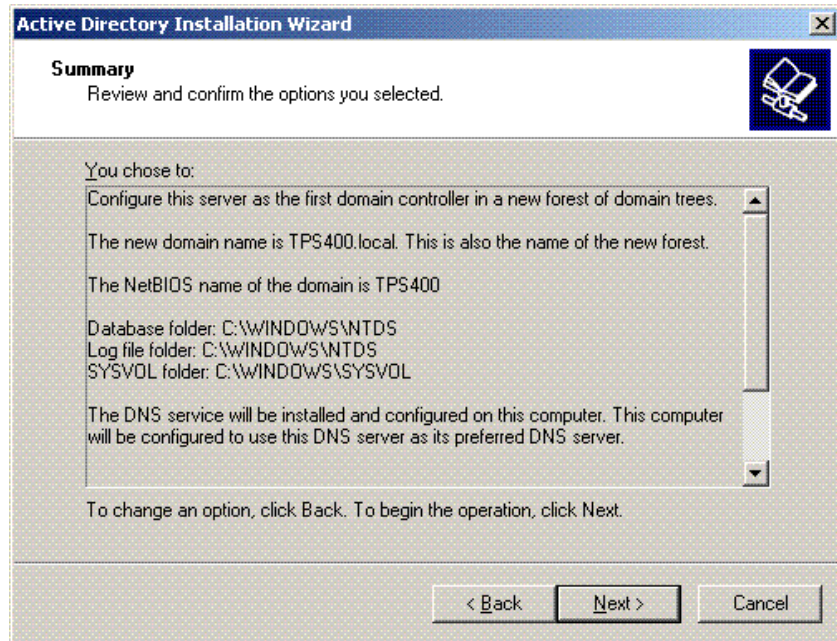
Step	Action
9	For correct TPS operation, select the option Permissions compatible only with Windows 2000 or Windows Server 2003 operating systems. This allows Windows 2003 to utilize "native mode" operation. Click Next to proceed.



Step	Action
10	Select an Active Directory Restore Mode Administration password.
	ATTENTION This is <u>not</u> your Windows 2000 Administrative password – it is the password for the account that you use to repair your system in the event of a failure with Active Directory services. Active Directory Help.' At the very bottom are three buttons: '< Back', 'Next >', and 'Cancel'." data-bbox="318 331 829 635"/>
11	Click Next .

3. Domain controller procedures

3.3. Installing Active Directory and DNS for root Domain Controller

Step	Action
12	Review your configuration settings on the summary screen. 
	Click Next on the summary screen to proceed with the installation. It may take a while to create the Active Directory and Dynamic DNS.
	 ATTENTION You may need to insert the Windows 2003 Server CD.
13	Click Finish on the Completing the Active Directory Installation Wizard dialog box, and then click Restart Now.
14	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

3.4 Adding a Reverse Lookup Zone to DNS



ATTENTION

This procedure is to be used in conjunction with the "New Domain Controller implementation checklist" or the "Migrate Windows 2000 Domain Controller to Windows 2003 Domain Controller checklist".

Step	Action
1	Open the DNS MMC: Start > Programs > Administrative Tools > DNS.
2	In the Console Tree (left-hand side of the MMC), expand your DNS server by selecting the plus sign (+).
3	Select the Reverse Lookup Zones folder.
4	Right-click Reverse Lookup Zones and select New Zone.
5	In the New Zone Wizard, click Next.
6	In the Zone Type dialog box, select Primary zone and click Next.
7	In the Active Directory Zone Replication Scope dialog box, select To all Domain Controllers in the Active Directory domain and click Next.
8	In the Reverse Lookup Zone dialog box, select the Network ID option, enter your Network ID (the first three octets of your IP address), and click Next.
9	In the Dynamic Update dialog box, select Allow only secure dynamic updates and click Next.
10	Verify the entered information and click Finish.
11	Close the DNS MMC.
12	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

3. Domain controller procedures

3.5. Installing Domain Controller package on DC for Honeywell High Security Policy

3.5 Installing Domain Controller package on DC for Honeywell High Security Policy

The Experion High Security Policy – Domain Controller Package applies security to several files and registry keys, including several domain groups, domain local groups, and domain users. Then, the package installs the TPS Group Policies, defining capabilities based on domain group membership. For example, the Operators group policy restricts all members of the Operators group from starting the Registry Editor.



REFERENCE – INTERNAL

Refer to the “TPS Security” section of the *TPS System Administration Guide* for additional information.



ATTENTION

This procedure is to be used in conjunction with one of the *TPS Implementation Checklists*.

Step	Action
1	Log on to the Domain Controller as the domain administrator.
2	Insert the TPS Software CD .
3	Select Domain Controller Package from the TPS Software CD menu (listed under the Experion PKS – High Security Policy menu item) to install the security configuration for Domain Controllers. Result: The package installation proceeds.
4	When the installation completes, click OK .
5	Close all windows and remove the CD from the drive.
6	Restart the machine.
7	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

3.6 Configuring domain users and groups for TPS



ATTENTION

This procedure is to be used in conjunction with one of the *TPS Implementation Checklists*.



REFERENCE - INTERNAL

Refer to the *TPS System Planning Guide*, section 3.8, for general information on users and security groups.

Customizing users, groups, and passwords for TPS

In environments where security requirements do not allow well-known user names and passwords, you may have to change user names and groups.



REFERENCE – INTERNAL

For instructions on customizing user names and groups, see TPS Security, in the *TPS System Administration Guide*.

TPS R420 changes to groups



ATTENTION

TPS R400 and greater use different groups than in previous releases:

global (domain) groups:

Previous TPS releases	TPS R400 and greater
TPS Domain Servers	DCS Domain Servers
TPS Administrators	DCS Administrators

local groups:

Previous TPS releases	TPS R400 and greater
TPS Local Administrators	Honeywell Administrators
TPS View Only Users	Local View Only Users
TPS Engineers	Local Engineers

...and so forth.

3. Domain controller procedures

3.6. Configuring domain users and groups for TPS

Note: In TPS R400 and greater, the local group 'TPS Local Administrators' is replaced by the local group 'Honeywell Administrators'. Members of 'TPS Local Administrators' previously got 'Engineer' privilege by default – in TPS R400 and greater, members of 'Honeywell Administrators' do not get 'Engineer' privilege.

Access to regedit tool

The Administrators group policy object (GPO) is applied last and will override the Engineers GPO - however, in the Engineers GPO access to the regedit tool is disabled, and in the Administrators GPO access to the regedit tool is NOT CONFIGURED. Consequently, a user who is a member of both the Engineers and DCS Administrators global (domain) groups will not be able to access the registry.

Note: In TPS R400 and greater, upon execution of linkdomaingroups.vbs on the GUS and APP nodes, the global group "DCS Administrators" becomes a member of "Administrators" and "Honeywell Administrators" local groups.

Registry access is disabled whenever a domain user is a member of any of the following global (domain) groups for TPS:

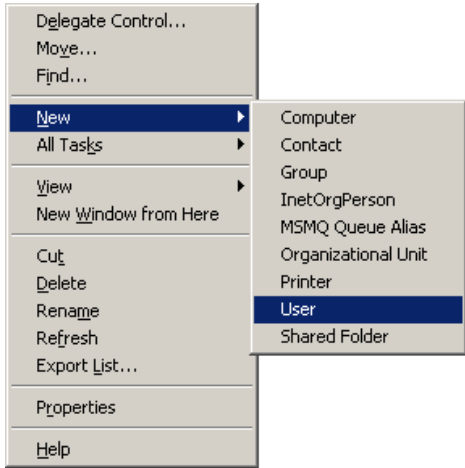
- Engineers
- Supervisors
- Operators
- View only users

By default, the domain administrator has access to the registry; however, if you add the domain administrator user to the 'Engineers' global (domain) group (to gain Native Window 'Engineer' access privilege), then the domain administrator will be prohibited from accessing the registry.

If access to the registry is needed, you can disable the **Prevent Access to Registry Editing Tools** setting in the Administrators group policy.

To create domain users for TPS

The following procedure describes how to create the "Operator" account and configure its group membership. The remaining accounts are listed in the table that follows:

Step	Action
1	Open the Active Directory Users and Computers MMC: Start > Programs > Administrative Tools > Active Directory Users and Computers.
2	In the Console Tree (left hand side of MMC), navigate to the Users container. It may be necessary to expand the domain to be able to access the Users container.
3	Right-click the Users container and select New > User . 
4	In the New Object – User dialog box, enter the following: First name: Operator User logon name: Operator
5	Click Next .
6	For the Operator account, specify a password. Make sure that only the following selections are checked: User cannot change password Password never expires
7	Click Next .
8	Click Finish .

3. Domain controller procedures

3.6. Configuring domain users and groups for TPS

Step	Action
To configure group membership	
9	Add the Operator user to the Operators Group: a) In the Console Tree (left hand side of MMC), click the Users container to display its contents. b) In the Details Pane (right hand side of MMC), right-click Operator and select Properties. c) Select the Member Of tab. d) Click Add. e) Click Advanced... f) In the 'Select Groups' dialog, click Find Now. g) Select the Operators group (you have to scroll down) and click OK.
10	Make the Operators group the primary group for the Operator user: a) In the 'Member of' field, select the Operators group. b) Click Set Primary Group.
11	Remove the Operator user from the Domain Users group: a) In the 'Member of' field, select the Domain Users group. b) Click Remove. c) Click Yes to confirm.
12	Click OK to close the Operator Properties.
To create the remaining domain accounts	
13	Perform similar steps to create the remaining accounts in the Users container.
 ATTENTION Refer to the table "New User Accounts on the Domain Controller" that follows for examples of the other common account names and group memberships that need to be configured using similar steps.	
14	After completing the configuration, if you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

3 Domain controller procedures
3.6 Configuring domain users and groups for TPS

New user accounts on the Domain Controller	
Create the following user accounts	Add the user accounts to the following domain user groups
Operator ^{1,2,3}	Operators
Supervisor ^{1,3}	Supervisors
Engineer ^{1,3}	Engineers
Other individually-named engineers ^{1,3}	Engineers
Individually-named DCS Administrators ^{1,3,4}	DCS Administrators
View Only ^{1,2,3}	View Only Users
Notes: ¹ Deselect (uncheck) the User Must Change Password at Next Logon option. Select the User Cannot Change Password and the Password Never Expires options. ² There should not be a password set on the Operator or View Only accounts. ³ Make the Assigned Group the Primary Group, and then remove the user from the Domain Users Group. ⁴ In TPS R400 and greater, upon execution of linkdomaingroups.vbs on the GUS and APP nodes, the global group "DCS Administrators" becomes a member of "Administrators" and "Honeywell Administrators" local groups. "Honeywell Administrators" is the replacement for the "TPS Local Administrators" local group that existed in previous releases; however "Honeywell Administrators" does not have the Engineer privilege.	

3.7 Editing the HOSTS and LMHOSTS files

Using Notepad to edit the HOSTS file

The HOSTS file contains the mappings of IP addresses to host names. The HOSTS file format is identical to the format used for host tables in the 4.3 Berkeley Distribution (BSD) UNIX/etc/hosts file.



ATTENTION

- This procedure is to be used in conjunction with one of the *TPS Implementation Checklists*.
- The HOSTS and LMHOSTS files must be accurate and consistent on every node. The files provide a backup for the name resolution normally provided by the DNS server, to ensure TPS software reliability in a failed network.

Step	Action
1	Open Windows Explorer.
2	If a site-specific HOSTS file is not already present in C:\WINDOWS\system32\drivers\etc\, then copy the sample file HOSTS.sam and paste the file into the following directory (Ensure that the file does not have any extension.): C:\WINDOWS\System32\Repl\Export\Scripts
	 TIP If file extensions are not visible (for example, HOSTS.sam and HOSTS), then configure the Windows Explorer Tools > Folder Options to <u>not hide</u> the file extensions.
3	Right-click > Open the following file in Notepad: C:\WINDOWS\System32\Repl\Export\Scripts\HOSTS

Step	Action
4	Modify the HOSTS file to include the Ethernet identification of all GUS nodes, APP nodes, and the Domain Controllers. For Example: The following are HOSTS file identification entries for an APP node and a GUS node: <pre>160.221.103.22 app1 # APP1 160.221.102.67 gus1 # GUS1 127.0.0.1 localhost</pre>
	ATTENTION Remove all non-essential host identifications. Do not remove the "localhost" identification entry.
5	Save the HOSTS file, and then Close Notepad.
6	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

Using Notepad to edit the LMHOSTS file

Microsoft TCP/IP for Windows NT 4.0 uses the LMHOSTS file. The file contains the mappings of IP addresses to Windows NT 4.0 Computer Names (NetBIOS names) for Windows-networking computers located outside the local subnet. The LMHOSTS file is read when WINS or broadcast name resolution fails, and resolved entries are stored in a local cache for greater access.



ATTENTION

- This procedure is to be used in conjunction with one of the TPS implementation checklists.
- The HOSTS and LMHOSTS files must be accurate and consistent on every node. The files provide a backup for the name resolution normally provided by the DNS server, to ensure TPS software reliability in a failed network.

3. Domain controller procedures

3.7. Editing the HOSTS and LMHOSTS files

Step	Action
1	Open Windows Explorer.
2	If a site-specific LMHOSTS file is not already available in C:\WINDOWS\system32\drivers\etc\., then copy the sample file LMHOSTS.sam and paste the file into the following directory and rename the file LMHOSTS. C:\WINDOWS\System32\Repl\Export\Scripts
3	Right-click > Open the following file in Notepad: C:\WINDOWS\System32\Repl\Export\Scripts\LMHOSTS
4	Modify the LMHOSTS file to include the Ethernet identification of all GUS nodes, APP nodes, Domain Controllers, and any other nodes that will be accessed. For Example: The following are LMHOSTS file identification entries for an APP node and a GUS node: 160.221.103.22 app1 # APP1 160.221.102.67 gus1 # GUS1 127.0.0.1 localhost
	ATTENTION Remove all non-essential host identifications—do not remove the “localhost” identification entry.
5	Save the LMHOSTS file, and then Close Notepad.
6	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

3.8 Editing OPERATOR.BAT file

Purpose of OPERATOR.BAT file

The accounts used by operators and supervisors are set up with restrictive desktop environments (accounts have been restricted by Group Policy Objects (GPOs). When the user logs on to the machine, the operator.bat file executes to start the applications required by the end user, such as Native Window, SafeView, and GUS displays.

There are two methods to implement the operator.bat file for TPS:

- User profile-specified logon script (**Note:** This is the preferred method for TPS.)
 - Can be used for local and domain accounts.
 - Is executed with or without the Domain Controller.
- GPO-specified logon script
 - Can be used for domain accounts only.
 - Is not executed if no Domain Controller is available.

User profile-specified logon script

The local logon script file is in the following location:

C:\WINDOWS\system32\repl\import\scripts

To attach a logon script file to an account, enter the name of the file in the **Profile** tab of the **Properties** dialog box for the account.

GPO-specified logon script

When a user logs on to a domain account, the appropriate GPO gets applied, and the associated logon script runs. When Operator (or View or Supervisor) logs on to a node, the authenticating Domain Controller provides the appropriate operator.bat logon script that gets executed at the GUS. The Engineer account by default has no logon script configured.

The logon script files are on the Domain Controller in the following location:

**C:\WINDOWS\SYSVOL\domain\Policies\{...group policy SID...}
\User\Scripts\Logon**

All of the logon script files have the same name: **operator.bat**.

GPOs can be filtered to apply to only specific groups or users. In the TPS system, for example, the Operators Group Policy applies only to the Operators global group. Each policy can have a different logon script; therefore, the policy that applies to the Supervisors group may have a different logon script than the policy that applies to the Operators group.



ATTENTION

A logon script attached to a GPO is not executed if no Domain Controller is available.

The preferred approach for TPS is to attach the logon script to the domain user account, so that the user profile-specified logon script runs with or without the Domain Controller.

Edit GPO-specified OPERATOR.BAT

To edit a GPO-specified Operator.bat file, perform the following steps to navigate through the GPO screens to the operator.bat file associated with the policy of interest (for example, Operators Group Policy).

Step	Action
1	Log on to the Domain Controller as administrator.
2	Open Active Directory Users and Computers: Start > Programs > Administrative Tools > Active Directory Users and Computers .
3	Display the properties of your domain by right-clicking the domain and selecting Properties .
4	In the domain properties dialog box, click the Group Policy tab, select the policy of interest (for example Operators Group Policy) , and click Edit .
5	In the Console Tree of the Group Policy window, expand User Configuration > Windows Settings , and click Scripts (Logon/Logoff) .
6	In the Details Pane, right click the Logon item and select Properties .
7	In the 'Logon Properties' dialog box, click Show Files . Note: If there are no files in the folder displayed, copy the operator.bat.sam file to this location and rename it operator.bat. (The sample file OPERATOR.BAT.SAM is installed on the Domain Controller by the Domain Controller Package.)
8	Right-click the operator.bat file and select Edit .
9	Modify the file as needed. Refer to "Edits for the OPERATOR.BAT file" for more information.
10	Save the file and close Notepad.
11	Close all windows associated with the GPO and Active Directory Users and Computers.
12	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

Edit and distribute user profile-specified OPERATOR.BAT

Step	Action
1	Log on to the Domain Controller as administrator.
2	Use Windows Explorer to navigate to the directory: C:\WINDOWS\system32\Repl\export\Scripts\
3	Open the following file in Notepad: Operator.bat Note: If there are no files in the folder displayed, copy operator.bat.sam and name it Operator.bat . (The sample file OPERATOR.BAT.SAM is installed on the Domain Controller by the Domain Controller Package.)
4	Modify the file as needed. Refer to "Edits for the OPERATOR.BAT file" for more information.
5	Save the file and close Notepad.
6	Distribute the operator.bat file to the GUS nodes by executing the following command in a command prompt window: C:\WINDOWS\system32\Repl\export\Scripts\Copy_Utility bat RESULT: The Operator.bat file is copied to the following location on the GUS nodes: C:\WINDOWS\system32\repl\import\scripts
7	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

Edits for the OPERATOR.BAT file

Listed below are modifications for the Operator.bat file for use on a GUS node:

Step	Action
1	<i>If the SafeView package is to be installed, perform the following modifications:</i> • Uncomment the SafeView line, and then enter the pathname for the Workspace file to use. • Add a line to start the following executable: C:\Program Files\Honeywell\TPS\Base\enablesetforeground.exe • In addition to modifying the operator.bat file, you must grant the affected users and groups Read and Execute permissions to the enablesetforeground.exe file.



ATTENTION

For SafeView to function properly the "enablesetforeground.exe" file must be added to the operator.bat file or other customized logon script and the appropriate permissions set for the file to be run. If the file is not run, then all SafeView calls issued after an initial call fail to display in the active window; instead, they open to the Task Bar.

As of GUS R350, GUS startup defaults to enableSetForeground. This option is the equivalent of including enableSetForeground.exe in a startup .bat file.

This new option can be turned off by the **Allow SetForeground during Startup** checkbox on the GUS Display Runtime Settings in the Configuration Utility.

Note: Unchecking this option only prevents enablesetforeground from running on future logins, it does NOT change the registry settings for existing users (that can still be done through the AllowSetForeground utility).

2	If GUS pictures are to be displayed at logon time, follow the instructions in the operator.bat file to start up a GUS picture as part of the logon sequence.
---	--

3. Domain controller procedures

3.8. Editing OPERATOR.BAT file

Step	Action
3	If Windows 2003 Offline Files are to be used to distribute GUS displays, then map a network drive that points to the 'scripts' share on the Domain Controller by changing the following lines: From this <pre>SET DISPLAYS=%SYSTEMROOT%\system32\replimport\HWIAC\ databases\displays SUBST G: /d SUBST G: %DISPLAYS%</pre> To this <pre>NET USE G: /delete NET USE G: \\<domain>\Scripts /persistent:YES</pre> Where: <domain> is the domain name (for example, mydomain.local)



TIP

NET USE vs SUBST

NET USE

- Must be used if using [Offline Files](#) to distribute GUS displays
- Can access a shared folder on the network (or on a local drive)
- Maps a network drive
- The mapped drive can be accessed only if the network is up (even if accessing a local share)

SUBST

- Can only access a local folder (does not have to be shared)
 - Creates a virtual drive
 - Can be accessed even if the network is down
 - For systems that are using methods of distributing GUS displays other than Offline Files
-

For additional details on using offline files to distribute GUS displays, refer to section 6.3, "Configuring Windows 2003 offline files and folders for replication".

For examples on configuration of the Operator.bat file for GUS nodes, refer to section 6.4, "Starting a GUS node for operations".

3.9 Editing the COPY_UTILITY.BAT file

Using Notepad to edit the COPY_UTILITY.BAT file

The COPY_UTILITY batch file is used to distribute the following files on the Domain Controller to members of the domain:

- OPERATOR.bat file
- HOSTS file
- LMHOSTS file



ATTENTION

This procedure is to be used in conjunction with one of the *TPS Implementation Checklists*.

Step	Action
1	Copy the COPY_UTILITY.BAT.SAM file from the following directory: C:\WINDOWS\System32\Repl\Export\Scripts\ And paste it into the same directory.
2	Rename the copy to be the following: COPY_UTILITY.BAT
3	Remove the read-only property from the COPY_UTILITY.BAT file. Right-click > Edit the file in Notepad. Find the line set nodegroup1 . Modify the file to include the Computer Names of all GUS nodes, APP nodes, and Domain Controllers. Observe the spaces between the names. Example 1: Added GUS node named "GUS24", APP node named "APP27", and domain controller named DC70: set nodegroup1 = GUS24 APP27 DC70



ATTENTION

Remove all non-essential host identifications.

Save the modified **COPY_UTILITY.BAT** file and then close Notepad.

If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

3. Domain controller procedures

3.9. Editing the COPY_UTILITY.BAT file

COPY_UTILITY operation on a TPS System

For instructions on using COPY_UTILITY to distribute the hosts, lmhosts, and operator.bat files, refer to section 4.9, "Distribute HOSTS, LMHOSTS and Operator.bat files".

The following table describes copy_utility batch file commands on a TPS system:

COPY_UTILITY Operation on TPS Systems	
Batch File Command	Action
copy_utility BAT	Copies all *.bat files (and therefore "operator.bat") from this directory on the Domain Controller: %windir%\system32\repl\export\scripts to these GUS and APP node directories: ADMIN\$\system32\repl\import\scripts
copy_utility CMD	Copies all *.cmd files from this directory on the Domain Controller: %windir%\system32\repl\export\scripts to these GUS and APP node directories: ADMIN\$\system32\repl\import\scripts
copy_utility NET	Copies all *HOSTS files (and therefore the HOSTS and LMHOSTS files) from this directory on the Domain Controller: %windir%\system32\repl\export\scripts to these GUS and APP node directories: ADMIN\$\system32\drivers\etc

3.10 Performing 'TPS domain' OU procedures

Purpose of 'TPS domain' OU

Creating a 'TPS Domain' OU allows you to organize and manage plant nodes of interest from a Honeywell System Management Display. Thus, Plant Control Network nodes can be organized along the lines of classic DCS systems into areas of interest or responsibility. A 'Console' OU provides a grouping of similar process control computers within a 'TPS Domain' OU.



REFERENCE – INTERNAL

For detailed information on 'TPS Domain' and 'Console' OUs, see the *System Management Configuration Guide*.



ATTENTION

These procedures are to be used in conjunction with one of the "TPS Implementation Checklists".

To install 'TPS domain/console configuration tool' on DC

This procedure installs an extension to the Active Directory Users and Computers snap-in. The extension is a "Tps Domain" tab in the Properties dialog box of an OU. The "Tps Domain" tab is used to assign the "TPS Domain" attribute to an OU which gives the OU's icon a different appearance.



TIP

To check if the TPS Domain/Console Configuration tool has been installed on the Domain Controller already, access Active Directory Users and Computers, then right-click any OU and select Properties. If the Tps Domain tab is present in the Properties dialog box, then the tool is installed and you do not have to perform this procedure.

Step	Action
1	At the Domain Controller with domain administration privileges, insert the TPS Software CD .
2	Select the TPS Domain/Console Configuration Tool from the TPS Software CD browser (menu).
RESULT: A DOS window displays with the following prompt: Do you want to continue? [Y,N]	

3. Domain controller procedures

3.10. Performing 'TPS domain' OU procedures

Step	Action
3	At the command prompt, type Y to continue. Result: The path displays and then the installation progress screen displays. After the installation completes, the progress screen and the DOS window go away.
4	Click Exit on the TPS Software CD browser, and then click OK. Remove the CD from the drive. RESULT: The "Tps Domain" property page is now available for each OU in Active Directory.
	ATTENTION The Tps Domain tab may not appear in the OU Properties dialog box until the Domain Controller is restarted.
5	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

To migrate existing legacy TPS domain structures to Active Directory OUs

This procedure creates an OU in Active Directory Users and Computers for each existing legacy TPS Domain. Using existing registry values, it sets the OU's display Name property to "TPSDomain" (where "TPSDomain" comes from the registry value SOFTWARE\Honeywell\TPSServer\<TPSDomain>\DomainName).

All computers (nodes) in that legacy TPS Domain are moved under the new OU (where the computer names come from the registry values SOFTWARE\Honeywell\TPSServer\<TPSDomain>\Nodes\<Node1>\<NodeName>).



ATTENTION

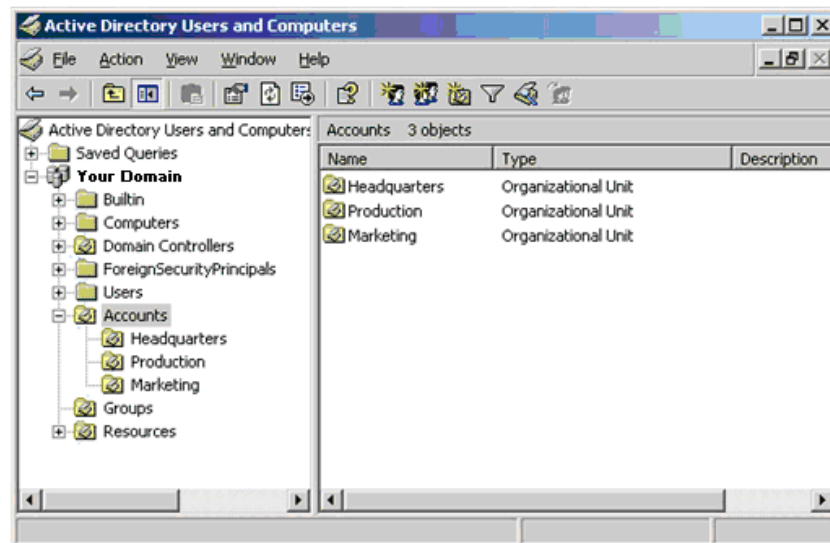
This procedure is required only if you need to migrate an existing legacy TPS domain structure to the new Microsoft organization unit (OU) structure.

Step	Action
1	At the Domain Controller with domain administration privileges, insert the TPS Software CD.
2	Select TPS Domain Structure Migration to OUs. RESULT: The legacy TPS Domain structures are converted to OUs in Active Directory Users and Computers.

Step	Action
3	At the Command Prompt, enter Y to continue. RESULT: TPS Domain OUs are created in Active Directory Users and Computers.
4	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

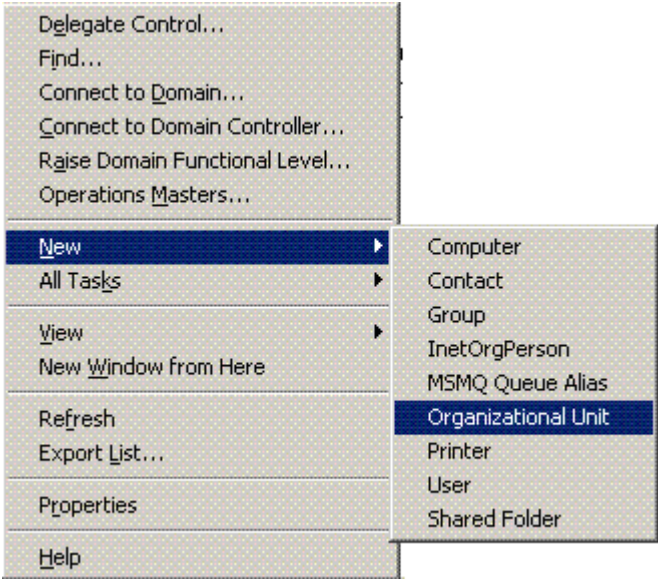
To create a new OU

Step	Action
1	Click Start > Programs > Administrative Tools > Active Directory Users and Computers . The Active Directory Users and Computers window appears.



3. Domain controller procedures

3.10. Performing 'TPS domain' OU procedures

Step	Action
2	In the left pane, right-click the domain name (for example, 'Your Domain' shown in the screen example in step 1), point to New, and then click Organizational Unit. 
3	Type the OU name in the Name box, and then click OK. The new OU is created and is displayed hierarchically below the domain (for example, 'Accounts' shown in the screen example in step 1).

To create a new OU within another OU

Step	Action
1	Right-click the parent OU and point to New, and then click Organizational Unit. The new OU is created and is displayed hierarchically below the parent OU (for example, 'Marketing' shown in the screen example in step 1 of the previous procedure).

To assign the 'TPS domain' attribute to an OU

To assign the 'TPS Domain' attribute to an OU

Step	Action
1	At the Domain Controller, with domain administration privileges, open Active Directory Users and Computers.
2	Create an OU. See section 3.10, "To create a new OU".
3	Right-click the new OU and select Properties .
4	Select the Tps Domain tab, and then select the TPS Domain option.
5	Click OK .
	RESULT: The OU now has the "TPS Domain" attribute.
6	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

To assign the 'console' attribute to an OU within a 'TPS Domain' OU

Step	Action
1	Click Start > Programs > Administrative Tools , and then click Active Directory Users and Computers . The Active Directory Users and Computers window appears.
2	Create an OU within the 'TPS Domain' OU. See "To create a new OU within another OU".
3	Right-click the new OU and select Properties . The Properties dialog box appears.
	 TIP The TPS Domain tab appears on an OU's Properties dialog box only if you have installed the TPS Domain/Console Configuration tool on the Domain Controller.
4	Click the TPS Domain tab. The TPS Domain Properties dialog box appears.
5	Click Console .
6	Click OK . The 'Console' OU is created in a 'TPS Domain' OU.

3. Domain controller procedures

3.10. Performing 'TPS domain' OU procedures

To move objects to an OU

Step	Action
1	Click Start > Programs > Administrative Tools , and then click Active Directory Users and Computers . The Active Directory Users and Computers window appears.
2	Right-click the object you want to move. You can select any one of the following: • Computers • Users • Groups
3	Select Move from the pop-up menu. A dialog box lists all the possible containers to which you can move the selected object.
4	Select the appropriate container and click OK . The selected object is moved.

To move a TPS GUS or APP node to a 'TPS domain' OU

Prior to performing this procedure, a 'TPS Domain' OU must have been created.

A TPS GUS or APP node that is to be added to a 'TPS Domain' OU must be a member of a Windows 2003 domain.



ATTENTION

GUS 3xx / APP 2xx nodes (TPS R1xx system) can share a single Windows domain with GUS 4xx / APP 4xx nodes (TPS R4xx system) only if

- the Domain Controller is installed with Windows Server 2003 Standard Edition
- TPS R1xx nodes occupy separate OUs from TPS R400 and greater nodes

If you have a single domain with both TPS 2xx and TPS R4xx nodes, create one or more OUs for the TPS R1xx nodes, and create separate OU(s) for the TPS R4xx nodes.

Step	Action
1	At the Domain Controller, with domain administration privileges open Active Directory Users and Computers .
2	Select the default Computers container to view all computers that are not yet assigned to an OU.
3	Select the computer (GUS or APP node) that you want to assign to a 'TPS Domain' OU.
4	Right-click the computer and select Move .
5	Select the 'TPS Domain' OU that is to contain the computer.
6	Click OK .
	RESULT: The GUS or APP node is moved to the 'TPS Domain' OU.
7	Restart the machine that was just moved.
8	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

Delegating control of an OU

The OUs can delegate the administrative control over them to a subset of users in the Active Directory. For instance, the domain administrator needs to designate one person in each department as the official Password Change Administrator. This reduces the administrative load. The domain administrator can delegate the authority to modify users' passwords to each user over only their respective OU. The administrator authenticates the users, but restricts them to certain areas of the Active Directory infrastructure.

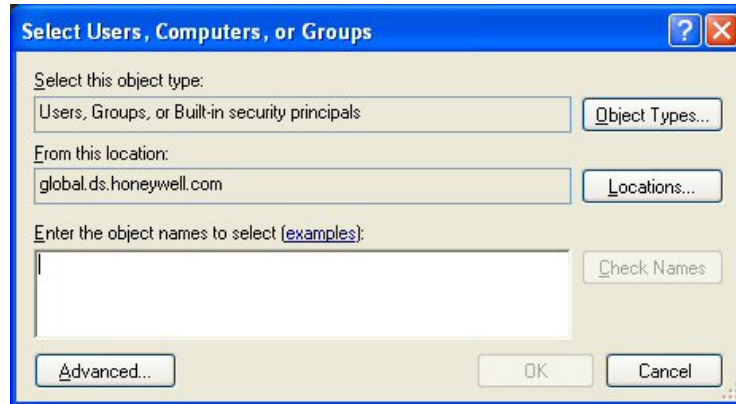
To delegate control of an OU

Step	Action
1	Click Start > Programs > Administrative Tools , and then click Active Directory Users and Computers . The Active Directory Users and Computers window appears.
2	In the left pane, right-click the OU which you want to delegate, and then click Delegate control . The Delegation of Control wizard appears.
3	On the Welcome screen, click Next . The Users or Groups screen appears.

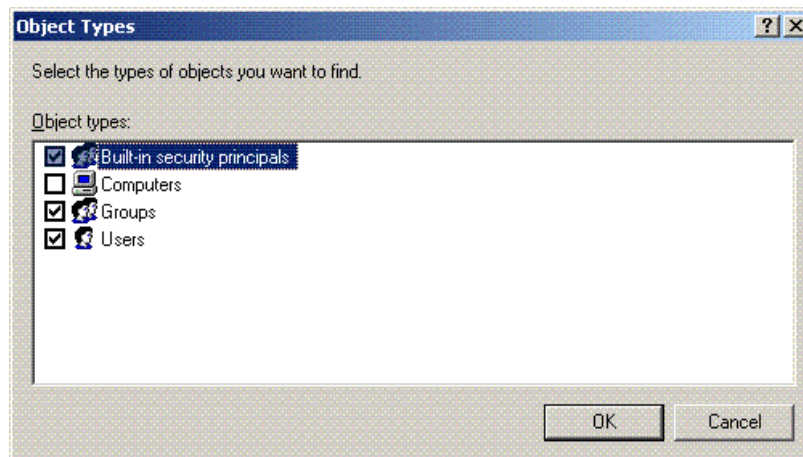
3. Domain controller procedures

3.10. Performing 'TPS domain' OU procedures

- | Step | Action |
|------|--|
| 4 | Click Add to select the users or groups to which control of the OU needs to be delegated. The Select Users, Computers or Groups dialog box appears. |

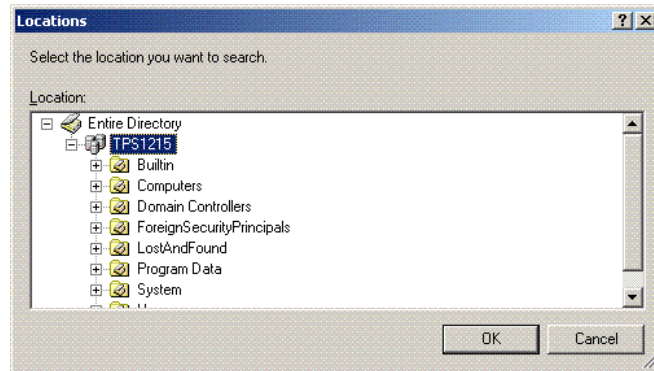


- | | |
|---|---|
| 5 | Click Object Types .
The Object Types dialog box appears. |
|---|---|

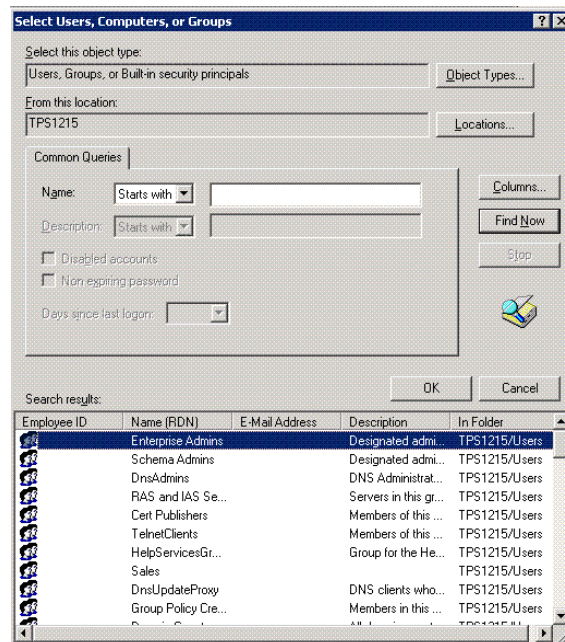


- | | |
|---|---|
| 6 | Select the object types you require and then click OK .
The Select Users, Computers or Groups dialog box appears. |
|---|---|
-

- | Step | Action |
|------|---|
| 7 | Click Locations . The Location dialog box appears. |



- | | |
|---|--|
| 8 | Select the location and then click OK . The Select Users, Computers or Groups dialog box appears. |
| 9 | Click Advanced . The Select Users, Computers or Groups dialog box appears. |



3. Domain controller procedures

3.10. Performing 'TPS domain' OU procedures

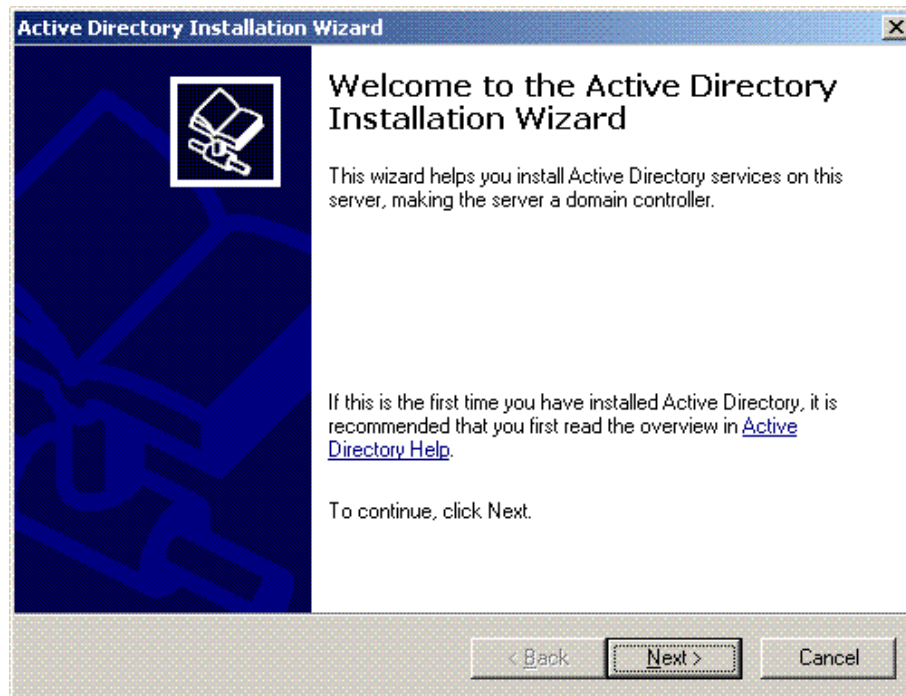
Step	Action
10	Select the objects from the search results box and click OK .
	TIP You can also type the name which you want to search in the Name box and click Find Now to search for that particular name.
11	Click OK until you encounter the Delegation of Control wizard.
12	Click Next . The Tasks to Delegate screen appears.
13	Select one of the following: • Delegate the following common tasks (Follow step 14) • Create a custom task to delegate(Follow step 15)
To delegate a set of common tasks	
14	Use Delegate the following common tasks option, if you want to delegate one or more commonly delegated tasks. a) Select the tasks you want to delegate. b) Click Next. c) Specify the details until you encounter the Finish option. d) Click Finish.
To delegate a set of custom tasks	
15	Use Create a custom task to delegate option if you want to delegate a set of custom tasks. a) Click Next. The Active Directory Object Type screen appears. b) Select the scope of delegation. c) Click Next. The Permissions screen appears. d) Select the permissions for delegation (Example: Full Control) e) Click Next. The summary screen appears. f) Click Finish.

3.11 Creating a new child domain in an existing domain tree

**ATTENTION**

This procedure is to be used in conjunction with one of the *TPS Implementation Checklists*.

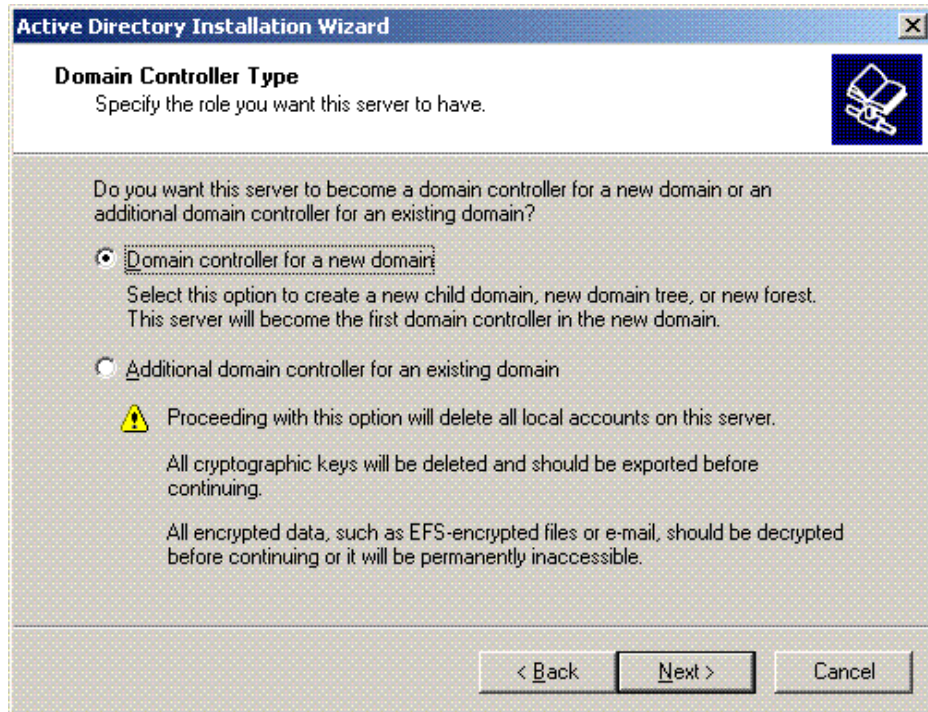
Step	Action
1	To initiate the Active Directory Installation Wizard, do one of the following, depending on the type of installation being performed: • If installing a new Root Domain Controller, then select Start > Run and enter DCPROMO. • If this is a migration of a Windows 2000 Root Domain Controller to Windows 2003 Root Domain Controller, then insert the Microsoft Windows 2003 Server CD and select Upgrade.



3. Domain controller procedures

3.11. Creating a new child domain in an existing domain tree

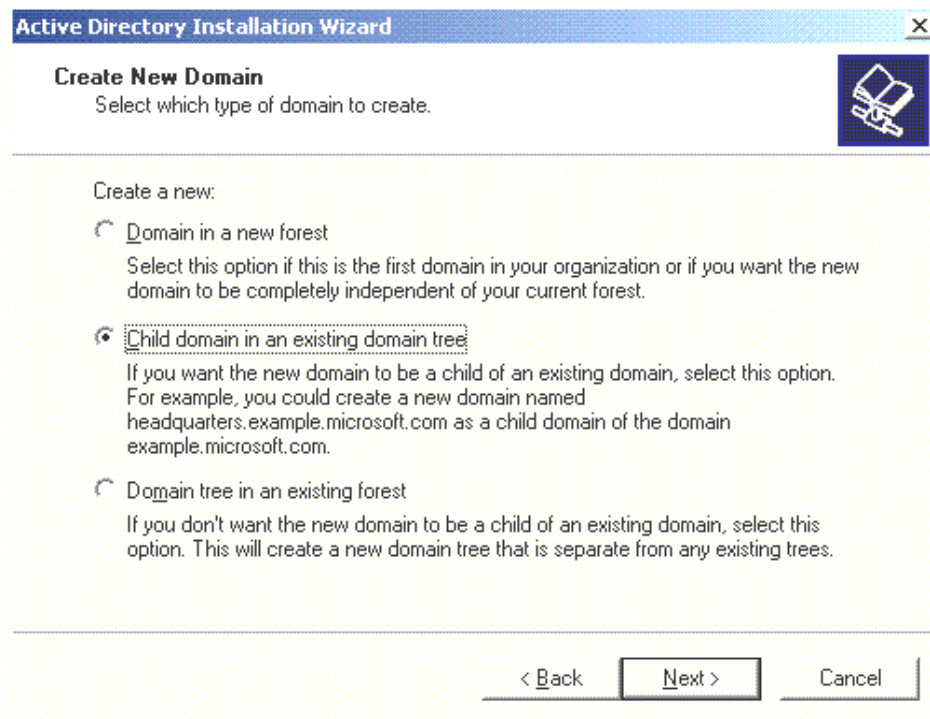
Step	Action
2	At the Active Directory Installation Wizard, select Domain Controller for a new domain .



3	Click Next .
---	---------------------

Step**Action**

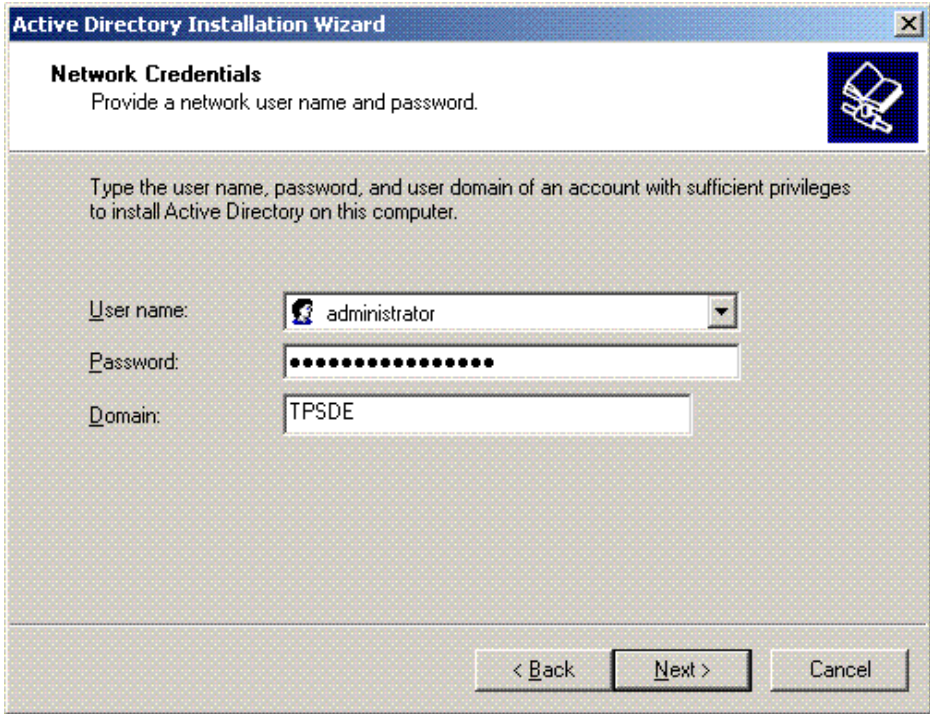
- 4 Select **Child domain in an existing domain tree**.

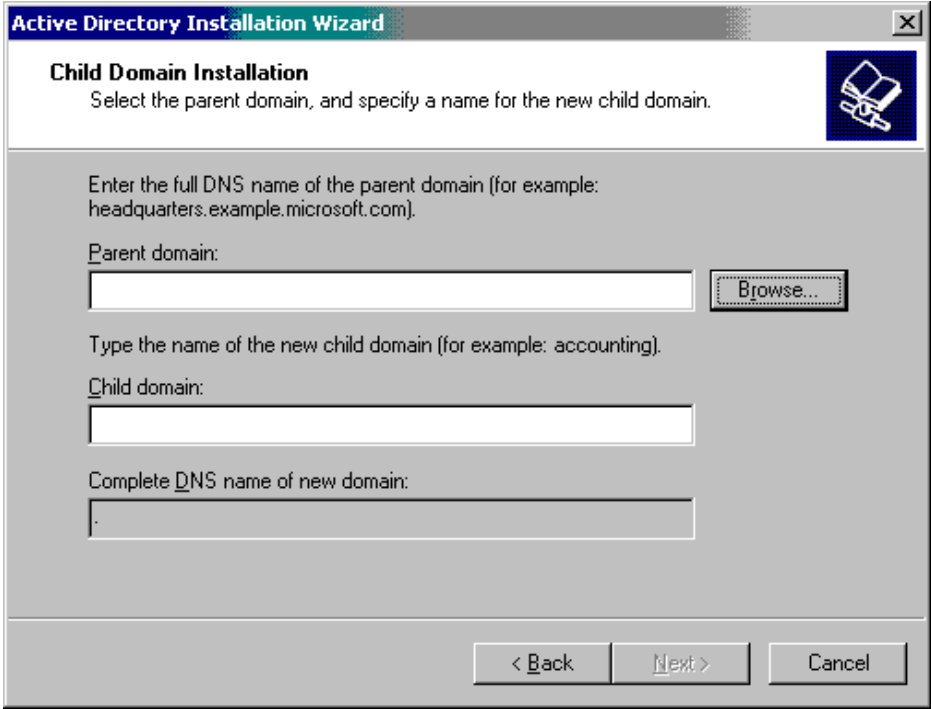


- 5 Click **Next**.
-

3. Domain controller procedures

3.11. Creating a new child domain in an existing domain tree

Step	Action
6	Type a user name and password that has privileges to add a child domain. Typically, this is an account in the Enterprise Administrators group.
	
 ATTENTION You may need your IT administrator to enter the admin username and password from the root domain.	
7	Click Next .

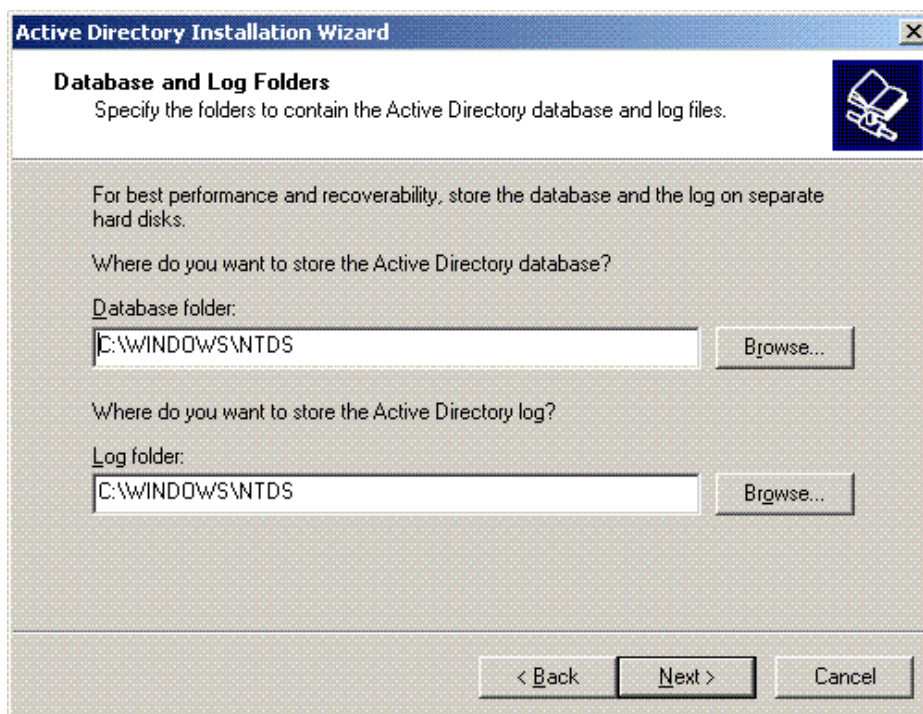
Step	Action
8	Type the <u>full DNS name</u> of the parent domain and, followed by the name of the child domain. EXAMPLE: • Parent domain: NorthAmerica.Honeywell.com • Child domain: ProcessControl1 In the above example, the full DNS name for the new child domain is ProcessControl1.NorthAmerica.Honeywell.com. 
9	Click Next .

3. Domain controller procedures

3.11. Creating a new child domain in an existing domain tree

Step	Action
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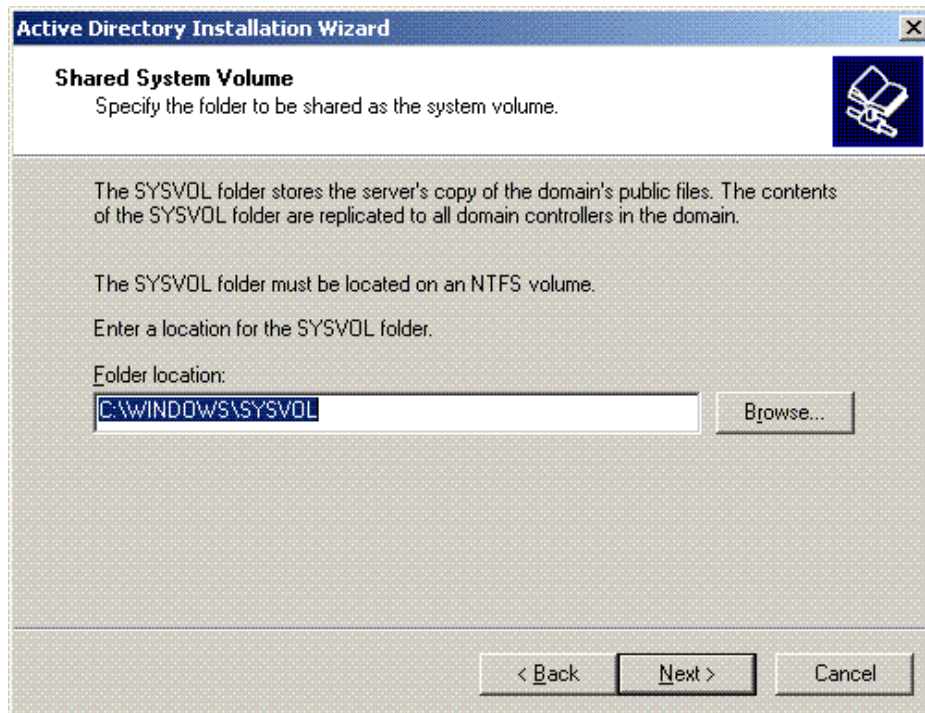
- | | |
|----|---|
| 10 | The following dialog box requests directory paths to store the Active Directory database and log files. Typically, the default directories are adequate. If disk space is an issue, select an alternate location. |
|----|---|



- | | |
|----|---------------------|
| 11 | Click Next . |
|----|---------------------|
-

Step**Action**

- 12** Click **Browse**. Select the directory location for **SYSVOL**. Typically, this directory is stored in C:\WINDOWS\SYSVOL.



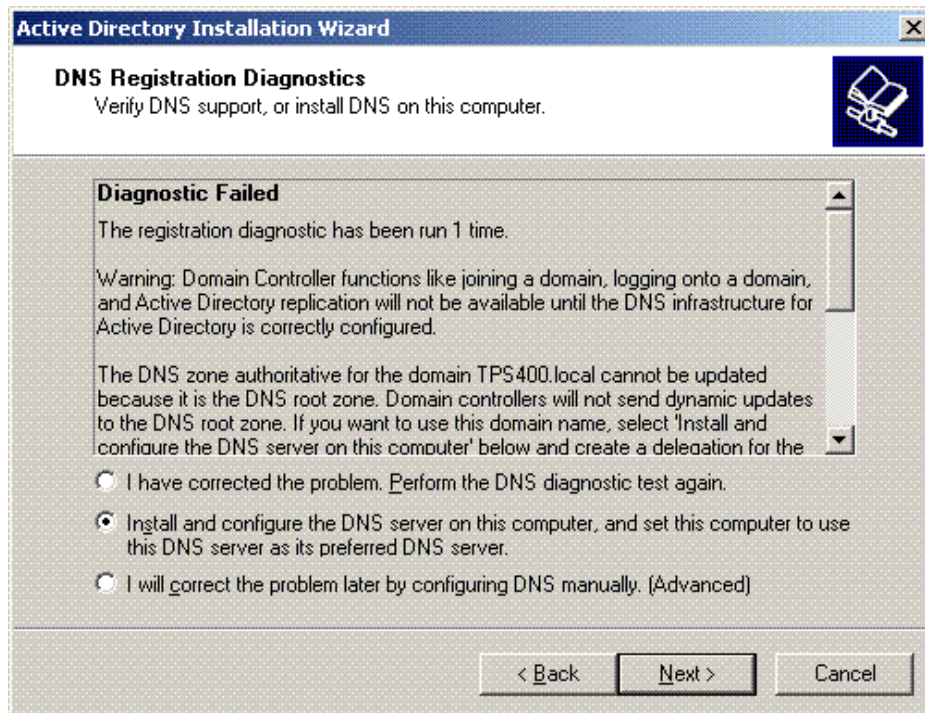
- 13** Click **Next**.
-

3. Domain controller procedures

3.11. Creating a new child domain in an existing domain tree

Step	Action
------	--------

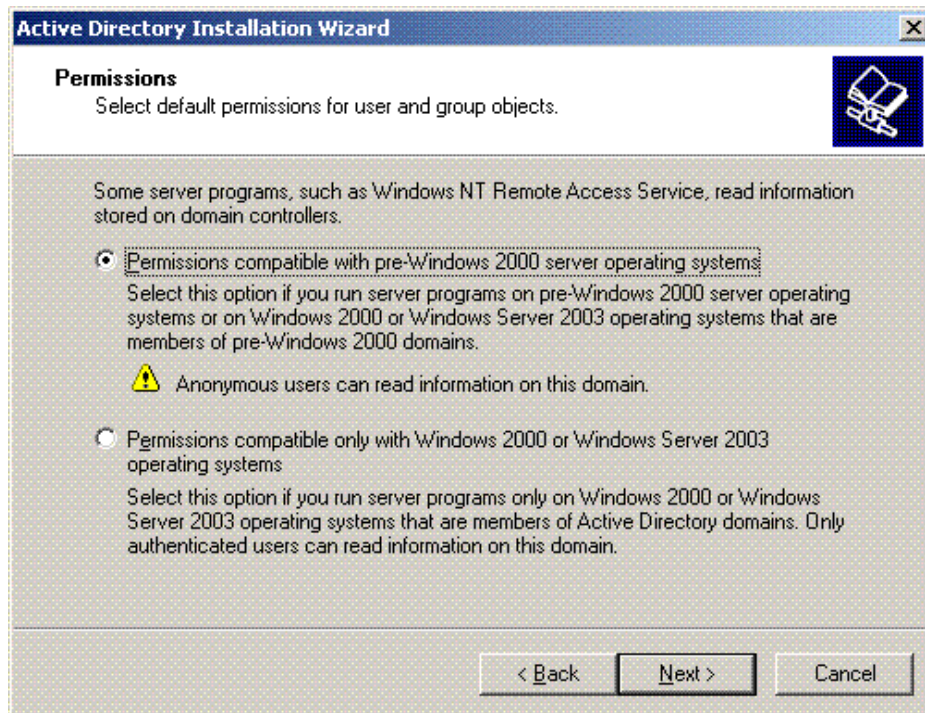
- | | |
|----|---|
| 14 | If you are prompted for a DNS installation, select Install and configure the DNS server on this computer . |
|----|---|
-



- | | |
|----|---------------------|
| 15 | Click Next . |
|----|---------------------|
-

Step**Action**

- 16** For correct TPS operation, select **Permissions compatible with pre-Windows 2000 server...**



- 17** Click **Next**.

3. Domain controller procedures

3.11. Creating a new child domain in an existing domain tree

Step	Action
18	Select an Active Directory Restore Mode Administration password.
	ATTENTION This is <u>not</u> your Windows 2003 Administrative password, but is an account used to repair your system in the event of a failure with Active Directory services.
	Active Directory Help.' At the bottom right are three buttons: '< Back', 'Next >', and 'Cancel'." data-bbox="229 331 798 671"/>
19	Click Next .
20	In the Summary screen, review your configuration settings and then click Next .
21	Click Finish on the Completing the Active Directory Installation Wizard dialog box, and then click Restart Now .
22	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

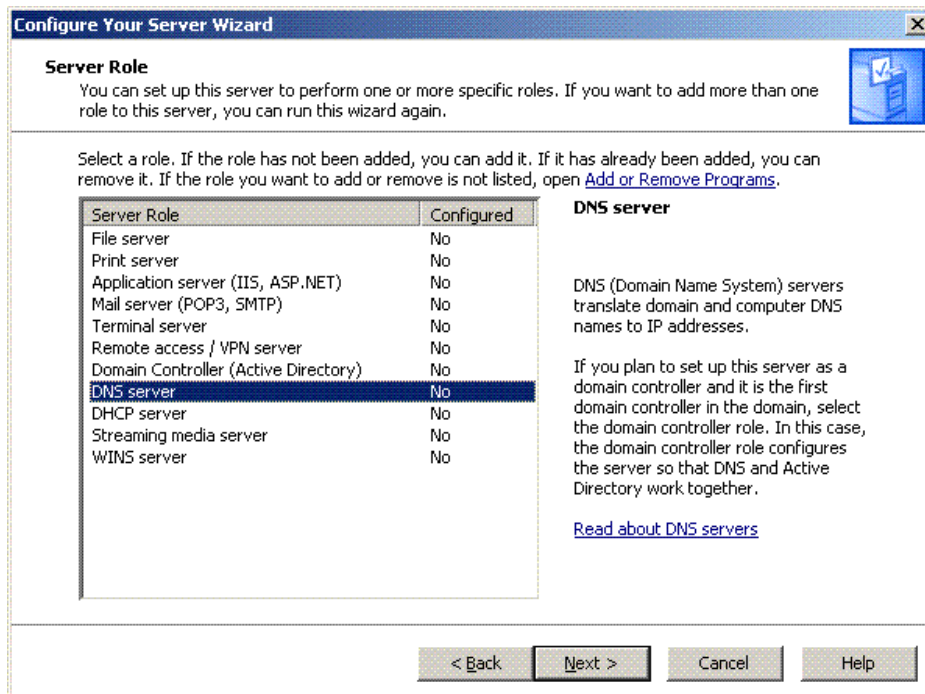
3.12 Installing DNS on child Domain Controller



ATTENTION

This procedure is to be used in conjunction with one of the *TPS Implementation Checklists*.

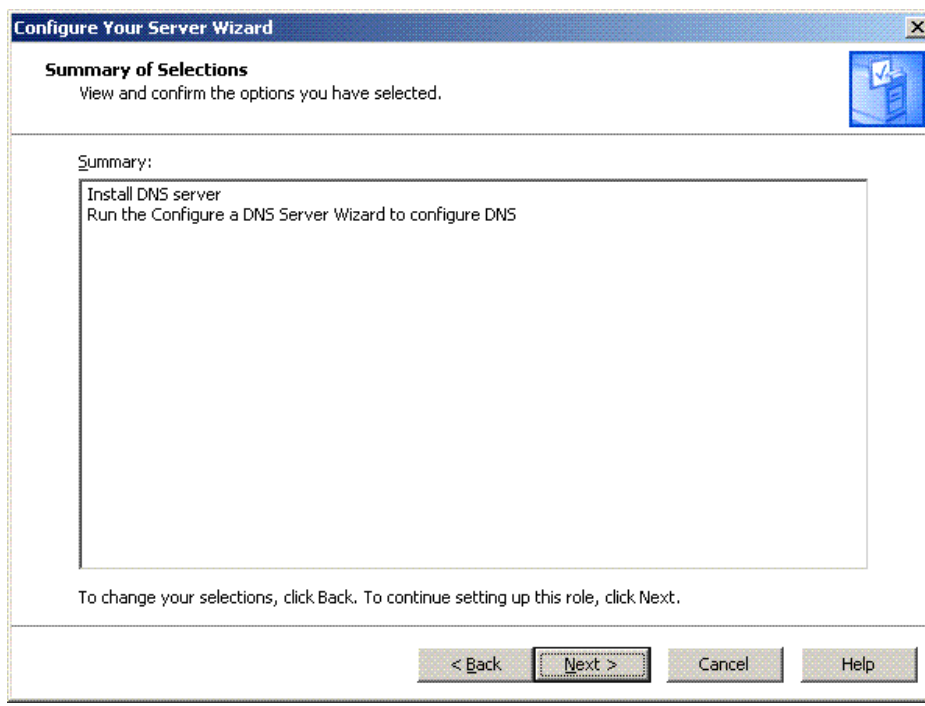
Step	Action
Install DNS Server	
1	Log on to the Domain Controller as the domain administrator.
2	Insert the Microsoft Windows 2000 Server CD .
3	Select the following: Start > Programs > Administrator Tools > Configure Your Server Wizard . In the Wizard, click Next in Welcome screens.
4	In the Server Roles dialog, select DNS Server .



3. Domain controller procedures

3.12. Installing DNS on child Domain Controller

Step	Action
5	The Configured column indicates No if DNS is not currently installed. Click Next to install DNS.
6	Click Next on the Summary screen to initiate DNS installation.

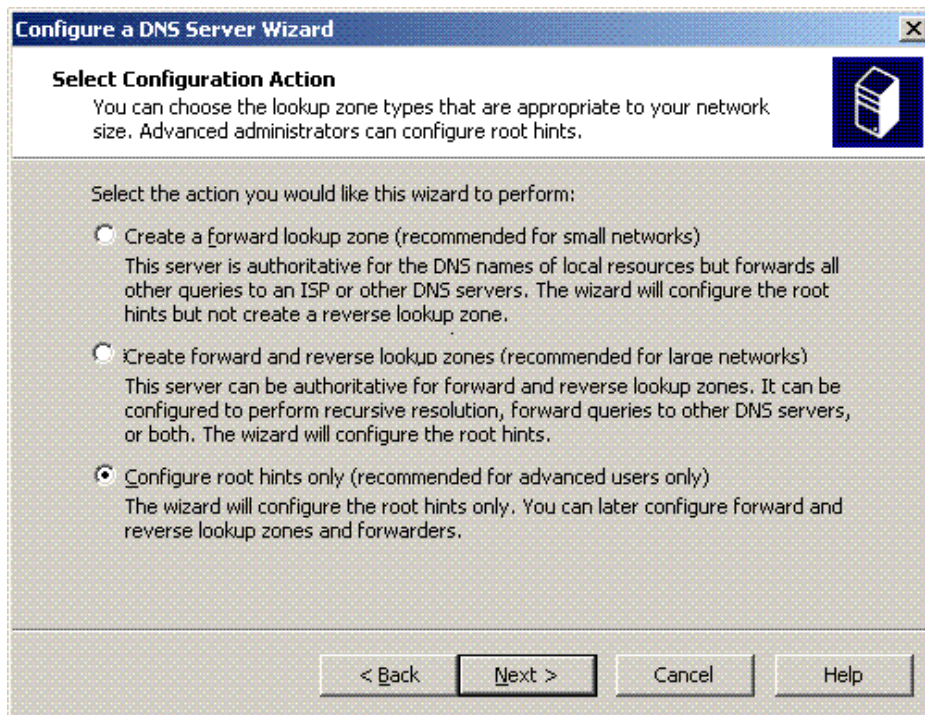


Step	Action
7	DNS installation is initiated. On completion, a Configure a DNS Server Wizard starts up.
	
8	Click Next .

3. Domain controller procedures

3.12. Installing DNS on child Domain Controller

Step	Action
9	In the following dialog, select Configure root hints only to configure forward and reverse lookup zones later. Result: DNS Server installs and configures.

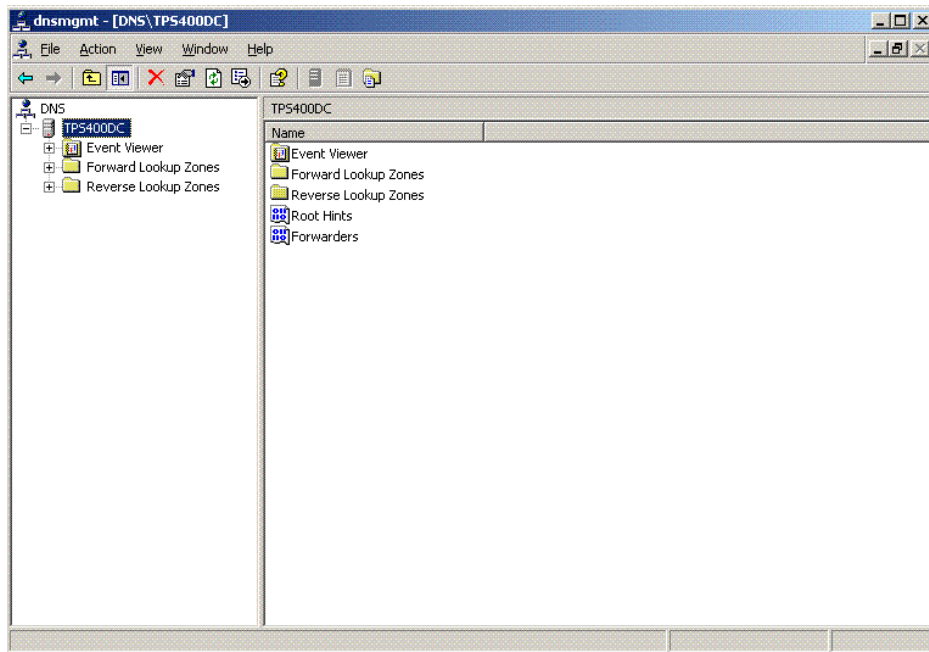


10	Click Next .
----	---------------------

Step	Action
------	--------

Configure Forward Lookup Zone

- 11** Log on to the Domain Controller as the domain administrator.
Select **Start > Programs > Administrator Tools > DNS**.



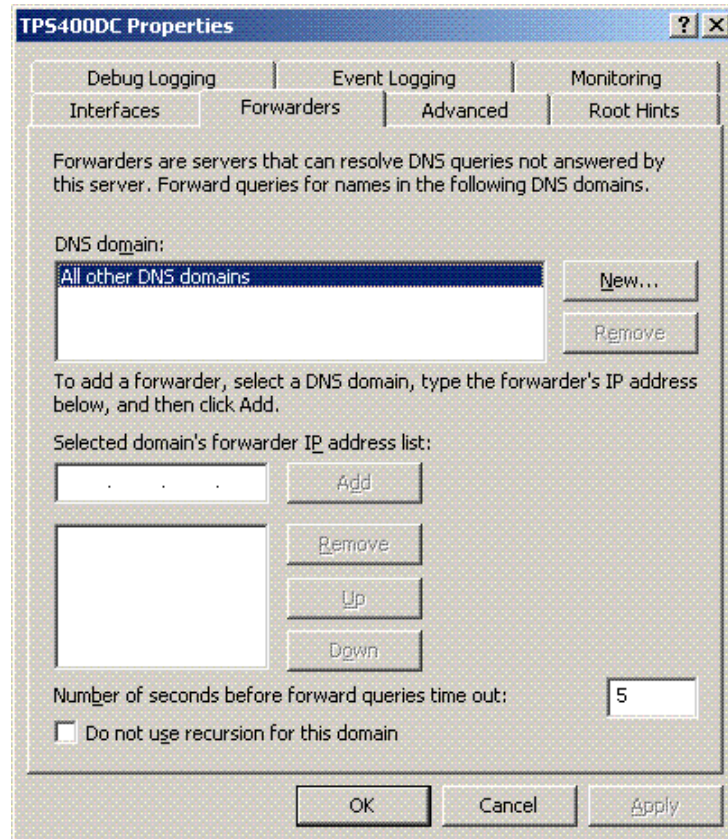
- 12** Select the icon for the Domain Controller.
13 Right-click the icon for the Domain Controller and select **Properties**.

3. Domain controller procedures

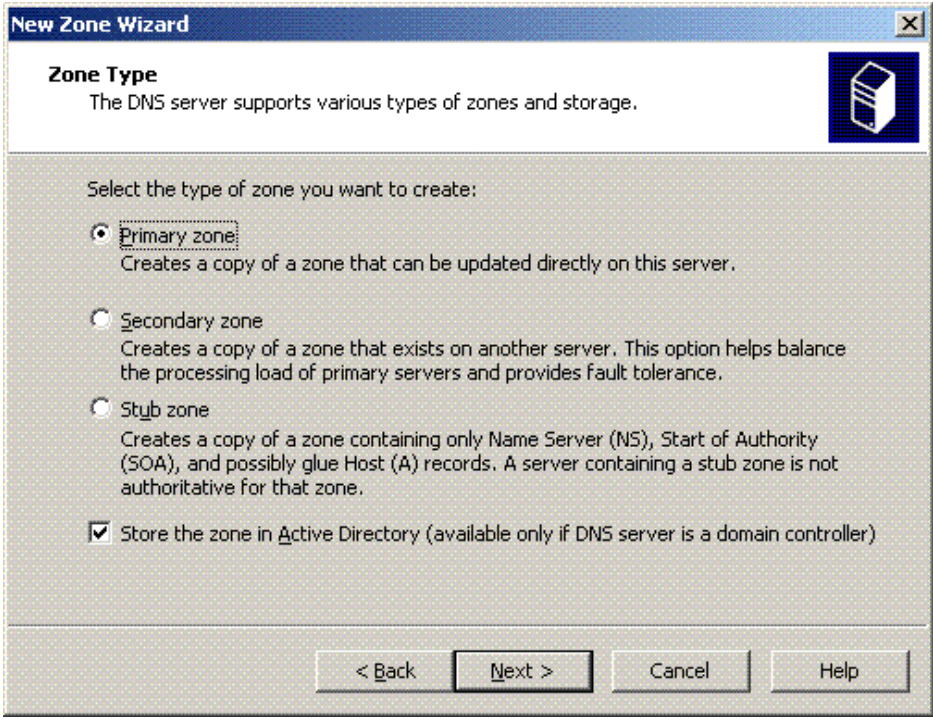
3.12. Installing DNS on child Domain Controller

Step	Action
------	--------

- | | |
|----|-----------------------------------|
| 14 | Select the Forwarders tab. |
|----|-----------------------------------|

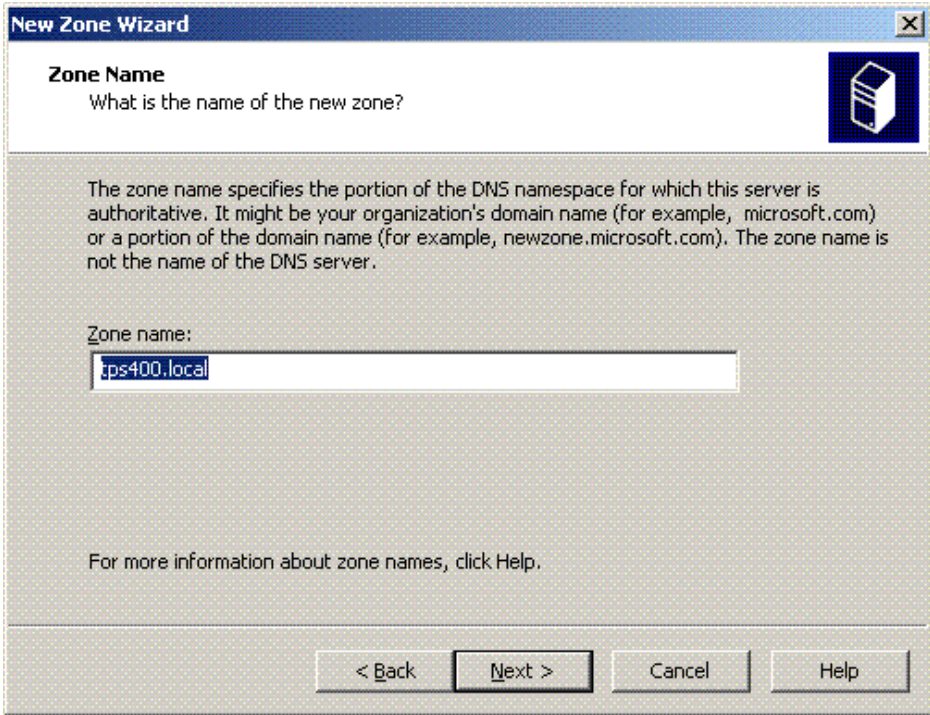


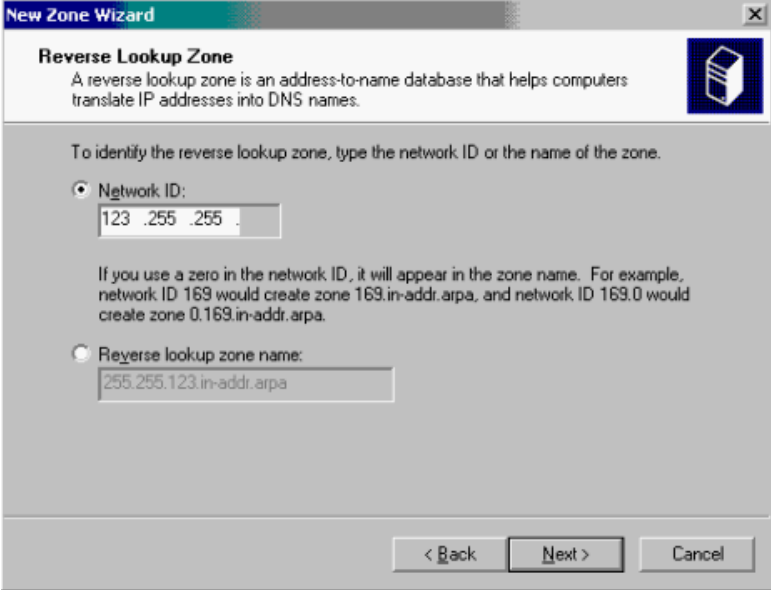
- | | |
|----|--|
| 15 | Add the IP address of the DNS Server you want to forward to (typically, the address of the enterprise DNS server). |
| 16 | Add the IP address of the secondary DNS Server, if any. |
| 17 | Click OK . |
| 18 | Right-click Forward Lookup Zone and select New Zone . |
| 19 | Click Next . |
-

Step	Action
20	Select the checkbox for Store the zone in Active Directory .
	
21	Click Next .

3. Domain controller procedures

3.12. Installing DNS on child Domain Controller

Step	Action
22	Enter the Zone Name. 
23	Click Next .
24	Click Finish .
Configure Reverse Lookup Zone	
25	Right-click Reverse Lookup Zone and select New Zone .
26	Select the option button for Active Directory .
27	Click Next .

Step	Action
28	Enter the Network ID of the local machine (first three octets of its IP address).
	
29	Click Next . Review the configuration, and then click Finish .
Configure TCP/IP Properties	
30	Select the following: a) Right-click My Network Places from Start menu and select Properties. b) Right-Click Local Area Connection and select Properties. c) Double-Click Internet Protocol.
31	Select the option button for Use the following DNS Server addresses .
32	Enter the IP address of the Preferred DNS server (the local machine). Enter the IP address of the Alternate DNS server .
33	Click OK .
34	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

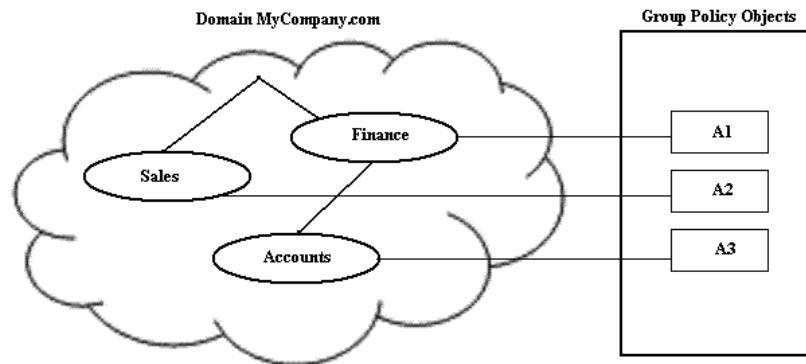
3. Domain controller procedures

3.13. Implementing Group Policy

3.13 Implementing Group Policy

An organization can have different types of users. You must deliver and maintain a customized desktop configuration for different types of users, including mobile users, information workers, or others assigned strictly to defined tasks. For example, one class of users does not have access to Internet Explorer, but the other class of users can access Internet Explorer. Group Policy helps in applying a customized configuration to a group of users.

The following figure shows the customized group policies assigned to the OUs within a domain:



You can infer the following from the preceding figure:

- The Group Policy A1 is applied to the Finance OU.
- The Group Policy A2 is applied to the Sales OU.
- The Group Policy A1 and A3 are applied to the Accounts OU.
- The members of the OUs receive the Group Policy from the respective OUs.

You can link the GPOs to the sites, domains, and OUs to implement the Group Policy settings. You can apply the settings to the organization or to a portion of the organization.

To create a Group Policy

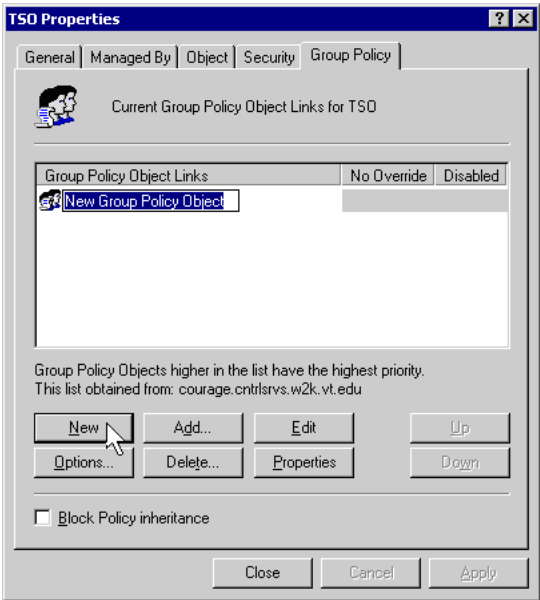
Step	Action
1	Click Start > Programs > Administrative Tools , and then click Active Directory Users and Computers . The Active Directory Users and Computers window appears.
2	In the left pane, right-click the container where you have to create the Group Policy. The pop-up menu appears.



TIP

You can create a GPO in the domain or in one of the OUs.

3	Click Properties . The Properties dialog box appears.
4	Click the Group Policy tab.
5	Click New to create a new GPO.



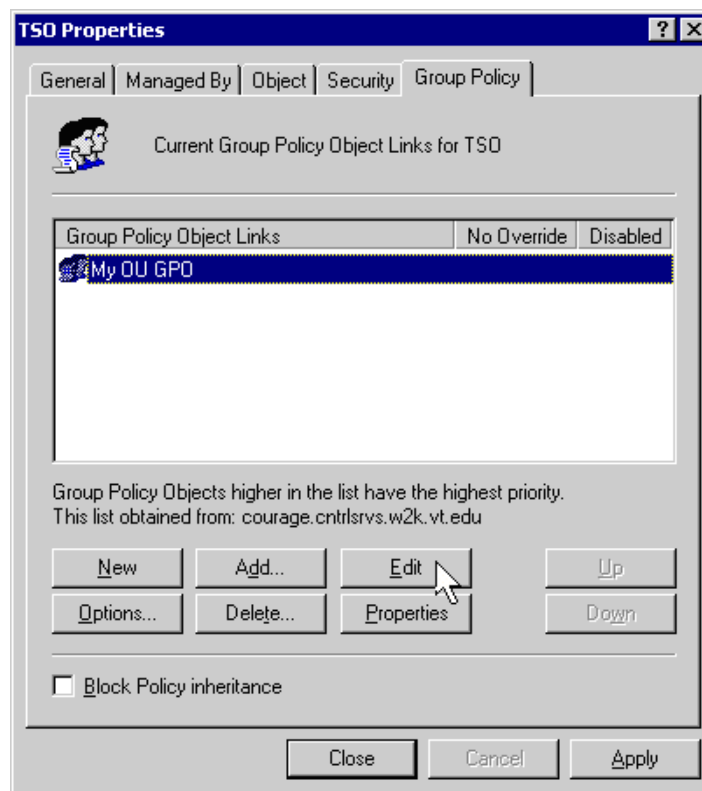
6	Rename the GPO. The new GPO is created. See "To edit a Group Policy".
---	--

3. Domain controller procedures

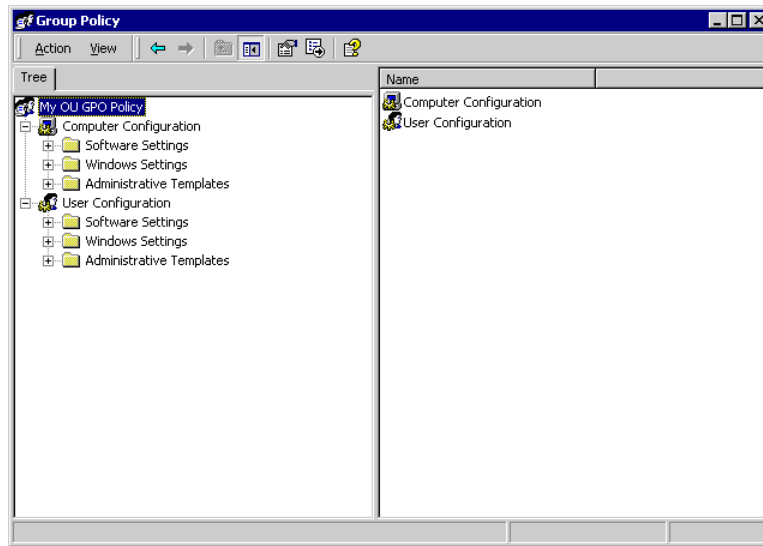
3.13. Implementing Group Policy

To edit a Group Policy

Step	Action
1	Click Start > Programs > Administrative Tools , and then click Active Directory Users and Computers . The Active Directory Users and Computers window appears.
2	In the left pane, right-click the container where you have the Group Policy. The pop-up menu appears.
3	Click Properties . The Properties dialog box appears.
4	Click the Group Policy tab.
5	Click to highlight the GPO you have to edit.



Step	Action
6	Click Edit . The Group Policy window appears. Policies can be viewed or modified by browsing the items within the scope pane. Most of the settings are applied within the User Configuration settings.



The settings under 'Computer Configuration' are applied to all computers and the settings under 'User Configuration' are applied to all users.

Applying a Group Policy to selected groups

The GPOs affect all users and systems that are contained in the domain or OU. Use DACLs (Discretionary Access Control Lists) to perform security configuration. Using this, a GPO can be modified to exclude or include the members of any security group.

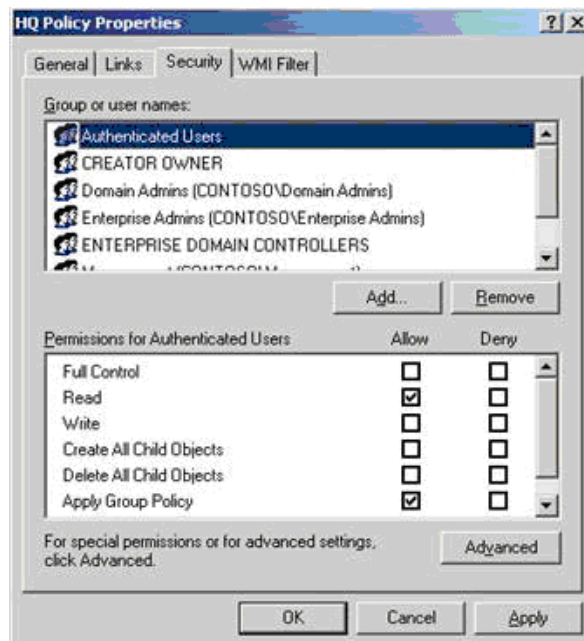
3. Domain controller procedures

3.13. Implementing Group Policy

To apply a Group Policy to selected groups

Follow the steps to apply a Group Policy to selected groups:

Step	Action
1	Click Start > Programs > Administrative Tools , and then click Active Directory Users and Computers . The Active Directory Users and Computers window appears.
2	In the left pane, right-click the container where you have the Group Policy. The pop-up menu appears.
3	Click Properties . The Properties dialog box appears.
4	Click the Group Policy tab.
5	Click to highlight the GPO you want to apply.
6	Click Properties to open the Properties page for the GPO.
7	Click the Security tab.



Step	Action
8	To apply Group Policy to a group or user or computer: - Select the group or user or computer you have to apply Group Policy. - Check Allow for the 'Apply Group Policy' ACL. - Click Apply and then click OK. Access Control List (ACL) is a list of permissions for an object. The list specifies who or what is permitted to access the object and what operations are permitted on that object.
	TIP If required user/group/computer is not available in the list, click Add to add the object to the list.
9	To exclude a Group Policy to a group or user or computer: - Select the group or user or computer. - Clear Allow for the 'Apply Group Policy' ACL. - Click Apply and then click OK.

To edit a GPO created by Honeywell Domain Controller security package

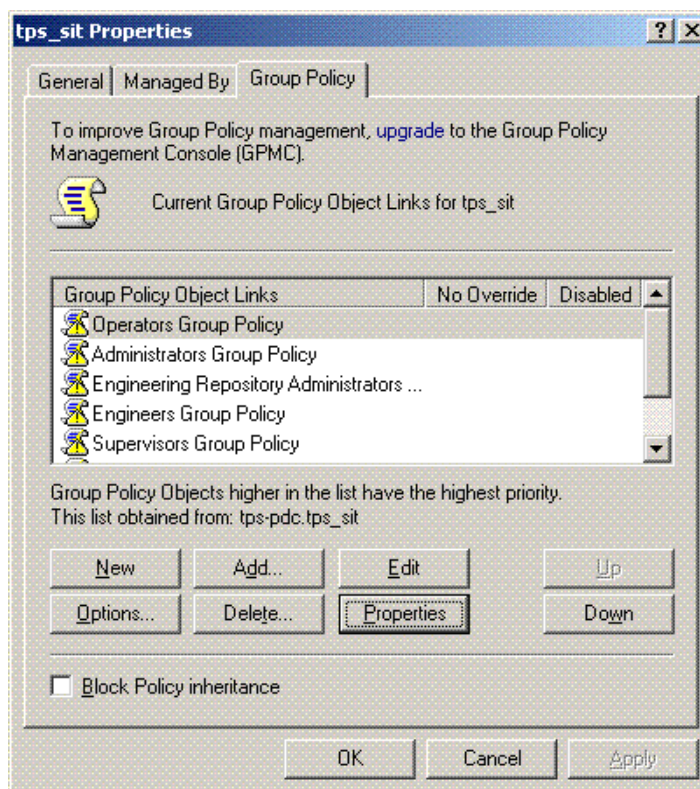
Follow the steps to edit a GPO created by the Honeywell High Security Domain Controller package:

Step	Action
1	Click Start > Programs > Administrative Tools , and then click Active Directory Users and Computers . The Active Directory Users and Computers window appears.
2	In the left pane, right-click the domain name and select Properties from the pop-up menu.
3	Click the Group Policy tab.
4	Click the GPO you want to view/edit.

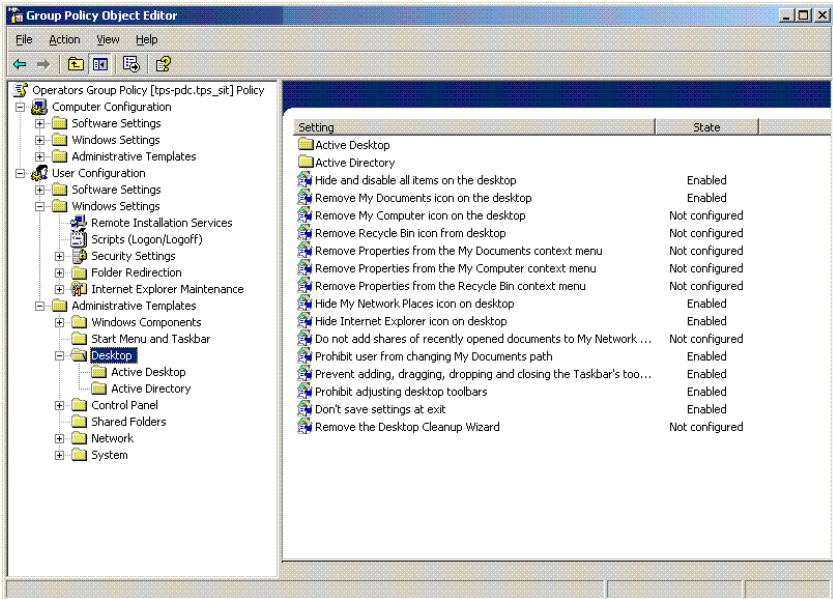
3. Domain controller procedures

3.13. Implementing Group Policy

Step	Action
------	--------



5	Click Edit . The GPO Editor appears.
---	---

Step	Action
	

6 View/Change the configurations as required.



TIP

The TPS related configuration is applicable only for User Configuration.

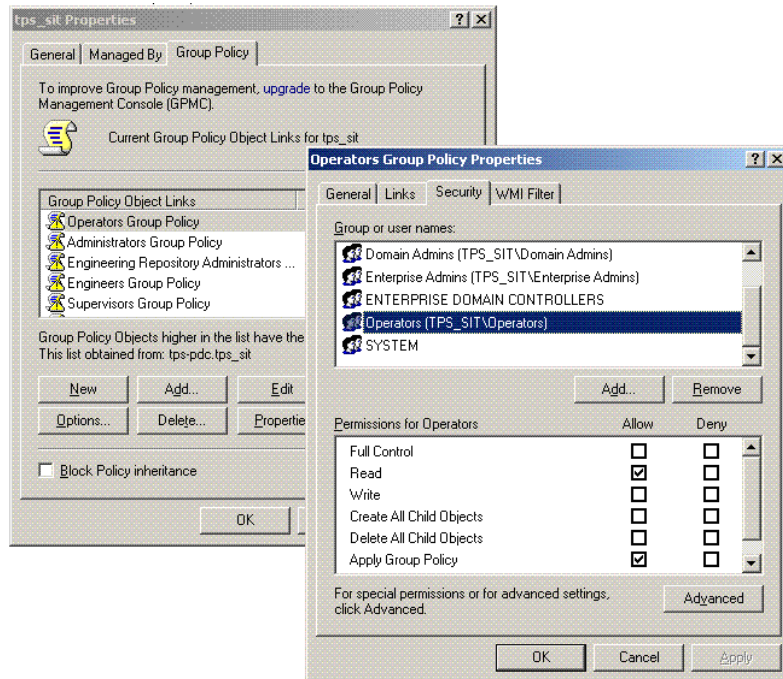
To edit a GPO-Group association

Step	Action
1	Click Start > Programs > Administrative Tools , and then click Active Directory Users and Computers . The Active Directory Users and Computers window appears.
2	In the left pane, right-click the domain name and select Properties from the pop-up menu.
3	Click the Group Policy tab.
4	Select a GPO.
5	Click Properties . The Properties dialog box appears.

3. Domain controller procedures

3.13. Implementing Group Policy

- | Step | Action |
|------|---|
| 6 | Select the Security tab in the GPO's Properties dialog box. |
| 7 | All Groups that have Apply Group Policy checked are associated with the GPO. |

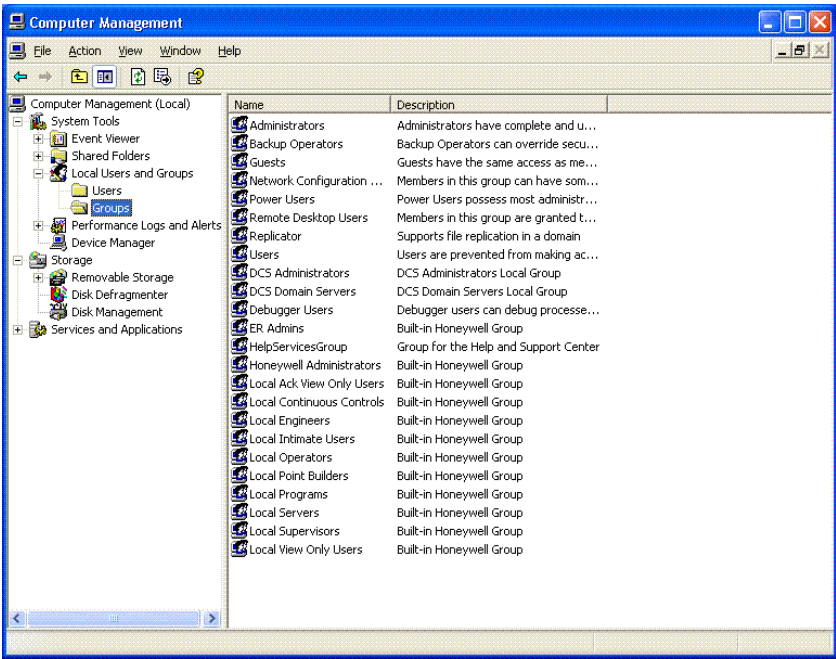


- | | |
|---|--|
| 8 | Select or clear the Apply Group Policy option for each users and group as required. |
| 9 | Click Apply and then click OK . The modified changes are applied. |

To view user accounts and groups created by Workstation security package

After you implement your TPS nodes, you can use the following steps to view the user accounts and groups created by the Honeywell High Security Workstation package.

Step	Action
1	Right-click My Computer icon from the Desktop. or Choose Start > My Computer , right-click My Computer . The pop-up menu appears.
2	Click Manage from the pop-up menu. Computer Management console appears.
3	Select 'Users' or 'Groups'. The accounts created by Workstation Security have the description 'Built-in Honeywell Group'.



3. Domain controller procedures

3.14. Implementing a Peer Domain Controller

3.14 Implementing a Peer Domain Controller

On Windows 2000, an additional Domain Controller can be created in an existing domain by replicating the entire directory database to a new Domain Controller. With low network bandwidth or a large directory database, this replication can take hours or days to complete.

With servers running Windows Server 2003, you can create an additional Domain Controller using a restored backup taken from a Domain Controller running Windows Server 2003. This backup can be stored on any backup media (tape, CD, or DVD) or a shared resource.

To implement a Peer Domain Controller

Step	Action
1	Perform the same steps that are required to implement any Domain Controller, except perform the following step during Active Directory installation: On the Domain Controller Type dialog box, click Additional Domain Controller for an existing domain.
2	On the Copying Domain Information dialog box, perform one of the following: - Click Over the network. Copies the database from the network, - Click From these restored backup files, and type the location of the restored backup files, or - Click Browse to locate the restored files, and then click Next. The files are restored.
3	Click Finish .



TIP

If you use the '**From these restored backup files**' option, specify the backup file for the Domain Controller (for which you are implementing a peer).

To generate a backup file, perform the following steps:

- 1 Click **Start > Programs > Accessories > System Tools > Backup**.
The Backup or Restore Wizard dialog box appears.
- 2 Click Advanced Mode.
The Backup Utility Window appears.
- 3 Click the Backup tab and then select System State from the left pane.
- 4 Click Browse to specify the path for the backup.
- 5 Click Start Backup.
RESULT: The backup files are saved in the specified path.

3. Domain controller procedures

3.14. Implementing a Peer Domain Controller

4. TPS node procedures

4.1 Using the TPS node procedures

The procedures in this section are to be used in conjunction with a TPS node checklist, and in most cases, you should have arrived in this section from one of those checklists. Those checklists guide you to procedures in this section, depending on the node type being implemented.

For instructions on using the checklists, refer to section 2, "TPS Implementation Checklists".



ATTENTION

This procedure is to be used in conjunction with one of the *TPS Node Implementation Checklists*.

4.2 Configuring TCP/IP settings



ATTENTION:

- This document does not include instructions for configuring TCP/IP settings of a Fault Tolerant Ethernet (FTE) node. For detailed instructions, refer the *FTE Planning, Installation, and Service Guide*.
 - This procedure is to be used in conjunction with one of the *TPS Node Implementation Checklists*.
-

Step	Action
1	From the desktop, right-click the My Network Places icon and select Properties .
2	Right-click the Local Area Connection and select Properties .
3	Double-click the Internet Protocol (TCP/IP) item.
4	Select the Use the following IP address option.
5	Enter the following from your <i>TPS Node Configuration Data Sheet</i> : IP Address, Subnet Mask, Default Gateway address
6	Enter the DNS Server addresses .

4. TPS node procedures

4.2. Configuring TCP/IP settings

Step	Action
	ATTENTION DNS configuration is required. Windows 2000/2003 requires domain members to use DNS servers supporting dynamic DNS. The addresses should be the same as the primary and backup Dynamic DNS Servers configured on the Domain Controller for this node's domain. Incorrect DNS configuration prevents proper communication with the Active Directory.
7	Click Advanced .
8	Select the WINS tab and enter the WINS servers, if applicable.
9	Make sure that the Enable LMHOSTS Lookup box is selected.
10	Click OK to close the Advanced TCP/IP Settings dialog box.
11	If you get a message "This connection has an empty primary WINS address. Do you want to continue?", Click Yes .
12	Click OK to close the Internet Protocol (TCP/IP) Properties dialog box.
13	Click OK to close the Local Area Connection Properties dialog box.
14	Close the Network and Dial-Up Connections window.
15	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

4.3 Configuring local user accounts for TPS



ATTENTION

This procedure is to be used in conjunction with one of the *TPS Node Implementation Checklists*.

Step	Action
1	Log on to the node as administrator.
2	Open the Computer Management MMC by right-clicking My Computer and selecting Manage .
3	Expand Local Users and Groups .
4	Click the Users folder to display its contents in the Details pane.
5	Double-click the TPSLocalServer account.
6	Check the following boxes: • User Cannot Change Password • Password Never Expires
7	Uncheck the following box: • Account is Disabled
8	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

For more information on creating a logon script file, refer to section 3.8, "Edit Operator.bat File".

4. TPS node procedures

4.4. Using AddBoard utility to add LCNP board to node

4.4 Using AddBoard utility to add LCNP board to node



ATTENTION

- This procedure is to be used in conjunction with one of the *TPS Node Implementation Checklists*.
 - Do not open the TPS Configuration Utility while the “AddBoard” utility is running. Similarly, running multiple instances of the Configuration Utility can cause inconsistent views of the information and other problems.
-

To do the following procedure, you need the LCNP board name of the node you are configuring. Refer to your *TPS Node Configuration Data Sheet*.

Step	Action
1	Log on as Administrator .
2	Exit all programs.
3	Select the following to display the “Add/Remove LCNP Hardware” dialog box: Start > Programs > Honeywell TPS > AddBoard.



- | | |
|---|---|
| 4 | If a configured board is present (for example, 0-Board 0 , as shown in the previous example), select it and then click Remove . Click Yes to confirm the deletion. |
|---|---|
-

Step	Action
5	Click Add to display the “Enter New board Information” dialog box.
6	Refer to your <i>TPS Node Configuration Data Sheet</i> and enter the pre-determined site-specific Board Name .
7	Select the appropriate option button in the “LCN Personality Supported” area, depending on your node type: • Select the GUS option button for TPS GUS, Standalone GUS, or Network GUS nodes. • Select the APP option button for APP nodes. • Select the Universal Station option button for US Replacement nodes. • Select the Application Module option button for AM Replacement nodes.

GUS Node
example:

The screenshot shows a dialog box titled "Enter New Board Information". It contains two input fields: "Board Number" with the value "0" and "Board Name" with the value "GUS14". Below these fields is a section titled "LCN Personality Supported" containing four radio buttons: "GUS" (selected), "Universal Station", "APP", and "Application Module". At the bottom are "OK" and "Cancel" buttons.

APP Node
example:

The screenshot shows a dialog box titled "Enter New Board Information". It contains two input fields: "Board Number" with the value "0" and "Board Name" with the value "APP14". Below these fields is a section titled "LCN Personality Supported" containing four radio buttons: "GUS", "Universal Station", "APP" (selected), and "Application Module". At the bottom are "OK" and "Cancel" buttons.

4. TPS node procedures

4.5. Installing TPN personality software on local drive

Step	Action
8	Click OK .
9	Click Yes to verify the board addition.
10	Click Close in the "Add/Remove LCNP Hardware" dialog box.
11	Shut down the node.
12	Switch power off the node, and then switch it on.
13	Log on to the machine as Administrator.
14	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

4.5 Installing TPN personality software on local drive



ATTENTION

This procedure is to be used in conjunction with one of the *TPS Node Implementation Checklists*.

Step	Action
1	To install the GUS Personality software on a GUS node, insert the GUS-TPN Software CD . To install the APP Personality software on an APP node, insert the TPN Application Module CD .



ATTENTION

Be sure that you use the proper CD for your TPN (LCN) software release.

2	If you are installing GUS personality software, use Windows Explorer to navigate to the following directory on the CD: \\GUS\\PERS_US\\DISK1 If you are installing APP personality software, use Windows Explorer to navigate to the following directory on the CD: \\APP\\PERS_AM\\DISK1
3	Double-click the SETUP.EXE file.

Step	Action
4	Read the "Welcome" dialog information.
5	Click Next .
6	Read the software license agreement.
7	Click Yes to accept the terms.
8	Type a Name .
9	Type your Company name.
10	Type 1 as the Serial number.
11	Click Next .
12	Type C: for the drive letter.
13	Click Next .
14	Select the Typical option button.
15	Click Next .
16	Click Next again to begin installation.
17	Click Finish .
18	Close the Windows Explorer.
19	Remove the CD.
20	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

4. TPS node procedures

4.6. Configuring LCNP (Board0) settings

4.6 Configuring LCNP (Board0) settings



ATTENTION

- This procedure is to be used in conjunction with one of the *TPS Node Implementation Checklists*.
 - Do not open the TPS Configuration Utility while the "AddBoard" utility is running. Similarly, running multiple instances of the Configuration Utility can cause inconsistent views of the information and other problems.
-

Edit LCNP settings

Step	Action
1	Display the Configuration Utility: Start > Programs > Honeywell TPS > Configuration Utility
2	Select the following to display the LCNP page: Configure > [Board 0 or your customized LCNP board name]
3	Refer to your <i>TPS Node Configuration Data Sheet</i> and enter the pre-determined LCN Address .
4	Refer to your <i>TPS Node Configuration Data Sheet</i> and enter the pre-determined NG Name , if any. If no NG name, enter a space.
5	Verify that the LCN Name (also known as the Board Name) is correct. Verify that the appropriate node Personality type is displayed.
6	If the appropriate node personality type is not displayed, you must remove and re-add the LCNP board configuration using the AddBoard utility. For detailed instructions to remove and re-add the LCNP board configuration, see Section 4.4, "Use AddBoard Utility to Add LCNP Board to Node".

Step	Action
7	The following example shows an LCNP board name of "Board 0" (the default name) and a Personality type of "GUS".



TIP

Click [Help](#) for definitions of each of the screen entries.

- | | |
|---|--|
| 8 | If you want to enable remote options such as Remote Native Window and Remote Displays, or to allow the LCNP to be reset from a remote node, select the LCNP Reset Controls Remote checkbox. |
|---|--|



REFERENCE – INTERNAL

For detailed information on Board0 screen entries, refer to the *TPS Configuration Utility User's Guide*.

- | | |
|---|---|
| 9 | If you want the TPN Personality to load automatically whenever the node transitions to the PWR_ON state, select Enable LCNP Auto Restart . |
|---|---|

4. TPS node procedures

4.6. Configuring LCNP (Board0) settings

Step	Action
10	Click OK .
11	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

Edit Native Window settings



ATTENTION

This procedure is to be used in conjunction with one of the *TPS Node Implementation Checklists*.

Step	Action
1	Select the Native Window tab to display the following:

Step	Action
2	To set the options, use the settings in Table 4 that follows.
	TIP Click Help for definitions of each of the screen entries.
	REFERENCE – INTERNAL For detailed information on Board0 screen entries, refer to the TPS Configuration Utility User's Guide .
3	After setting the options, click Apply .
4	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

Table 4 Native Window Page Settings

Option	Option setting
Enable Local Connections	It is recommended that this box be checked.
Enable Remote Connections	Check to enable a Native Window connection to the local LCNP board by remote users.
Enable Local Connection Override	It is recommended that this box be checked.
Allow Disconnect	Check to allow any user account to disconnect a local or remote Native Window connection.
Confirm Disconnects	It is recommended that this box be checked.
Clear Display on Disconnect	It is recommended that this box be checked.
Enable LCNP Auto Restart	Check to display the LCNP Auto Restart menu item in the Access menu of the Native Window.
Local LCNP Reset	Check to allow a reset only from the local Native Window or Emulators Status display.
Remote LCNP Reset	Check to enable remote options such as Remote Native Window and Remote Displays, or to allow the LCNP to be reset from a remote node.
Maintain 1:1 Aspect Ratio	It is recommended that this box be checked.

4. TPS node procedures

4.6. Configuring LCNP (Board0) settings

Option	Option setting
Font Smoothing	It is recommended that this box be checked.
Full Screen	It is recommended that this box be unchecked.
Show Menu/Title Bars	It is recommended that this box be checked.
Show Status Bar	It is recommended that this box be checked.
Mouse Button Function Selection Options	It is recommended to use the default settings. Click Help for function information.
Disable Print Trend	Check to disable the Print Trend functionality from GUS Native Window.

Edit Keyboard settings



ATTENTION

This procedure is to be used in conjunction with one of the *TPS Node Implementation Checklists*.

Step

Action

- 1 Select the **Keyboard** tab to display the following:

Board 0 : TPSDE23

Printer | Emulated Disks | Data Access Configuration
LCNP | Common | Native Window | TPN Time Sync | **Keyboard**

Description
This page configures the Emulated Keyboard behavior.

Labels for User Defined Keys

1	&AM STATUS	2	&COM NTWK STATUS
3	&PRC NTWK STATUS	4	&ORG SUMMARY
5	PERF&MENU	6	CLEAR &SCREEN

Annunciator Attributes

Contact #	Pitch	Duration	Repeat Rate	Enabled	
3	High	Short	Fast	☒	Test 3
2	Medium	Medium	Medium	☒	Test 2
1	Low	Long	Slow	☒	Test 1

Settings...

OK Cancel Apply Help

4. TPS node procedures

4.6. Configuring LCNP (Board0) settings

Step	Action
2	If you want the PC's speakers to announce alarms (thus enabling audible alarming to remote Native Window users), then do the following steps. a) Check the Enabled boxes. b) Click Settings. c) Enter the desired Pitch, Duration, and Repeat Rate parameters. d) Click OK to close the Settings dialog box. e) Click Test to test your settings. Or, If you do not want the PC's speakers to announce alarms, then clear the Enabled checkboxes to disable the Annunciator Attributes.
3	Click Apply .
	TIP Click Help for definitions of each of the screen entries.
	REFERENCE – INTERNAL For detailed information on Board0 screen entries, refer to the TPS Configuration Utility User's Guide.
4	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

Configure TPN (LCN) printer settings

Definition of printer configuration choices

The three printer configuration choices are described below:

- If the node does not connect to a printer, select the **No Printer Selected** option button. Select this mode if the printer function of the Universal Station is not to be used. If this mode is selected and the TPN NCF has a printer configured for the node, then the Console Status display indicates that the printer for this node is off-line.
- If the node connects to a shared printer, select the **Batch Printer Selection** option button. This mode has been available on GUS stations in past releases. This mode allows selection of local or remote printers, and allows the selected printer to be shared with other applications and/or other nodes. Printed output from the US is collected until the configured time or line limit is reached, and printed as a "batch". In this mode, the true status of the printer is hidden by the Windows operating system, and most often shows as OK on the Native Window Console Status display, even when the printer is off-line or absent.
- If the node connects to a dedicated printer, select the **Continuous Printer Selection** option button. This mode most closely emulates the printing behavior of the classic Universal Station. The printer must be connected to a serial port or parallel port of this node and it cannot be used across the network. With TPS R360, the printer can be connected to an USB port. The current offering is an Epson printer that can be connected to either the parallel port or the USB port.

When used in this mode, the printer is dedicated to this function, and cannot be shared with any other application on this or any other node. In this mode, print output from the Universal Station is sent immediately to the printer. This is particularly useful for printing alarms. To configure a dedicated printer to a serial /parallel/USB printer, follow the procedure.

Procedure to configure TPN printer

To begin: Select the **Printer** tab and perform one of the following procedures, depending upon the printer requirements for the node:

- Node does not connect to a printer (**No Printer Selected**)
- Node connects to a shared printer (**Batch Printer Selection**)
- Node connects to a dedicated printer (**Continuous Printer Selection**)

4. TPS node procedures

4.6. Configuring LCNP (Board0) settings

Node does not connect to a printer (No Printer Selected)

- | Step | Action |
|------|---|
| 1 | Select the No Printer Selected option button and the following displays: |

Board 0 : TPSDE23

LCNP | Common | Native Window | TPN Time Sync | Keyboard

Printer | Emulated Disks | Data Access Configuration

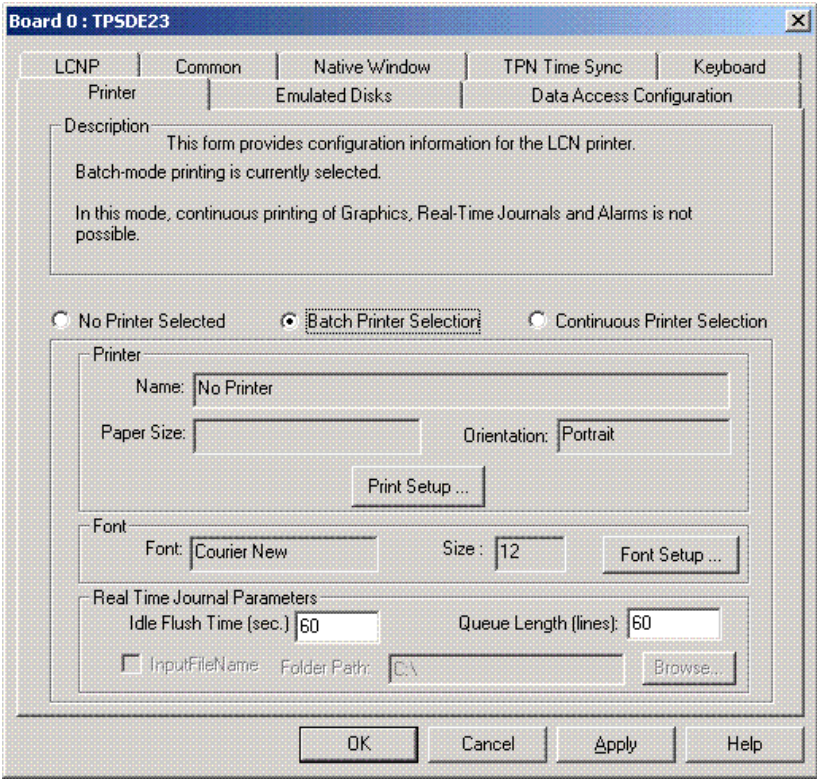
Description
This form provides configuration information for the LCN printer.
No printer is currently selected.
Select Batch Print or Continuous Print below to configure an LCN printer for this node.

☒ No Printer Selected ☐ Batch Printer Selection ☐ Continuous Printer Selection

OK Cancel Apply Help

- | | |
|---|---|
| 2 | Click OK to complete LCNP board configuration. |
| 3 | If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist. |
-

Node connects to a shared printer (Batch Printer Selection)

Step	Action
1	Select the Batch Printer Selection option button to display the following: 
2	Complete the Windows XP printer setup, and then execute the next step. For detailed instructions to configure a Windows XP printer (only if the local printer is not the alarm printer), see Section 6.2, "Local or Network Printer Configuration".
3	Click Print Setup to display the standard Windows XP Print Setup dialog box.
4	Select the desired options and then click OK.
5	Click Font Setup to change the font used for printing.

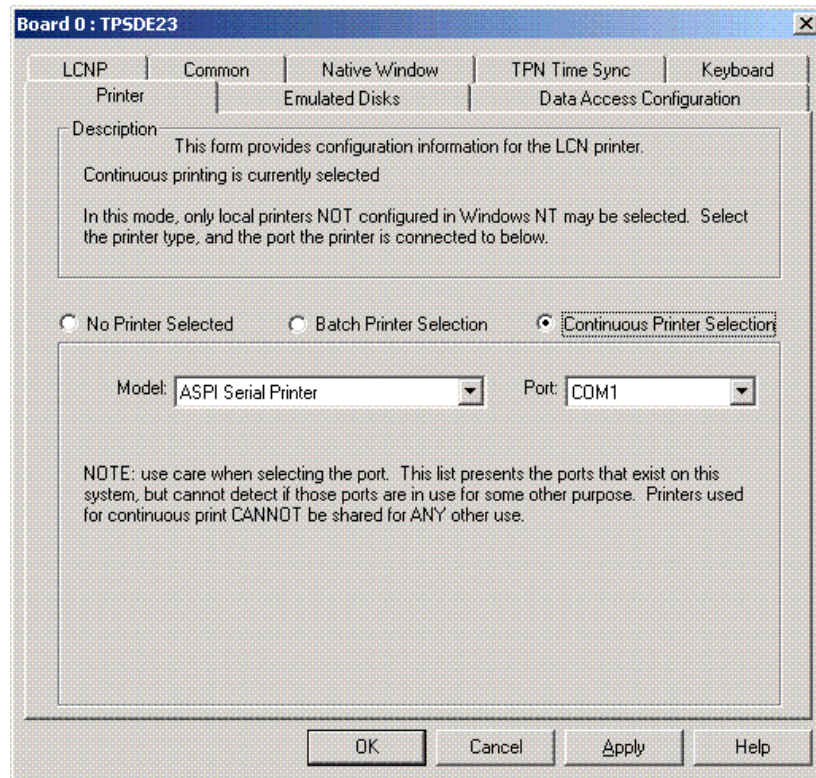
4. TPS node procedures

4.6. Configuring LCNP (Board0) settings

Step	Action
6	Change the Idle Flush Time field to the desired number of seconds.
7	Change the Queue Length field to the desired number of lines.
	TIP Idle Flush Time is the maximum period of time in seconds that RTJ entries are stored before they are automatically flushed from the RTJ print queue and sent to the printer. The Idle Flush Time setting controls printing only when the Idle Flush Time maximum is reached before the Queue Length maximum. Queue Length is the maximum number of lines allowed in the RTJ print queue before the lines are automatically sent to the printer. The Queue Length setting controls printing only when the Queue Length maximum is reached before the Idle Flush Time maximum. Idle Flush Time and Queue Length If RTJ events occur at infrequent intervals, the Idle Flush Time parameter can be set to over a year (32 million seconds). This forces the Queue Length parameter to activate printing when the desired number of lines is reached, instead of when the idle time runs out. If the Idle Flush Time and Queue Length are both set to 60, then after every 60 seconds or every 60 lines (whichever comes first) the LCNP board sends the RTJ print queue to the printer.
8	Click OK to complete LCNP board configuration.
9	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

Node connects to a dedicated printer (Continuous Printer Selection)

Step	Action
1	Select the Continuous Printer Selection option button. The following two options are available in the Model field: - ASPI Serial Printer - Generic Parallel Printer
2	Select ASPI Serial Printer as the Model, to display the following:



TIP

If your system has additional printer filters installed, select the correct printer type in the Model field. If some print filters display a Protocol field; set that selection to match the configuration of the printer.

4. TPS node procedures

4.6. Configuring LCNP (Board0) settings

Step	Action
------	--------

3	Select the correct Port for the printer.
---	---



TIP

The printer must not be configured as a shared Windows printer. If the printer is currently configured as a shared printer, then the printer's port is not selectable. Use the Printer Control Panel to deconfigure the printer, so it can be used as a continuous mode printer.

4	Click OK to complete LCNP board configuration OR Select the Generic Parallel Printer as the Model to display the following:
---	--

Step	Action
5	Select the correct Port for the printer.
6	Click OK to complete LCNP board configuration.

	ATTENTION Signum 2043 printer On the Signum 2043 printer itself, set the parameter "LENGTH" to be one line for every path (manual, front push, rear push), allowing the printer to print a real-time alarm line, followed by a one line "page feed".
---	--

4. TPS node procedures

4.7. Connecting the transceiver and TPN cables



ATTENTION

This procedure is to be used in conjunction with one of the *TPS Node Implementation Checklists*.

Step	Action
1	Select the following: Start > Shut Down . Select Shutdown from the pull down menu.
2	Click OK to verify the shutdown.
3	If necessary, turn off the power to the node.
4	Connect the AUI cable from the LCNP board to the Transceiver (MAU).
5	Connect the TPN (LCN) cable to the Transceiver .
6	Power up the node.
7	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

4.8 Configuring Signon Manager

Signon Manager functional overview

Signon Manager maintains the current keylevel for a TPS node. As applications that require sign-on start (such as GUS pictures and Native Window), they register with Signon Manager, which notifies them of the current keylevel and any subsequent keylevel changes. Several system elements can initiate a keylevel change: the physical keyswitch, the software keyswitch, and scripting operations within GUS pictures. All of these elements notify Signon Manager when the keylevel changes. Signon Manager then notifies sign-on-aware applications that the change occurred; letting each application implement its security policy as is needed.

Signon Manager must be running on all R220 and later GUS nodes, plus any nodes running Remote Native Window and/or Remote Displays.



REFERENCE – INTERNAL

For additional information on TPS security, refer to the subsection “Users and Groups” in the *TPS System Administration Guide*.

Perform the following procedures to configure Signon Manager. For more information, refer to "Suggested keyswitch configuration settings" and "Suggested keyswitch access permissions".

With TPS R420/TPN R682 and later, the Native Window and TPN access levels are synchronized with the current Signon Manager login and/or Windows login. For the Native Window, the access level is controlled by the Signon Manager and/or Windows login, irrespective of the CURRENT DEFAULT value in the ACCESS CHG target of the CONSOLE STATUS.

- The access level in the Native Window and the TPN changes to View-Only when the View-Only user logs in using Signon Manager and/or Windows login.
- The access level in the Native Window and the TPN changes to Operator when the Operator logs in using Signon Manager and/or Windows login.

There is a consistency in behavior and visual representation of access levels between the Native Window and the TPN.

Establish keylevel access permissions

To enable a user or group of users to access the TPN (LCN) through a specific node's

4. TPS node procedures

4.8. Configuring Signon Manager

LCNP, the user or group must be in a TPS local group. The best practice is to add your global users to the global groups at the Domain Controller. For example, add JoeEngineer to the global Engineers group. The global Engineers group is a member of the Local Engineer's group and has Engineer keylevel access by default.

Default keylevel access permissions		
<i>This Default Keylevel...</i>	<i>Is granted to Members of this TPS Local Group...</i>	<i>This Global Group is a member of the TPS Local Group</i>
Engineer	Local Engineers	Domain\Engineers*
Supervisor	Local Supervisors	Domain\Supervisors*
Operator	Local Operators	Domain\Operators*
View Only	Local View Only	Domain\View Only*
* "Domain" represents the name of the user domain.		



ATTENTION

In TPS R400 and greater, the local group "Honeywell Administrators" is the replacement for the original "TPS Local Administrators" local group. However, "Honeywell Administrators" does not have Engineer privilege.

Configure Signon Manager in configuration utility

After performing the procedure "Establish Keylevel Access Permissions", perform the following steps:

Step	Action
1	Log on to the node as Administrator.
2	Display the Configuration Utility: Start > Programs > Honeywell TPS > Configuration Utility
3	Select the following in the Configuration Utility: Configure > Signon Manager
4	Check the desired settings for the Signon Manager as described in the following table Signon Manager Settings In Configuration Utility .
5	Click OK , and do not close the Configuration Utility.

Step	Action
6	Select the following in the Configuration Utility: Configure > Devices/Services
7	Check the TPS Signon Manager Service box.
8	Click OK .
9	Exit the Configuration Utility.
10	If you made changes, reboot the node. RESULT: The Signon Manager is now ready for use.
11	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

4. TPS node procedures

4.8. Configuring Signon Manager

Signon Manager settings in configuration utility

The **Signon Manager** page includes the **Options**, **KeyLevel**, and **Timeouts** tabs as described in the following table.

Signon Manager settings	
Options tab	
Signon Display Colors (Default = Customize Colors with blue text on cyan (light blue-green) background)	This option allows the user to make the Signon Bar either more obvious or less obvious, depending on site preference. To change the colors of the window, select: • Dialog Colors: the Signon bar uses system appearance settings to set text and background colors consistent with dialog boxes. • Taskbar Colors: the Signon Bar adopts the coloring of the Windows taskbar, as controlled by the system appearance settings. • Customize Colors: Click the Text and/or Background buttons to use standard Windows color selection dialog boxes to select specific colors. The preview window below the option buttons shows the selected color combination.
Auto-Hide Signon bar (Default = on)	By default, this check box is selected, causing the Signon bar to remain hidden until you move the cursor to the edge of the display occupied by the Signon bar. Note that high priority windows can cover the small area of the bar that remains visible when the bar is auto-hidden, preventing the bar from becoming visible when the mouse moves to that display edge.
Support Terminal Server sessions (Default = on)	This option is disabled unless the computer supports Terminal Server. When the option is on, all sessions logged in to terminal server on the computer have their own Signon bar, and all Signon-aware applications running in the session use the signed on user for the session. When the option is off, only session 0 (the session attached to the display of the computer) has a Signon bar, which controls the security settings for all Signon-aware applications in that session. Signon-aware applications for all other sessions operate with security appropriate to the user logged in to that session.

Signon Manager settings	
Keylevel tab	
Trust keylevel changes from IKB (Default= on)	When this option is on, LCN keylevel changes originating from a physical keyswitch are used even if the currently signed on user is not authorized at this level. For sites not using the physical keyswitch, this option is not used. For sites that do use the physical keyswitch, this option allows a quick keylevel change without the delay of entering a user id and password.
Treat physical key "OPR" position as ViewOnly (Default = off)	This checkbox determines whether GUS displays respond to a change in the IKB key to "OPR" by switching to OPR or VIEW. For the Native Window, this behavior is controlled by the NCF. With TPS R420/TPN R682 and later, this option determines the access levels (OPR and VIEW) applied to the Native Window based on the change in the IKB key.
Automatically raise keylevel to maximum allowed for Signon user (Default = off)	When this option is on, TPS applications that are Signon Manager-aware (Native Window and GUS displays) automatically raise the LCN keylevel for a new Signon to the highest keylevel allowed for the new user. When this option is <u>not</u> set, TPS applications do not automatically raise the keylevel for a new Signon, but the TPS applications always lower the key setting if the new user is not authorized at the current level.
Default Workstation Keylock (Default = View Only)	This option allows a site to decide the default keylevel for a workstation. This default keylevel is set whenever a Windows login or logout occurs. The default provides the initial key setting for a workstation at the time a user logs onto the machine. If the "automatically raise" option is enabled, this setting is of little concern.
Timeouts Tab	
Sign users off if there is no activity for: n minutes (Default= off)	When this option is on (box is checked), the users are signed off if the period of inactivity exceeds the configured limit.
Sign override users off after: n minutes (Default= off)	When this option is on (box is checked), the override users are signed off if the period of inactivity exceeds the configured limit. Select the option 'User may select the timeout of up to: n minutes' to specify a limit for user selectable timeout.

4. TPS node procedures

4.8. Configuring Signon Manager

Signon Manager settings	
Timeout warning	The following configurable options are available for timeout warnings: a) Provide a timeout warning n seconds before signing users off – Select this option to warn users before the configured number of seconds before actually signing off a timed out user. b) Show user interface if hidden – Select this option to display the user interface (signon bar) if it is hidden at the time of warning. c) Play sound – Select this option to configure a sound file that needs to be played during a timeout warning.

Suggested keyswitch configuration settings



ATTENTION

Indications of the keylock access level may be different between the Native Window, GUS pictures, and the physical keyswitch. Users should use the indication in the status bar of the GUS or Native Window to determine the access currently in effect for that particular window.

The following tables describe how to implement different options for initial keyswitch configuration.

OPTION 1

To “Disable” logon-based keyswitch security and use the physical keyswitch:

Step	Action
1	Check “Auto-hide”.
2	Uncheck “Automatically raise keylevel to maximum allowed for SignOn user”.
3	Check “Trust keylevel changes from IKB”.
4	Select “View Only” as the Default Workstation Keylock.

OPTION 2

To “disable” logon-based keyswitch security using the software keyswitch and disable the physical keyswitch:

Step	Action
1	Perform steps in Option 1 above, plus, do the following steps.
2	On each GUS and GUS Remote Displays Server node, add “Everyone” access to the “Engineer” security file. (In a standard installation, this file is located at C:\Program Files\Honeywell\TPS\Emulators\Security\TDCBoard0\Engineer.)
3	On GUS Remote Displays Client stations, select the appropriate Default Workstation Keylock level or check “Automatically raise keylevel to maximum allowed for SignOn user”. (This option allows any valid Windows user to select any level of LCN access.)

4. TPS node procedures

4.8. Configuring Signon Manager

OPTION 3

To use keyswitch security based on current Windows logon only (use security, but not the ability to change sign-on through Signon Manager window):

Step	Action
1	Check "Auto-hide".
2	Check or uncheck "Automatically raise keylevel to maximum allowed for SignOn user" as desired. Recommended checked on GUS Remote Displays Client stations.
3	Check or uncheck "Trust keylevel changes from IKB" as desired.
4	Select desired level as the Default Workstation Keylock (not applicable if "Automatically raise keylevel to maximum allowed for SignOn user" is checked).
5	Refer to the following subsection <i>Setting Keyswitch Access Permissions</i> .

OPTION 4

To use keyswitch security based on current Signon Manager logon (can change access without Windows logout/logon):

Step	Action
1	The preferred setting is to check "Auto-hide".
2	Perform steps 2-5 as in Option 3 above.

Suggested keyswitch access permissions

The following tables describe ways to assign various levels of TPS keyswitch access based on Windows IDs. Use these security configurations in conjunction with options 3 or 4 as described in the previous section, "Suggested keyswitch configuration settings".

OPTION 1 - Keyswitch Access Permissions

To use keyswitch security based on Windows user ids within a single domain or between trusted domains (minimally, the domain of the server trusts the domain of the client):

Step	Action
1	Add any user who should have Engineer keylevel access to the global group "Engineers".
2	Add any user who should have Supervisor keylevel access to the global group "Supervisors".
3	Add any user who should have Operator- keylevel access to the global group "Operators".
4	Add any user who should have View Only- keylevel access to the global group "View Only".



TIP

Membership in each group also provides access to all lower levels.

4. TPS node procedures

4.8. Configuring Signon Manager

OPTION 2 - Keyswitch Access Permissions

To use keyswitch security based on Windows user ids from a non-trusted domain:

Step	Action
1	Create local accounts on each GUS for every user that accesses that GUS remotely. This account must exactly match the remote account in both name and password.
2	Add any user who should have Engineer keylevel access to the local group "Local Engineers" on each GUS node.
3	Add any user who should have Supervisor keylevel access to the local group "Local Supervisors" on each GUS node.
4	Add any user who should have Operator keylevel access to the local group "Local Operators" on each GUS node.
5	Add any user who should have View Only keylevel access to the local group "Local View Only" on each GUS node.

OPTION 3 - Keyswitch Access Permissions

To use keyswitch security based on Windows user ids on a stand-alone GUS:

Step	Action
1	Create local accounts on each GUS for every user that use that GUS.
2	Add any user who should have Engineer keylevel access to the local group "Local Engineers" on each GUS node.
3	Add any user who should have Supervisor keylevel access to the local group "Local Supervisors" on each GUS node.
4	Add any user who should have Operator keylevel access to the local group "Local Operators" on each GUS node.
5	Add any user who should have View Only keylevel access to the local group "Local View Only" on each GUS node.

Invoking the Signon Manager

After TPS software is installed, Signon Manager should start whenever a user logs onto Windows.

To activate the Signon dialog, click the Signon Bar's sign-on button, or click the Signon Manager icon in the taskbar.

If the Signon Manager icon is not present in the taskbar, make the following selections to start the Signon Manager utility: **Start > Programs > Honeywell TPS > Signon**.

The dropdown list for the user name in the Signon Manager user window is the last five names accepted by Signon Manager, so it is populated as Signon Manager signons succeed.



REFERENCE – INTERNAL

For more information on the Signon Manager user window, refer to the *TPS System Operations Guide*, TP07.

4. TPS node procedures

4.9. Distributing HOSTS, LMHOSTS and Operator.bat files

4.9 Distributing HOSTS, LMHOSTS and Operator.bat files

Prerequisite

To use Copy Utility to distribute files, the files must be copied to the
\\WINDOWS\\System32\\Repl\\Export\\Scripts directory on Domain Controller.

To use offline files to distribute files, refer to section 6.3, "Configuring Windows 2003
offline files and folders for replication".



ATTENTION

This procedure is to be used in conjunction with one of the *TPS Node
Implementation Checklists*.

From DC, Distribute HOSTS and LMHOSTS files

Step	Action
1	Open a Command Prompt window on the Domain Controller.
2	Change directories to the following directory: \\WINDOWS\\System32\\Repl\\Export\\Scripts
3	Enter the following command on the Domain Controller to distribute the HOSTS file and the LMHOSTS file to all of the TPS nodes: Copy_Utility NET
4	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

From DC, Distribute Operator.bat file

Step	Action
1	Enter the following command at the Domain Controller to distribute the operator.bat file to all specified TPS nodes: Copy_Utility BAT
2	Log off of the Domain Controller.
3	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

4.10 Installing TPS software packages



ATTENTION

- These procedures are to be used in conjunction with one of the *TPS Implementation Checklists* (see section 2.1, "Selecting the correct checklist".)
- Running multiple instances of the TPS Configuration Utility can present inconsistent views of the information and may cause problems. Similarly, the TPS Configuration Utility should not be open while the "Add/Remove Board" utility is being run.

Record current devices/services settings

Perform these steps if upgrading or migrating an existing node.

Step	Action
1	Log on to the node as Administrator.
2	Display the Configuration Utility: Start > Programs > Honeywell TPS > Configuration Utility.
3	Select Configure > Devices/Services.
4	In the following table, record the node's current checkbox settings. Use this table later to reactivate the devices and services.

Current devices and services settings		
Checkbox	Checked	Unchecked
TPS Signon Manager Service (NT 4: TPS Console Operator)		
Device Driver (tdcaint.sys)		
Services (Tdcaserver)		
Time Synchronization Service Enable Note: Leave this checkbox cleared unless you are instructed by Honeywell to do otherwise.		
LCN File Service		
GUS Remote LXS		
CarbonCopy		
TPN Communication Service		
Integrated Keyboard Driver		

4. TPS node procedures

4.10. Installing TPS software packages

Deactivate devices/services

Step	Action
1	Log on to the node as Administrator.
2	Display the Configuration Utility: Start > Programs > Honeywell TPS > Configuration Utility
3	Select Configure > Devices/Services .
4	Clear all of the checkboxes.
5	Click OK .
6	Close the Configuration Utility .
7	Restart the machine. (This restart is necessary to make sure all services are not running.)
8	Hold down the Shift key and log on to the machine as Administrator. Hold down the Shift key as you press the Enter key. Continue holding down the Shift key until the hourglass disappears and the cursor becomes the normal cursor. RESULT: The AutoRun items are locked out, preventing TPS software from starting automatically.
9	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

Installation prerequisite

These procedures are to be used in conjunction with one of the *TPS Implementation Checklists* (see section 2.1, "Selecting the correct checklist").

As described in the checklists, you must use Microsoft Add/Remove Programs to remove all TPS, GUS, and SafeView, packages before installing the new software.



ATTENTION

Certain TPS software packages require that HCI Runtime software be installed from the TPS Software CD prior to installing the TPS package. TPS software packages with this dependency are indicated with an (H) in section 4.11. The TPS software Installer informs you if HCI Runtime is required.

Procedure to Install TPS software packages

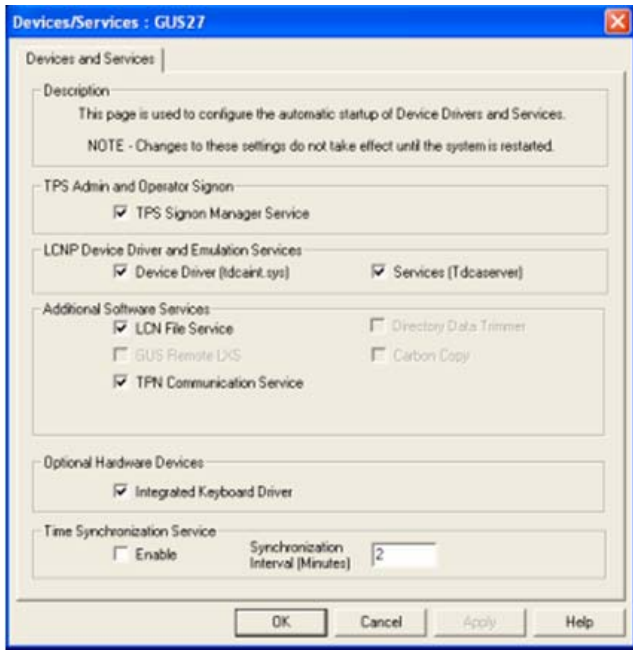
Step	Action
1	Insert the TPS Software CD .
2	Click Install Packages from the menu of items.
3	When the <i>Welcome</i> dialog displays, read the details and if you agree, click Next .
4	Click Next to accept the software license agreement terms.
5	Note the third party software information, and then click Next .
6	If this dialog has been used before, the information fills in; if not, enter the Name , Company , License No. , and Authorization No. information, and then click Next . RESULT: After the License No. and Authorization No. have been validated, the "Package Selection" dialog shows a list of available licensed packages.
7	Select every mandatory package for your node type and any optional package you need for this node that is authorized by your license. See Section 4.11, "TPS Software Package Usage by Node Type" to determine the mandatory packages for your node type.
8	Click Install Package .
9	Select the Typical (Default) radio option for Installation Type .
10	Click OK .

4. TPS node procedures

4.10. Installing TPS software packages

Step	Action
11	Click Finish .
	ATTENTION If you select one or more TPS software packages that depend on the HCI Runtime component, the following message displays: "Please ensure that the correct HCI Runtime package has been installed before performing this install." If the installer detects that HCI Runtime is not registered in the registry, the install does not continue and the following message displays: "The HCI Runtime package MUST be installed before performing this install." For TPS 400 and greater, the HCI Runtime package is on the TPS Software CD.
	ATTENTION Carbon Copy is not a licensed installable package on the R4xx TPS Software CD. Carbon Copy must be installed from a separate Honeywell Carbon Copy CD.
	ATTENTION If the installation encounters a read-only file during TPS software packages installation, a message displays asking if it is OK to overwrite the read-only file. In this case, the install is not successful and you must do the following: 1. Verify that the TPS services are disabled by logging on to the machine while holding down the Shift key. 2. Reinstall the Honeywell Repackaged Redistributables package from the TPS Software CD. 3. Reboot the machine and try again.
12	When all selected packages are installed, click Exit .
13	Click Yes to verify the exit.
14	Select Exit , and then click Yes .
15	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

Activate devices/services

Step	Action
1	Display the Configuration Utility: Start > Programs > Honeywell TPS > Configuration Utility .
2	Select Configure > Devices/Services . The following screen is displayed:
	
3	Refer to the Devices and Services Page Settings table that follows to determine which boxes to check. In addition, use the Current Devices and Services Settings table that you filled out earlier to recheck the appropriate Devices/Services page checkboxes with the following exceptions.
4	Click OK .
5	Close the Configuration Utility .
6	Restart the machine.
7	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

4. TPS node procedures

4.10. Installing TPS software packages

Devices and services page settings	
Checkbox	Checkbox setting
TPS Signon Manager Service	Select this box on all nodes unless you are instructed by Honeywell to do otherwise.
Device Driver (tdcaint.sys)	Select this box (if not shaded) on all nodes unless instructed by Honeywell to do otherwise.
Services (Tdcaserver)	Select this box (if not shaded) on all nodes unless instructed by Honeywell to do otherwise.
Time Synchronization Service Enable	Clear this box (if not shaded) unless instructed by Honeywell to do otherwise.
LCN File Service	This service is used for viewing LCN files on a GUS node. Select this box (if not shaded), unless instructed by Honeywell to do otherwise. Must be cleared (if not shaded) on APP nodes.
GUS Remote LXS	For Network GUS and TPS GUS nodes, this box appears shaded unless the "GUS Remote Displays Server" package has been installed. Select the box only if you have installed the "GUS Remote Displays Server" package. This box is shaded for the following nodes: APP and AMR nodes, Standalone GUS and USR nodes.
Carbon Copy	Select this box on all nodes containing the licensed software, unless instructed by Honeywell to do otherwise.
TPN Communication Service 	Select this box on all nodes to start the TPN Communication Service. ATTENTION: The TPN Communication Service is mandatory for monitoring HCI components from the TPN.
Integrated Keyboard Driver	Select this box on GUS and US Replacement nodes that have either of the following: • an Integrated Keyboard (IKB), or • an Operator Entry Panel (OEP) on a Honeywell Icon Series Console. This box must be cleared (if not shaded) on APP and AM Replacement nodes.

4.11 TPS Software package usage by node type

Key to node type information in the table that follows	
B	The package includes TPS Base Components (provides the TPS Configuration Utility which is accessible through Start > Programs > Honeywell TPS).
H	For TPS R130 and later (GUS R330 and later and APP R211 and later), the package requires that the HCI Runtime component be installed prior to installing the package.
M	The package is Mandatory for the node type.
N	The package is Not Allowed or Not Applicable for installation on the node type.
O	The package is Optional for the node type.

TPS Software package usage by node type					
<i>TPS software packages</i>	<i>APP</i>	<i>TPS GUS</i>	<i>Network GUS</i>	<i>Standalone GUS</i>	<i>Non-TPN (LCN) connected nodes</i>
APP Base Software (B)	M	N	N	N	N
APP Solution Package (H)	O	N	N	N	N
CL/NT Application Server (Note 12)	O	N	N	N	N
Fault Tolerant Ethernet Driver (FTE)	O	O	O	N	O
File Transfer	O	O	O	O	N
GUS Base Software	N	M	M	M	N
GUS Display Builder (Note 1)	N	O	O	O	O
GUS Display HCI Client Add-in (H)	N	M	N	N	O (Note 2)
GUS Display Runtime	N	O (Note 17)	O (Note 17)	O	N
GUS Multiple Displays	N	O	O	O	N (Note 4)
GUS PRO	O	O	O	O	O

4. TPS node procedures

4.11. TPS Software package usage by node type

TPS Software package usage by node type					
<i>TPS software packages</i>	<i>APP</i>	<i>TPS GUS</i>	<i>Network GUS</i>	<i>Standalone GUS</i>	<i>Non-TPN (LCN) connected nodes</i>
GUS Remote Displays Client (B)	N	O (Note 17)	O (Note 17)	N	O (Notes 3, 4)
GUS Remote Displays Server (Note 5)	N	O	O	N	N
GUS Remote IKB Client	N	O	O	N	O (Note 3)
HCI Client Connectivity (H)	O	O	N	N	O
HCI Client Developer Toolkit (H) (Note 6)	O	O	N	N	O
HCI Server Developer Toolkit (H) (Note 6)	O	O	N	N	O
Honeywell System Management Display (Note 7)	M	M	O	N	O
Native Window Client (B) (Note 8)	O	O	O	O	O
Redirection Manager (Note 9)	O	O	O	N	O
SafeView	N	O	O	O	O
System Management and HCI Runtime					
System Management Runtime	(Note 10)	(Note 10)	O	N	O
HCI Runtime	(Note 11)	(Note 11)	O	N	O
TPN Server (H) Note 12)	O	O (Note 13)	N	N	N
TPS Open Layer Software (B) (H) (Note 14)	N	N	N	N	O
Version Management Tool	(Note 15)				
WebGUS	N	N	N	N	O (Note 16)

4 TPS node procedures
4.11 TPS Software package usage by node type

The following notes apply to the previous table	
Note	Comments
1	GUS Offline Display Builder is used only for GUS R312 and earlier software releases. On GUS R320 and later, there is only one GUS Display Builder.
2	On a non-TPN connected node, the "GUS Display HCI Client Add-in" package requires the installation of the "GUS Remote Displays Client" package. In addition, the non-TPN connected node must be in the same domain and the same TPS Domain OU (that is, the same OU with the same multicast addressing) as the TPN-connected node.
3	"GUS Remote IKB Client" and "GUS Remote Displays Client" must not be installed on the same non-TPN connected node.
4	Starting with GUS R311, GUS Multiple Displays may be installed on the same node as Remote Displays Client.
5	Use of the remote operations features of the "Carbon Copy (Full)" package can adversely affect the performance of the "GUS Remote Server" therefore, Honeywell recommends against loading these two packages on the same node. GUS Remote Displays Server package is included with the "TPS Open Layer Software" package.
6	Toolkit packages should only be installed on nodes that are used to develop applications. They are not required to run applications.
7	The System Management Display package or the System Management Runtime package must be installed on OPC and HCI Client nodes, including those that need to use Redirection Manager.
8	The Native Window Client package is included with the "GUS Base Software" package.
9	Redirection Manager should be installed on all nodes that run an OPC or HCI client requiring redirection. Install the Honeywell System Management Display before installing Redirection Manager. Refer to Redirection Manager documentation for detailed installation instructions.

4. TPS node procedures

4.11. TPS Software package usage by node type

The following notes apply to the previous table	
Note	Comments
10	System Management Runtime is included in each of these licensed common component packages: Fault Tolerant Ethernet, Honeywell System Management Display, and Redirection Manager. For OPC or HCI server nodes (if no licensed common component package is already installed on the node), System Management Runtime must be installed to be able to communicate with Redirection Manager. For HCI client nodes (if no licensed common component package is installed), System Management Runtime is required only if the client node needs to run an HCI client application that calls the HCI runtime routine GetComponentInfo() to retrieve the hostname and ProgID of the target server.
11	These licensed common component software packages include HCI Runtime: Fault Tolerant Ethernet, Honeywell System Management Display, and Redirection Manager. Several licensed TPS Software packages require HCI Runtime. Section 4.11 designates those packages with an (H).
12	The TPN Server and CL Server packages install the actual servers themselves. If you only intend to connect to, monitor, or configure a TPN Server or CL Server that is hosted by a remote node, and then install the "HCI Client Connectivity" package.
13	TPN Server has limited throughput when installed on a TPS GUS node. The TPN Server is required on a GUS node that has TPS Builder installed. The public TPN Server is available with the purchase of the GUS TPN Server package. The public GUS TPN Server allows an application to access small amounts of OPC data from a GUS node. Primary use is for non-critical applications where the user accepts reduced GUS performance.
14	The "TPS Open Layer Software" package is usually installed on non-TPN connected nodes.
15	The Version Management Tool is installed as a part of TPS Base (B).
16	WebGUS PCs require installation of System Management Runtime, Experion PKS High Security – Workstation Package, and execution of Linkdomaingroups.vbs.
17	"GUS Display Runtime" and "GUS Remote Displays Client" must not be installed on the same node.

Documentation for configuring installable TPS software packages

The following table lists TPS software that can be installed on a system. The table specifies whether configuration needs to be performed following installation, and where you can find the configuration instructions.

Installable TPS system software packages (depends on license purchased)	User configuration required?	For configuration instructions, refer to the TPS Knowledge Builder or to the specific document/help shown below
APP Base Software		
APP Solution Package		
CL Server (APP)	Yes	<i>CL Server User's Guide (TP13)</i> This guide describes the purpose and uses for CL Server. Includes CL Server requirements as well as installation configuration and operation procedures.
Fault Tolerant Ethernet Driver	Yes	<i>FTE Planning, Installation, and Service Guide, FE05</i> <i>FTE Status Server and Auxiliary Display User's Guide, EX22</i>
File Transfer (All Nodes)	Yes	<i>File Transfer Installation and Use manual, FT05</i>
GUS Display Builder		
GUS Display HCI Client Add-in	Yes	<i>GUS HCI/OPC Data Access User's Guide</i>
GUS Display Runtime		
GUS Multiple Displays		
GUSPRO	Yes	<i>ProMenu User Guide</i> <i>ProTrend User Guide</i>
GUS Remote Displays Client	Yes	<i>GUS Remote Displays User's Guide</i>
GUS Remote Displays Server	Yes	<i>GUS Remote Displays User's Guide</i>

4. TPS node procedures

4.11. TPS Software package usage by node type

Installable TPS system software packages (depends on license purchased)	User configuration required?	For configuration instructions, refer to the TPS Knowledge Builder or to the specific document/help shown below
GUS Remote IKB Client	Yes	<i>Remote GUS User's Guide</i>
HCI Client Connectivity	Yes	<i>GUS HCI/OPC Data Access User's Guide</i>
HCI Client Developer Toolkit	Yes	<i>GUS HCI/OPC Data Access User's Guide (TP60)</i> This guide describes tools, techniques, and procedures for developing HCI client applications. Also describes the facilities available to the client application developer and the API for client applications.
HCI Server Developer Toolkit	Yes	<i>GUS HCI/OPC Data Access User's Guide (GU2B)</i> This guide details HCI/OPC client/server data access support for GUS Display Builder. Describes the syntax and definition of Named Data Access from the GUS Display Builder to OLE for Process Control (OPC) servers, including Honeywell Communications Interface (HCI) servers. It also discusses the GUS Node-Wide Display Database concept, which is based on a GUS-specific HCI/OPC server.
Native Window Client	Yes	<i>GUS Remote Displays User's Guide</i>
NWDDDB Server (installed as part of GUS Display HCI-client add-in package)	Yes	<i>GUS HCI/OPC Data Access User's Guide</i>
SafeView (GUS)	Yes	<i>SafeView User's Guide</i>
System Management Display	Yes	<i>System Management Configuration Guide, EX08</i> <i>System Management Operations Guide, EX07</i>

4 TPS node procedures

4.11 TPS Software package usage by node type

Installable TPS system software packages (depends on license purchased)	User configuration required?	For configuration instructions, refer to the TPS Knowledge Builder or to the specific document/help shown below
Redirection Manager (RDM)	Yes	<i>Redirection Manager User's Guide (EX20)</i> This guide provides an overview of Redirection Manager including uses, planning, and how it supports OPC clients and OPC servers. Includes installation and configuration procedures.
TPN Server (APP)	Yes	<i>TPN Server User's Guide (TP11X)</i> This guide describes using the TPN Server primarily from the server perspective, although it does include some information about name forms and data types from the client perspective. Contains installation, configuration and operation procedures for the TPN Server.
TPS Open Layer Software		
Version Management Tool		<i>TPS System Configuration Utility User's Guide</i>
WebGUS	Yes	<i>WebGUS Admin Help</i> (available as online Help within the WebGUS application), <i>WebGUS Planning and Implementation Guide</i> , <i>WebGUS Authoring and Runtime Guide</i>
Additional installable licensed software		
TPS Builder (TPS Builder CD)	Yes	Software Change Notice, <i>TPS Builder Installation, Implementation, and Administration Guide</i>
Carbon Copy (Carbon Copy CD)	Yes	<i>Remote GUS User's Guide</i>

4. TPS node procedures

4.11. TPS Software package usage by node type

Installable TPS system software packages (depends on license purchased)	User configuration required?	For configuration instructions, refer to the TPS Knowledge Builder or to the specific document/help shown below
Experion/TPS Backup and Restore		<i>Experion/TPS Backup and Restore User's Guide (BR20)</i> This guide describes how to use the optional Experion/TPS Backup and Recovery (EBR) tool to backup TPS and Experion nodes within a Honeywell process control system. Additionally, the guide provides hardware and software requirements, as well as suggested backup and recovery strategies.

4.12 Migrating GUS displays



ATTENTION

Before you perform this procedure, check the TPS SCN to determine if you need to migrate your GUS displays. Also check the TPS SCN for the version of Migrator appropriate for your release.

Step	Action
1	Insert the TPS Software CD , and use Windows Explorer to navigate to the MIGRATOR directory on the CD.
2	Open the folder(s) until you see a series of release folders and select the folder for the release you are migrating FROM. (Ignore the numbers following the first three digits of the release).



ATTENTION

If the required Migrator is not on the current CD, you must locate the media containing that Migrator.

3	Select the Disk1 folder, and then double-click the setup.exe file.
4	Read the Welcome dialog text, and then click Next .
5	Click Yes to accept the software license agreement terms.
6	Verify that the Name and Company information is correct, and then enter 1 as the Serial number.
7	Click Next .
8	Accept the default for all remaining prompts to complete the install
9	You may now use the node to migrate GUS displays.
10	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

4. TPS node procedures

4.13. Installing System Management display software package

4.13 Installing System Management display software package



ATTENTION

These procedures are to be used in conjunction with one of the *TPS Implementation Checklists*.

For System Management Display usage (optional, mandatory, or not allowed) on each node type, refer section 4.11, "TPS Software Package Usage by Node Type".

Step	Action
1	Insert the TPS Software CD .
2	To install the System Management Display software package, select the Install Packages menu item.
3	Review the information about installing software, license agreements, and third-party compatibility on the next several screens. Click Next to continue.
4	Enter your license and authorization numbers. Click Next to continue.
5	Select Honeywell System Management Display . (If Honeywell System Management Display does not appear in the list, verify the licensing and authorization numbers for this package.)



TIP

Installation of System Management Runtime (included in the Honeywell System Management Display software package) upgrades the node to remove legacy HCI component configuration not associated with the local node. For nodes running GUS R312 and greater or APP R210 and greater, the HCI component configuration is maintained on the host node only.

The installation of Honeywell System Management Display also installs the following components:

- Honeywell Repackaged Redistributables
 - HCI Runtime
 - System Management Runtime
-

6	Click Install Package .
---	--------------------------------

Step	Action
7	Click OK . The installation process starts and usually takes a few minutes.
8	After a successful installation, the screen in the previous step reappears. Click Exit and OK to end the installation process.
9	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

4.14 Setting system BIOS boot order

Set boot order for WS360-based Honeywell workstation

Step	Action
1	Restart the machine.
2	Press the F2 key before complete boot up to access the BIOS menu.
3	Move the highlight selection to Boot Sequence and press the Enter key.
4	Under the boot sequence, make sure the boot order is Hard Drive, Floppy Diskette, and then CD. Press Enter key to save the boot sequence.
5	Press <ESC> and Select Save Changes and Exit .
6	Restart the machine.

Set boot order for WS370/470/490-based Honeywell workstation

Step	Action
1	Restart the machine.
2	Press the F2 key before complete boot up to access the BIOS menu.
3	Move the highlight selection to Boot Sequence and press the Enter key.
4	Under the boot sequence, make sure the boot order is Hard Drive, Floppy Diskette, and then CD. Press ESC key and select Save to save the boot sequence.
5	Press the ESC key and select Save/Exit to save the settings and exit setup.
6	Restart the machine.

4. TPS node procedures

4.15. Network install of TPS software

Set boot order for WS1600/1800/1900/2850/2900/2950-based Honeywell workstation

Step	Action
1	Restart the machine.
2	Press the F2 key before complete boot up to access the BIOS menu. Wait until the BIOS settings screen is displayed.
3	Move the highlight selection to Boot Sequence and press the Enter key.
4	Under the boot sequence, make sure the boot order is Hard Drive, Floppy Diskette, and then CD. Press Enter key to save the boot sequence.
5	Press the ESC key and select Save Changes and Exit .
6	Restart the machine.

4.15 Network install of TPS software

Purpose

The Network Install method of installing TPS software allows the target machine to pull the installation from a network distribution point.



ATTENTION

These procedures are to be used in conjunction with one of the *TPS Implementation Checklists*.

Prerequisites

Before you perform a Network Install, ensure the following prerequisites have been met:

- 'Administrative Installation' of TPS software has been performed.
- Existing TPS software has been uninstalled from the target computer (see section 4.16 of this guide).



REFERENCE - INTERNAL

For instructions to perform the prerequisite 'Administrative Installation', refer to the section "Administrative Installation" in the *TPS System Administration Guide*.

To perform network install of TPS software on target machine

Step	Action
1	NOTE: You can skip this step (step 1) if you initialized the target (client) machine from the EXPPlus DVD and then installed the TPS base software (as described in section 4.17, "Initializing TPS node using EXPPlus DVD"). a) On the target machine, log on as Administrator. b) From the TPS Software CD, install i386\Packages\hwredist\setup.exe (Honeywell Repackaged Redistributable Files). c) From the TPS Software CD, install i386\Packages\wksecurity\setup.exe (Workstation Security Package).
2	Restart the machine and log on as a user with Administrator privileges in the domain (for example, domain Administrator).
3	In the Windows Explorer, browse to Program Files\ Honeywell\ WkStaSecurity. Run the script LinkDomainGroups.vbs. For a detailed procedure, refer to section 4.19, "Performing LinkDomainGroups". RESULT: This script file identifies standard global groups existing on a Domain Controller (such as Engineers, Supervisors, and Operators), and adds them to the equivalent TPS local groups.
4	Log off the target machine. On the target machine, log on as Administrator while holding down the SHIFT key.



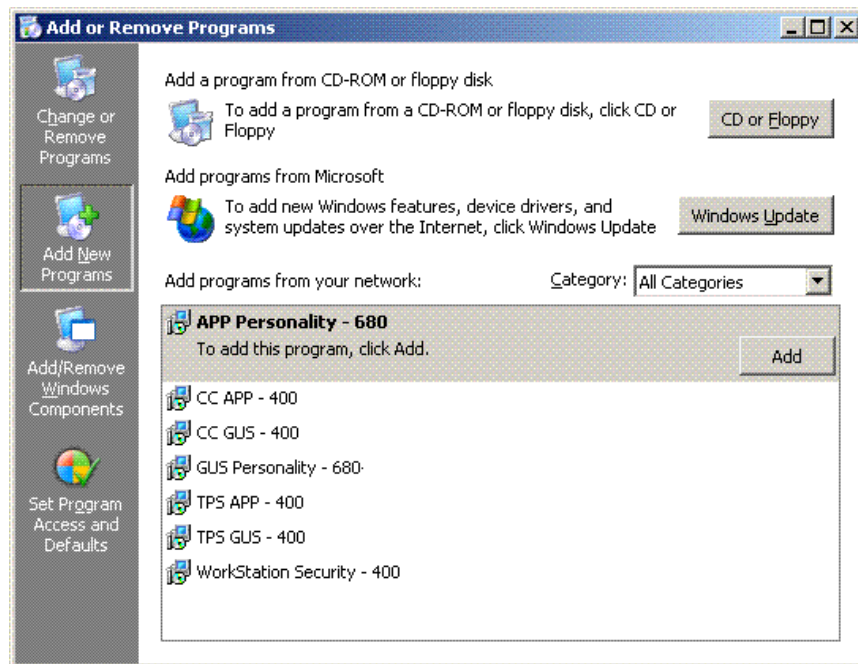
ATTENTION

- Do not log on as an Engineer because the Engineer is not a member of the Administrator group, by default.
- Do not log in as a local Administrator because the local Administrator is not a domain user and you cannot see the published software.
- In TPS R400 and greater, upon execution of linkdomaingroups.vbs, the global (domain) group "DCS Administrators" becomes a member of "Administrators" and "Honeywell Administrators" local groups. "Honeywell Administrators" is the replacement for the "TPS Local Administrators" local group that existed in previous releases; however "Honeywell Administrators" does not have the Engineer privilege.

4. TPS node procedures

4.15. Network install of TPS software

- | Step | Action |
|------|--|
| 5 | Display the Control Panel: Start > Settings > Control Panel . |
| 6 | Double click Add/Remove Programs . |
| 7 | Click Add New Programs on the left panel. |



Step	Action
8	<u>For a GUS node:</u> Click Add on each package in the following order: a) CC_GUS b) TPS_GUS c) GUS Personality d) WorkStation Security <u>For an APP node:</u> Click Add on each package in the following order: a) CC_APP b) TPS_APP c) APP Personality d) WorkStation Security
	 ATTENTION The published software installs on the client machine in an automated manner. The client installation automatically uses the captured administrator interaction that was recorded in the install.ini file. When the installation is complete, a dialog box appears to indicate the components that have been installed.
9	After the installation completes, if this is the first time installation of common component packages, restart the machine to set up the registry.
10	Display the Configuration Utility: Start > Programs > Honeywell TPS > Configuration Utility .
11	In the Configuration Utility page, click Configure > Devices/Services .

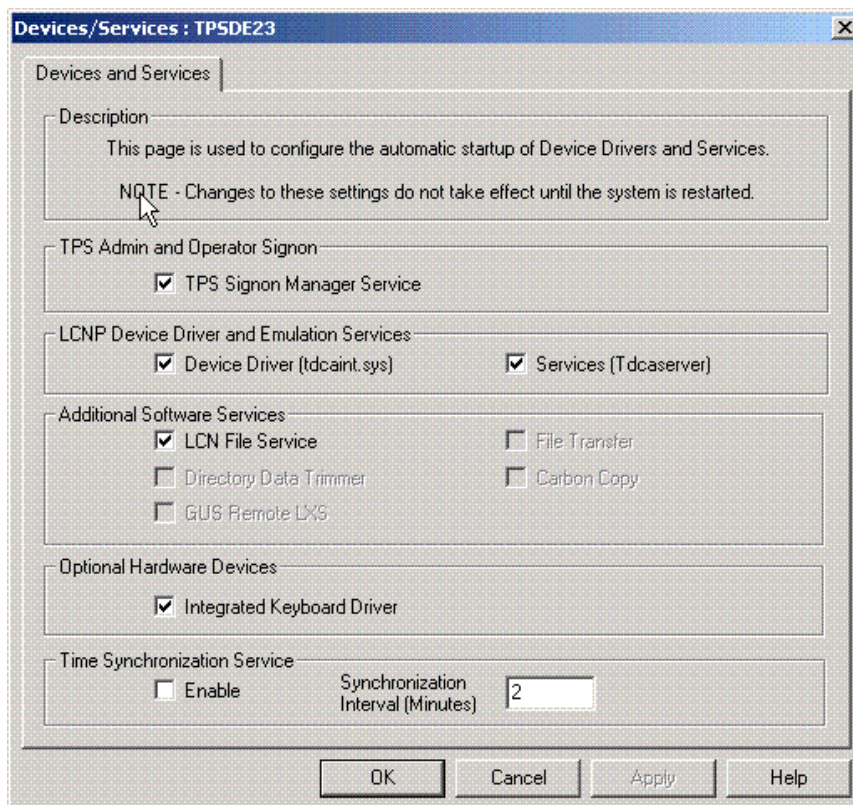
4. TPS node procedures

4.15. Network install of TPS software

Step

Action

- 12 Select all the boxes except **Time Synchronization Service**, and then click **OK**.



- 13 In the Configuration Utility page, click **Configure > Board0**.

Step	Action
------	--------

- | | |
|----|--|
| 14 | In the LCNP tab, select Enable LCNP Auto Restart and then click OK . |
|----|--|

Board 0 : UIS-GUS-GX5JF

Printer	Emulated Disks	Data Access Configuration
LCNP	Common	Native Window
		TPN Time Sync
		Keyboard

Description

The LCNP adapter card supports the execution of a standard TDC Personality and provides connection to the LCN.

This form provides configuration information for the adapter's identity on the LCN network.

LCN Address: NG Name:

LCN Name:

Personality: GUS

LCNP Reset Controls Local Remote

From LCNP Status Display ☒ ☐

LCNP Auto Restart

☒ Enable LCNP Auto Restart

OK Cancel Apply Help

- | | |
|----|-----------------------------|
| 15 | Restart the machine. |
|----|-----------------------------|

4. TPS node procedures

4.16. Un-installation of GUS or APP software

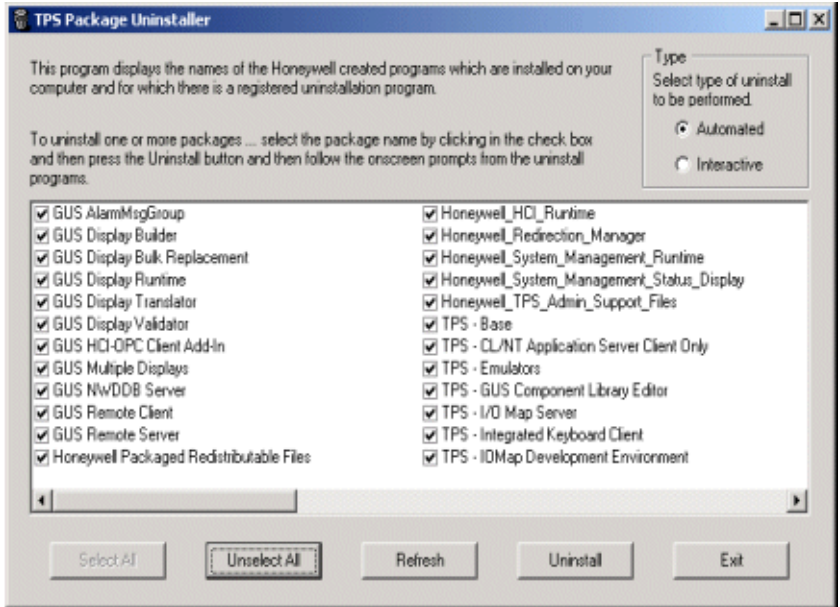
4.16 Un-installation of GUS or APP software

Uninstallation guidelines

For an overview of the Honeywell uninstaller, refer to section 1.2, "TPS software un-installation guidelines".

To uninstall GUS or APP on the target machine

Step	Action
1	Log on to the target machine as Administrator .
	ATTENTION Do not log on as an Engineer because Engineer is not a member of the Administrator group by default.
2	Display the Configuration Utility: Start > Programs > Honeywell TPS > Configuration Utility .
3	In the Configuration Utility page, click Configure > Devices/Services .
4	Clear all the boxes, and then click OK .
5	In the Configuration Utility page, click Configure > Board0 .
6	Clear all the boxes in the LCNP tab, and then click OK .
7	Restart the machine.
8	Log on as Administrator while holding down the SHIFT key.
9	Do one of the following: - Using Windows Explorer, browse to and double-click i386\packages\Uninstaller\tpsuninstaller.exe - Or - Insert the TPS Software CD and select Uninstall Packages from the menu.
	NOTE: Copy tpsuninstaller.exe from the CD to C:\users\uninstaller directory, so that you require the CD only for the first time.

Step	Action
10	In the Package Uninstaller page, click Select All. 
	NOTE: A pop-up appears if some packages cannot be selected at this time.
11	In the pop-up dialog box, click OK. NOTE: If more than one package cannot be uninstalled, you have to manually remove those using Add/Remove Programs.
12	Click Uninstall .
13	On the Confirm page, click OK .
14	When the un-installation is complete, in the Package Uninstaller page, click Exit .
15	Display the Control Panel: Start > Settings > Control Panel .
16	Double click Add/Remove Programs .

4. TPS node procedures

4.16. Un-installation of GUS or APP software

Step	Action
17	Select all the packages that are prefixed with the following labels and remove them. GUS ... Honeywell ... TPS -... NOTE: You do not need to remove the TPS-Workstation Security Package if it is not changed in the current release.
18	Restart the machine.

4.17 Initializing TPS node using EXPPlus DVD

Overview of node initialization

This initialization procedure sets a TPS node to the Honeywell factory-default state. A TPS node at the factory-default state contains the following:

- Windows 2003 server or Windows XP with and the appropriate service pack as defined in the TPS R4xx Software Change Notice, and
- TPS Base software.

CAUTION The initialization procedure reformats the entire hard drive - all data currently on the hard drive is lost. Before proceeding, save any needed application and configuration data.

Before you begin node initialization

Prior to initialization, you should refer to the TPS R4xx software change notice for any known problems related to initialization.

Prior to initialization, you should backup node data files.



REFERENCE – INTERNAL

- For details on backing up a TPS node, refer to the *Experion/TPS Backup and Restore User's Guide*.
 - For additional details on TPS node initialization, refer to the *TPS System Administration Guide*.
-

Initialization media

To perform a TPS node initialization, you need the following media:

- EXPPlus Initialization DVD - Installs the Windows operating system and hardware drivers with Honeywell factory defaults. Refer to the TPS R4xx software change notice for the applicable DVD release for your TPS release.
- Microsoft Windows XP or Windows 2003 Server CD.
- TPS Software CD - Installs TPS and common component software.

4. TPS node procedures

4.17. Initializing TPS node using EXPPlus DVD

To install operating system

Step	Action
1	Go to the BIOS menu and ensure that the EXPPlus DVD is auto-bootable: a. Restart the machine and press the F2 key before complete boot up to access the BIOS menu. b. In the BIOS settings screen, move the highlight selection to Boot Sequence and press the Enter key. c. Ensure that the boot order is configured such that the DVD/CD-ROM device appears first in the boot Sequence. d. Save any changes and exit the BIOS menu.
2	Insert the EXPPlus Initialization DVD and restart the machine. Result: The node boots from EXPPlus DVD.
3	In Step 1 of the EXPPlus Installation wizard , select TPS and click Generate .
4	In the TPS Configuration Tool that is launched, select appropriate options/configurations in each of the tabs. While the defaults may be appropriate in most of the cases, you may need to specifically enter the following: - Network configuration/IP address - System Details: - Computer Name - Company Name and other details - Windows Product Key - Disk Partition Information If this is an initialization for Windows 2003 Server, then specify the following information: - Server License Details - Server Services

Step	Action
5	Click Generate Configuration File on the Summary tab. Click either Yes or No to respond to Do you want to backup the Configuration files? If you click Yes, you need to select the root of the floppy disk or USB flash drive to save the files to that media. Note: If you save the files to the root of the media, then you can use the media during any subsequent initializations, so that you do not have to reenter the information. To use the media in subsequent initializations, insert the floppy disk or USB flash drive and then insert the EXPPLUS DVD.
6	Click Continue on the EXPPlus Installation wizard.
7	Click I Agree on the Windows Preinstallation Environment EULA page.
8	In a later Step, when a dialog box in the EXPPlus Installation wizard requests a Windows XP/Windows 2003 Server CD, insert an appropriate CD. RESULT: Windows XP/Windows 2003 Server installation begins and the machine reboots.
9	Click Continue . Accept the EULA agreement when prompted.
10	Specify an administrator password when prompted.
11	The machine restarts on completion of installation. RESULT: Windows XP Professional or Windows 2003 Server installation completed successfully. The node is at the Honeywell factory default state, minus the factory-default TPS software.
12	Click Finish. Remove the media from their drives. The system is now ready for TPS software installation.

To install Workstation security package

Perform this procedure to establish a set of default Honeywell local groups and accounts.

Note: This step must be performed before installing APP Base software.

Step	Action
1	Insert the TPS Software CD and click Workstation Package .
2	Follow the instructions to complete the installation.

4. TPS node procedures

4.17. Initializing TPS node using EXPPlus DVD

To install factory-default TPS software (GUS Base or APP Base)

Step	Action
1	Insert the TPS Software CD .
2	Click Install Packages from the dialog.
3	When the Welcome dialog displays, read the details and then click Next .
4	Accept the software license agreement terms by clicking Next .
5	Note the third party software information and then click Next .
6	If this dialog has been used before, the information fills in; if not, enter the Name, Company, License No., and Authorization No. information, and then click Next . RESULT: After the License No. and Authorization No. have been validated, the "Package Selection" dialog displays a list of the available licensed packages.
7	<u>For GUS nodes:</u> Install GUS Base Software . Click Install Package . <u>For APP nodes:</u> Install APP Base Software . Click Install Package .
8	In the Installable Packages dialog, retain the default selection and click OK .
9	Restart the machine. The node is now at the Honeywell factory-default state.

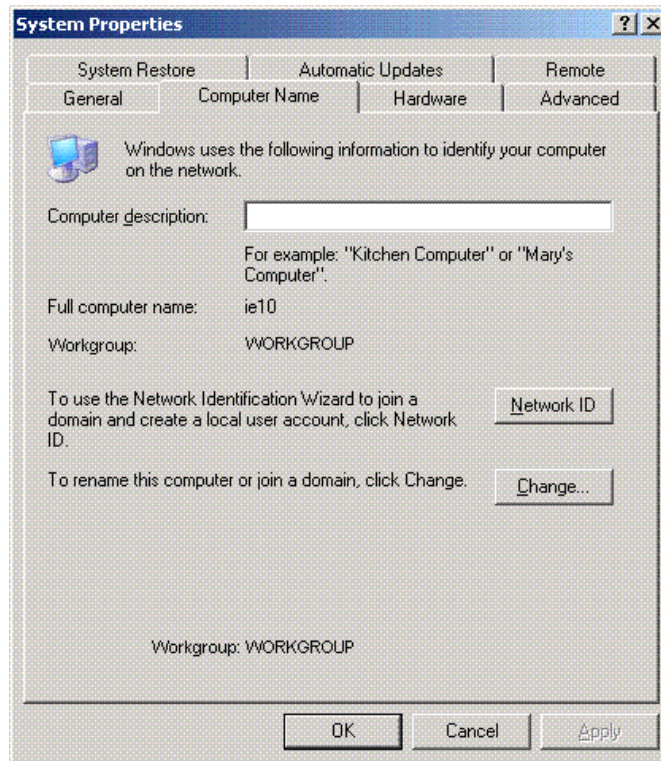
Next step

After completing the initialization procedure, you can perform either of the following tasks to complete the TPS node implementation:

- "New TPS node implementation checklist"
- "Network install of TPS software"

4.18 To add a computer to a domain

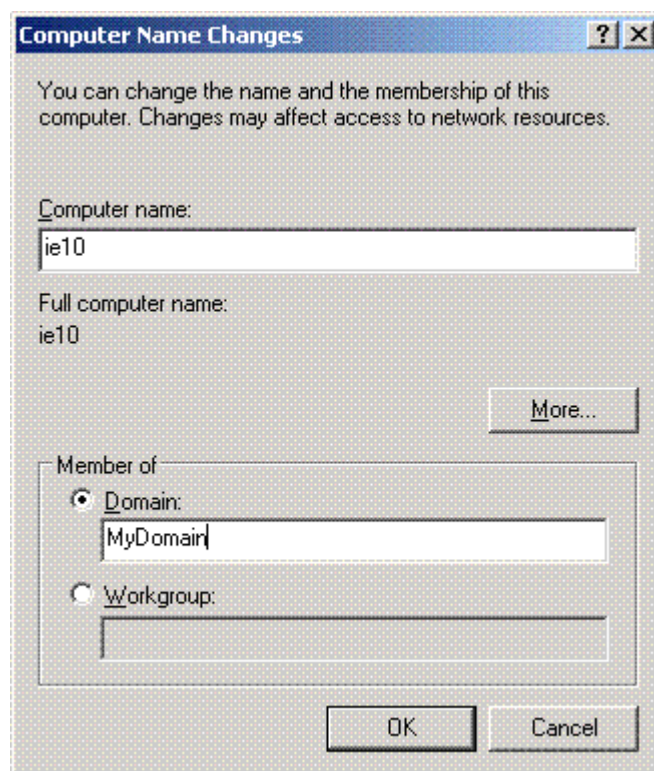
Step	Action
1	Log on to the computer.
2	Right-click My Computer icon from the Desktop. or Choose Start > My Computer , right-click My Computer . The pop-up menu appears.
3	Click Properties . The System Properties dialog box appears.
4	Click the Computer Name tab.



4. TPS node procedures

4.18. To add a computer to a domain

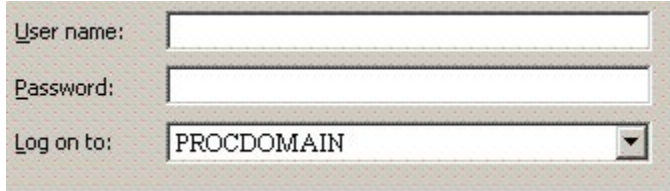
Step	Action
5	Click Change . The 'Computer Name Change's dialog box appears.
6	Click the Domain option button, and then type the name of the domain.



Step	Action
7	Click OK. The 'Computer Name Changes' dialog box appears. 
8	Type the user name and password of the user account that has enough privileges to join a computer to the domain.
	TIP This user account is typically a member of the Domain Admins user group in the Domain Controller.
9	Click OK. Wait until you receive a message welcoming you to the domain.
10	Click OK in the message box. The system requests you to restart.
11	Click Yes to restart. After restarting you can log on to the domain.
12	To log on to the Domain, press Ctrl-Alt-Del, then Options >>> on the log on screen.

4. TPS node procedures

4.19. Performing LinkDomainGroups

Step	Action
13	Select the domain from the Log on to drop-down box and specify a user account of the domain. 

4.19 Performing LinkDomainGroups



ATTENTION

LinkDomainGroups must be performed after installing the Workstation Security package from the TPS Software CD and before installing TPS packages.

The Honeywell High Security Workstation package installs a script file 'LinkDomainGroup.vbs'.

Execution of this script adds the Honeywell High Security Global (domain) groups to the Honeywell High Security Local groups (For example, the "Operators" Global (domain) group is added to the "Local Operators" Local group).



REFERENCE - INTERNAL

For information on the Global Users and Global Groups for TPS, refer to "Applying TPS Security Model" in the *TPS System Administration Guide*.

To perform LinkDomainGroups

Step	Action
1	At the TPS node, log on as a user with Administrator privileges in the domain (for example, domain Administrator).
2	Double-click the script file LinkDomainGroup.vbs in \Program Files\Honeywell\WkStaSecurity directory. Result: The LinkDomainGroups script file is performed.



TIP

If you need to localize the TPS security for a non-domain TPS system, refer to the *TPS System Administration Guide*.

4.20 Installing Microsoft .NET Framework



ATTENTION

This procedure is to be used in conjunction with one of the *TPS Implementation Checklists*.

Step	Action
1	Log on to the target machine as Administrator.
2	Insert the TPS Software CD .
3	Select Install Package from the TPS Software CD menu.
4	Review the information about installing software, license agreements, and third-party compatibility on the next several screens. Click Next to continue.
5	Enter your license and authorization numbers. Click Next to continue.
6	Select .NET Framework Redistributables .
7	Click Install Package .
8	For a GUS node - Check the first and second checkboxes. For an APP node - Check the first and third checkboxes.
9	Click OK . The installation process starts and usually takes a few minutes.
10	After a successful installation, the screen in the previous step reappears. Click Exit and OK to end the installation process.
11	Restart the machine.
12	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

4. TPS node procedures

4.20. Installing Microsoft .NET Framework

5. TPN (TDC) procedures

5.1 Loading GUS as US from emulated disks

Step	Action
Setup Emulated Disks	
1	Log on to the GUS as Administrator.
2	Insert the GUS-TPN Software CD .
	ATTENTION Ensure that the GUS base software is installed prior to GUS personality installation.
3	Copy all of the following files from the CD: disk_&?.lcn to the following folder on the local hard disk: \\HWIAC\Databases\EmulatedDisks
	ATTENTION If the \\HWIAC\Databases\EmulatedDisks directory does not exist, create it.
4	Display the LCNP Status display: Start > Programs > Honeywell TPS > LCNP Status .
	ATTENTION Wait for the window to appear to ensure that the board status passed the self-test so that you can monitor the status of the LCNP board while starting the GUS.
5	Display the Native Window: Start > Programs > Honeywell TPS > Native Window .
6	Select Access > Mount/Dismount Emulated Disks .
7	Connect the left drive to the following file: C:\\HWIAC\\Databases\\EmulatedDisks\\disk_&Z6.lcn
	ATTENTION For all remaining steps in this procedure, \$F1 refers to the left drive.
8	Connect the right drive to the following file: C:\\HWIAC\\Databases\\EmulatedDisks\\disk_&Z1.lcn

5. TPN (TDC) procedures

5.1. Loading GUS as US from emulated disks

Step	Action
	ATTENTION For all remaining steps in this procedure, \$F2 refers to the right drive.
9	Click OK .
<i>Load Native Window With US Personality</i>	
10	Click LOAD on the bottom menu of the Native window.
11	When the prompt "W,N, ... ?" appears, type 2 and then press Enter .
12	When the prompt "OPR or UNP ?" appears, type UNP and then press Enter .
13	When the prompt "NCF ?" appears, type 1 and then press Enter .
14	When the prompt "ABST ?" appears, type 2 and then press Enter .
15	When the prompt "BUTT ?" appears, type 1 and then press Enter .
16	When the node finishes loading, dismount the right-hand drive.
17	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

5.2 Using Native Window to initialize the TPN HM



ATTENTION

This Procedure is for New Systems Only.

This procedure is only needed when the directory structure and volumes are not yet initialized on the TPN History Module.

This procedure assumes that the user is familiar with using emulated disks through the Native Window on a GUS.

This procedure need only be done once per TPN.

Step	Action
Create Directory on Emulated Disk for NCF Backup File	
1	Create a new drive of 20 MB called "backup-asy.lcn".
 ATTENTION This local drive is a backup of the NCF on the TPN History Module.	
2	Mount this new drive on to the right hand drive.
3	Using the Native Window Command Processor, format the new drive as a Bernoulli drive.
4	Create an "&ASY" directory.
5	Copy the contents of the &ASY directory from \$F1 (the left drive) to the &ASY directory on \$F2 (the right local backup drive of 20 MB).
 ATTENTION The Node is now usable as a Universal Station on the LCN for configuring the NCF.	
Configure TPN and Initialize History Module	
6	Configure the TPN System.
 REFERENCE – INTERNAL For detailed instructions, refer to the <i>TPN System Startup Guide</i>.	
7	Initialize the History Module and configure the LCN network.

5. TPN (TDC) procedures

5.2. Using Native Window to initialize the TPN HM

Step	Action
	REFERENCE – INTERNAL For detailed instructions, refer to the <i>TPN System Startup Guide</i> .
8	If you need a Fast Load Cartridge on the node, create the disk now.
	REFERENCE – INTERNAL For detailed instructions to create a Fast Load cartridge, refer to the "Create Fast Load Emulated Disk" topic in the <i>TPN System Startup Guide</i> .
9	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

5.3 Using Native Window to copy GUS load modules to HM

Requirements

For the GUS software to operate properly, certain system files are required in the CUS directory on the TPN system History Module. Use the following procedure to copy the system files to the TPN system History Module, if they do not already exist.

Special considerations



ATTENTION

- If the GUS is being added to a system that already has GUS nodes running GUS pictures, then this procedure is not required.
- **This procedure needs to be performed only once per TPN, not per node.**
- **By default, administrators do not have Engineer key access level as they did in earlier releases.**

If you have VIEW ONLY access to the Native Window (that is, you log on as administrator and the administrator does not have permission to change the keylevel of the Native Window), then you have to do one of the following to get permission to copy files to the History Module:

- Obtain the physical key for the GUS keyboard and set it to the Engineering keylock access level, or
- Sign on to Signon Manager as an engineering user, that is, a member of the Local Engineers group. For detailed instructions, refer to section 4.8, "Configure Signon Manager".
- For GUS R331 and later, the GUS Load Modules are on the Utilities and Load Modules Software CD. For prior GUS releases, the GUS Load Modules were on the TPS Software CD. The Load Module UPBASE.LO is still on the GUS-TPN Software CD.
- Load modules typically increase in size as newer versions are released. TPN tool kit schematic CBREV can be displayed in the Native Window to determine the version of load modules currently in use. When installing the newer versions, you should ensure that your NCF configuration has allowed sufficient space for the larger load modules.

5. TPN (TDC) procedures

5.3. Using Native Window to copy GUS load modules to HM

Copy GUS load modules to HM

Step	Action
1	Display the Native window: Start > Programs > Honeywell TPS > Native Window.
2	Go to the Console Status display.
3	Locate this node (marked by an asterisk *).
4	Determine and write down the Drive Numbers for this station.
5	Insert the Utilities and Load Modules Software CD.
6	From the Access menu on the Native Window menu bar, select Mount/Dismount Emulated Disks.
7	Click Create.
8	Navigate to the Multiple Schematics (MSCHEM) folder on the CD and open the appropriate folder, depending on the current TPN release of your system: <drive>:\TPN_R53X (If the TPN release is R533 or later) <drive>:\TPN_R620_R6xx (If the TPN release is R620 or later)
9	Select the following .lcn system file: Disk_&z14.lcn
10	Click Open.
11	Accept the default information on the Emulated Disk Information dialog box and click OK.
12	Select the disk_&z14.lcn file in the Available (Dismounted) Emulated Disks port of the Mount/Dismount Emulated Disks window.
13	Click Mount. Select the Left Drive. Click OK.
14	Select the following .lcn system file: Disk_&z12.lcn
15	Click Open.
16	Accept the default information on the Emulated Disk Information dialog box and click OK.

Step	Action
17	Select the disk_&z12.lcn file in the Available (Dismounted) Emulated Disks port of the Mount/Dismount Emulated Disks window.
18	Click Mount . Select the Right Drive . Click OK .
19	On the Native Window menu bar, select the Engineering menu, and click Menu .
20	On the Engineering Main Menu, select the Command Processor target.
21	Enter the LSV NET command to verify that CUS directory exists. Use the Create Directory command to create the CUS directory if it does not exist.
22	Execute the following command: EC \$Fn>&EC>MSCHEMZ.EC \$Fn -D , where: • \$Fn represents the drive number that has Disk_&Z14.lcn mounted (left is always the odd number). • \$Fx represents the drive number that has Disk_&Z12.lcn mounted (right is always the even number) RESULT: The command file performs commands (including those shown below) to update the load module files on the System History Module. UNPT NET>&CUS>*. * CP \$Fn>&CUS>MSCHEM.LO NET>&CUS>= CP \$Fn>&CUS>CSCHEM.LO NET>&CUS>= CP \$Fx>&CUS>YGOCX.LO NET>&CUS>= CP \$Fx>&CUS>GUSALA.LO NET>&CUS>= CP \$Fx>&CUS>GUSALAG.LO NET>&CUS>= PT NET>&CUS>*. *
23	From the Access menu on the Native Window Menu Bar, select Mount/Dismount Emulated Disks . Dismount the emulated disk: Click Dismount (next to the drive containing the Disk_&z14.lcn file and the Disk_&z12.lcn file).
24	Remove the CD.

5. TPN (TDC) procedures

5.3. Using Native Window to copy GUS load modules to HM

Step	Action
25	Insert the GUS - TPN Software CD .
26	Using steps similar to the previous operation, mount the emulated Disk_&Z1.lcn
27	From the Native Window Command Processor, perform the following copy command, where \$Fn can be either drive: UNPT NET>&CUS>*.* CP \$Fn>&CUS>UPBASE.LO NET>&CUS>= PT NET>&CUS>*.*
28	Dismount the Emulated Drive and remove the CD.
	ATTENTION
	Each GUS node that is to use the updated load modules must have its LCNP card shutdown and reloaded for it to begin using the updated load module software.
29	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

5.4 Using Native Window to copy APP load modules to HM

For the APP software to operate properly, certain system files are required in the CUS directory and the CLX directory on the TPN System History Module.



ATTENTION

The load module OPC_AE.LO is required *only* if the APP node is acting as an Integration (IN) node between a PlantScape or Experion system and the TPN. OPC_AE.LO is not required by other APP nodes or LCN-connected Server nodes.

The load module ESVT.LO is required by APP nodes acting as LCN-connected Server nodes. (For installation procedures, refer to document EX23, *Integrated Experion-TPS Nodes (ES-T, ESVT, and ACE-T) User Guide*.)



ATTENTION

- The following procedure needs to be performed **only once per TPN, not per node**.
- By default, administrators do not have the Engineer key access level (for Native Window access) as they did in earlier releases.

If you have VIEW ONLY access to the Native Window (that is, you log on as administrator and the administrator does not have permission to change the keylevel of the Native Window), then you have to do one of the following to have permission to copy files to the TPN History Module:

- Obtain the physical key for the GUS keyboard and set it to the Engineering keylock access level, or
- Sign on to Signon Manager as an engineering user, that is, a member of the Local Engineers group. For detailed instructions, refer to section 4.8, "Configuring Signon Manager".

Step	Action
1	Log in to an existing functional GUS as Administrator.
2	Insert the Application Module Software CD .

5. TPN (TDC) procedures

5.4. Using Native Window to copy APP load modules to HM

Step	Action
	ATTENTION Be sure to use the appropriate CD for your TPN software release.
3	Select Programs > Honeywell TPS > Native Window .
4	From the Access menu on the Native Window menu bar, select Mount/Dismount Emulated Disks .
5	Mount the following LCN system file on the CD into either the left or the right emulated drive: DISK_&Z3.LCN
6	From the Engineering menu on the Native Window menu bar, select Menu .
7	From the Engineering Main Menu, select the Command Processor target.
8	Enter the LSV NET command to verify that the CUS and CLX directories exist. If they do not exist, use the Create Directory command to create the directories.
9	Copy the files from the CD using the following commands (replace "\$Fx" with "\$F1" for the left emulated drive or "\$F2" for the right emulated drive): UNPT NET>&CUS>*.* UNPT NET>&CLX>*.* CP \$Fx>&CUS>AMCL06_2.LO NET>&CUS>= CP \$Fx>&CUS>XACCES.LO NET>&CUS>= CP \$Fx>&CUS>XOPTN.LO NET>&CUS>= CP \$Fx>&CLX>AMCL06.SF NET>&CLX>= PT NET>&CUS>*.* PT NET>&CLX>*.*
10	Dismount the drive.
11	Remove the CD.
12	If you arrived at this procedure from a checklist for node implementation, migration, or upgrade, return to the checklist.

Loading APP nodes without using system HM

With Release R650, the APP nodes can be reloaded without using a system HM. When the system HM is not available or when the APP node loses communication with the system HM, the APP node can be loaded with the APP personality from its hard drive and obtain the required support files using the new fast-load cartridge mounted on a US/GUS.

5.5 Using Native Window to update an existing NCF for GUS

This procedure updates the TPN NCF with the new GUS node information. This can be done from an existing Universal Station or from the Native Window on a GUS.

**ATTENTION**

- This procedure requires the Engineering keylock access level.
- If you have VIEW ONLY access to the Native Window (that is, you are logged on as administrator and the administrator does not have permission to change the keylevel of the Native Window), then you have to do one of the following to have permission to copy files to the TPN History Module:
 - Obtain the physical key for the GUS keyboard and set it to the Engineering keylock access level, or
 - Sign on to Signon Manager as an engineering user, that is, a member of the local TPS Engineer group. For detailed instructions, refer to section 4.8, "Configuring Signon Manager".

Step	Action
1	Display the Native Window: Start > Programs > Honeywell TPS > Native Window.
2	Select Engineering > Menu > Support Utilities > Modify Volume Paths.
3	Ensure that the NCF backup path points to a disk that has an &asy directory.
4	Select Engineering > Menu > LCN Nodes.
5	Select the relevant node number.

5. TPN (TDC) procedures

5.5. Using Native Window to update an existing NCF for GUS

Step	Action
6	On an existing node: Select the relevant node number, select Modify Node and then go to the next step. On a new LCN node: Refer to the appropriate manuals in the TPN bookset (<i>Network Form Instructions</i> and <i>Network Data Entry</i>) for node configuration instructions for a Universal Station (GUS), and then go to next step.
7	Page forward to the External Load Module page. Add the following external load modules and their corresponding personalities (UP denotes Universal Personality): • MSCHEM UP • CSCHEM UP • UPBASE UP • YGOCX UP • GUSALA UP • GUSALAG UP
8	Clear the value in the MAXIMUM EXTERNAL MODULE MEMORY UP port.
9	Enter 15000 in ADDITIONAL MODULE MEMORY (WORDS)...UP port.
10	Press ENTER . The total amount of memory required is recalculated and displayed as the TOTAL (MODULES PLUS ADDITIONAL MEMORY).
11	The MAXIMUM must be 10,000 greater than the TOTAL. Enter the appropriate value into the MAXIMUM EXTERNAL MODULE MEMORY UP port.
12	Enter, Check, and Install the NCF change.
13	If you use fast load disks, then copy the new NCF to all of the fast load disks and emulated fast load disks.
14	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

5.6 Using Native Window to update an existing NCF for APP

This procedure updates the TPN NCF with the new APP node information. This can be done from an existing Universal Station or from the Native Window on a GUS.



ATTENTION

- This procedure requires the Engineering keylock access level.
- If you have VIEW ONLY access to the Native Window (that is, you are logged on as administrator and the administrator does not have permission to change the keylevel of the Native Window), then you have to do one of the following to have permission to copy files to the TPN History Module:
 - Obtain the physical key for the GUS keyboard and set it to the Engineering keylock access level, or
 - Sign on to Signon Manager as an engineering user, that is, a member of the local TPS Engineer group. For detailed instructions, refer to section 4.8, "Configuring Signon Manager".

Step	Action
1	Log in to an existing functional GUS as the local Administrator.
2	Display the Native Window: Start > Programs > Honeywell TPS > Native Window .
3	Select Engineering > Support Utilities > Modify Volume Paths .
4	Ensure that the NCF backup path points to a disk that has an &asy directory.
5	Select Engineering > Menu > LCN Nodes .
6	On an existing node: Select the relevant node number, select Modify Node and then go to the next step. On a new LCN node: Refer to the appropriate manuals in the TPN bookset (<i>Network Form Instructions</i> and <i>Network Data Entry</i>) for node configuration instructions for an Application Module (APP or AMR), and then go to the next step.

5. TPN (TDC) procedures

5.6. Using Native Window to update an existing NCF for APP

Step	Action																
7	Use the recommendations in the following table to set up the node. <table><tr><th>Setting</th><th>Recommendation</th></tr><tr><td>Number of Background CL Tasks</td><td>10</td></tr><tr><td>Concurrent Data Accesses from Background CL</td><td>4</td></tr><tr><td>Background Task Stack Size</td><td>25000</td></tr><tr><td>CVB Size for Fast and Slow Point Processors</td><td>Use "2000" or refer to <i>Application Module Implementation Guidelines</i> manual for information on how to calculate the value.</td></tr><tr><td>Redundancy Buffer Increase</td><td>0</td></tr><tr><td>Include Internet Point Processor</td><td>Select YES if this APP node needs to access points on another LCN through a Network Gateway. After selecting YES, you are prompted for the CVB Size for IPP - Enter "2000" or refer to <i>Application Module Implementation Guidelines</i> manual for information on how to calculate the value.</td></tr><tr><td>User Memory Reserved</td><td>0</td></tr></table>	Setting	Recommendation	Number of Background CL Tasks	10	Concurrent Data Accesses from Background CL	4	Background Task Stack Size	25000	CVB Size for Fast and Slow Point Processors	Use "2000" or refer to <i>Application Module Implementation Guidelines</i> manual for information on how to calculate the value.	Redundancy Buffer Increase	0	Include Internet Point Processor	Select YES if this APP node needs to access points on another LCN through a Network Gateway. After selecting YES , you are prompted for the CVB Size for IPP - Enter "2000" or refer to <i>Application Module Implementation Guidelines</i> manual for information on how to calculate the value.	User Memory Reserved	0
Setting	Recommendation																
Number of Background CL Tasks	10																
Concurrent Data Accesses from Background CL	4																
Background Task Stack Size	25000																
CVB Size for Fast and Slow Point Processors	Use "2000" or refer to <i>Application Module Implementation Guidelines</i> manual for information on how to calculate the value.																
Redundancy Buffer Increase	0																
Include Internet Point Processor	Select YES if this APP node needs to access points on another LCN through a Network Gateway. After selecting YES , you are prompted for the CVB Size for IPP - Enter "2000" or refer to <i>Application Module Implementation Guidelines</i> manual for information on how to calculate the value.																
User Memory Reserved	0																
8	Add the following External Load Modules for the AMW personality <u>in this order</u> : 1. AMCL06_2 AMO 2. XACCES AMO 3. XOPTN AMO																
9	Enter, Check, and Install the NCF change.																
10	If you use fast load disks, then copy the new NCF to all of the fast load disks and emulated fast load disks.																
11	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.																

5.7 Using Native Window to update display search path in Area Database

To enable the GUS runtime to locate a GUS schematic, a search path must be given to the system. This search path is currently configured in the Area Database that resides on the TPN side of the GUS system. The reason for using the Area Database is to allow a system that also has older TPN schematics to use GUS schematics.



REFERENCE - INTERNAL

For detailed instructions on configuring Area Database Pathname Catalog, refer to the *TPN System Startup Guide*.



ATTENTION

- This procedure requires the Engineering keylock access level.
- If you have VIEW ONLY access to the Native Window (that is, you are logged on as administrator and the administrator does not have permission to change the keylevel of the Native Window), then you have to do one of the following to have permission to copy files to the TPN History Module:
 - Obtain the physical key for the GUS keyboard and set it to the Engineering keylock access level, or
 - Sign on to Signon Manager as an engineering user, that is, a member of the local TPS Engineer group. For detailed instructions, refer to section 4.8 *Configure Signon Manager*.

Step	Action
1	Display the Native Window: Start > Programs > Honeywell TPS > Native Window .
2	Through the Native Window select the Main Menu display.
3	From the Main Menu display select Area Data Base .
4	Press the [COMND] key and select the SELECT AREA target.
5	Select the Area number of interest.
6	Press the [CANCEL] key.
7	Select the Pathname Catalog area entity.

5. TPN (TDC) procedures

5.7. Using Native Window to update display search path in Area Database

Step	Action
8	If you are modifying an existing Pathname Catalog, press ENTER , and then press [CTL] [F7] to reconstitute the configuration.
9	The Pathname Catalog is to contain the schematic search paths to be used for display access. Enter all the search paths that are to be used, and then press ENTER. Example: G:\ Where: G:\ denotes a drive containing GUS pictures. For the details on locating GUS displays on a mapped network drive and using Windows 2003 Offline Files and Folders for replication, refer to section 6.3, "Configuring Windows 2003 offline files and folders for replication".
10	Press [CTL] [F12] to Load the configuration to a work file.
11	Press the [COMND] key.
12	Select INSTALL AREA then press ENTER to copy the work file to the Area Database file.
	ATTENTION After the new Area Database is loaded into a GUS node (by loading the node's Native Window Personality or by performing an <i>Area Change</i> if the node's Native Window is already loaded), the GUS software searches the paths for displays when invoked.
13	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

5.8 Loading LCNP board with TPN personality

Load Native Window (GUS or US) Personality

Step	Action
1	Log on to the node as Administrator.
2	Display the Native Window: Start > Programs > Honeywell TPS > Native Window
3	Click the LEDs: xx button to display the LCNP Status display, where xx is the LCN address of your node.
4	Click Details .
5	Click Reset LCNP , and then click Yes to the "Are you sure..." prompt.
6	When the State shows PWR_ON and the Native Window shows the "greater than" symbol (>), click Load on the Native Window status bar.
7	Type W and press Enter in response to the W, N, 1, 2, 3, 4, X? prompt in the Native Window.
8	Type N and press Enter in response to the NCF? N, 1, 2, 3, 4, X? prompt in the Native Window. RESULT: When the loading completes, the Native Window System Status display appears.
9	Select the node, and then select Node Status . RESULT: The Node Status display should indicate the appropriate node type. • GUS node appears as node type GUS • US Replacement node appears as node type US
10	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

5. TPN (TDC) procedures

5.8. Loading LCNP board with TPN personality

Load the APP or AM Personality when a checkpoint exists



ATTENTION

The following procedure is only required for a new LCN installation where an existing AM checkpoint exists.

Step	Action
1	From the System Status display (on a Native Window or US), select the node of interest and then select Node Status .
2	Select the node of interest and perform Shut down as necessary to set the node to the PWR_ON or FAIL state.
3	Load the LCN-side of the node with the AMW personality using the AUTOLOAD NET target. RESULT: When the loading completes, the Node Status display (Native Window) should indicate the appropriate node type: • AM Replacement node appears as node type AM • APP node appears as node type APP
4	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

Load the APP/AM Personality when a checkpoint does not exist



ATTENTION

The following procedure is required for a new LCN installation or adding an APP node to an existing TPN system, where an existing APP/AM checkpoint does not exist.

Prerequisite—Create & Znn directory

If directory & Znn (where nn is the AM/APP LCN node number) does not already exist on the HM, create the directory in the AM/APP checkpoint volume &5np (where np is the HM node pair number).

If volume &5np is not on the HM, and you want to wait to do the HM configuration and initialization at a later time, you may do the following until the HM is initialized to contain the checkpoint volume:



ATTENTION

If the HM is not initialized to contain the checkpoint volume after loading AM or APP personality, the AM or APP Autocheckpoint functionality will not be supported.

- Find an HM user volume large enough to contain the &Znn directory plus the total number of units configured on the AM/APP node (LSV NET). A volume cannot have more than 63 directories.
- Create the &Znn directory in the HM user volume of your choice (CD NET>uservol> &Znn).

RESULT: The System creates an &Fxx directory for each unit (where xx is the Unit ID's index number (position) in the Unit Names NCF configuration). Now the HM contains directories allowing the AM/APP to be MANUALLY checkpointed to the HM. This does NOT allow the AM/APP node to be automatically checkpointed to the HM, as volume &5np is required for automatic checkpointing.

Load/Save procedure

This procedure requires the Engineering keylock access level.

Step	Action
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5. TPN (TDC) procedures

5.8. Loading LCNP board with TPN personality

Step	Action
1	From a GUS node, open the Native Window, use the Access > Mount Emulated Disks menu selection to mount the disk_&z6.lcn emulated disk. This file is on the GUS TPN Software CD .
2	Display the Native Window System Status display.
3	Select your node and display the Application Module Node Status display.
4	The node should display PWR_ON. If it doesn't please check the configuration and all cable connections.
5	a) Perform a Manual Load of your APP node from the AM node status display. b) Load the AM personality (program) from the NET. c) Load the Null database from the Alternate Source disk_&Z6.lcn emulated disk.
6	Once the AM node is in the proper state (AM OK), checkpoint the AM database (Save Data) to the History Module (Net).
7	After the save is complete, Shut down the LCN-side of the APP node TWO times to place it into the FAIL state.
8	Load the LCN-side of the APP node with the APP personality using the AUTOLOAD NET target. (The node gets the personality from its own Local hard drive (PC) and gets the NCF from the HM.) RESULT: When the loading completes, the Native Window node status display should indicate the appropriate node type: • AM Replacement node appears as node type AM • APP node appears as node type APP
9	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

Loading GUS or APP nodes using a fast-load cartridge

Prior to release R650, the APP node could not be loaded without the system HM. With Release R650, the APP or AM node personality can be loaded from the system HM or from removable media (fast-load cartridge) mounted on a GUS/US. The fast-load cartridge can be created on any available physical or emulated media.

When the system HM is not available or when the APP node loses communication with the system HM, the APP node can be loaded using the fast-load cartridge mounted on a GUS/US.

When the APP node is to be reloaded with the APP personality without the system HM, then the personality is loaded from its local disk and the support files are obtained from the fast-load zip disk mounted on the GUS/US. If the APP node is to be reloaded with the AM personality, the personality and support files are obtained from the fast-load zip disk mounted on the GUS/US.



ATTENTION

The fast-load cartridge can contain only one personality; either the APP personality or the AM personality. If both personalities are present in the cartridge, by default the AM personality loads. To load the same node with both the personalities, two separate fast-load zip disks are required, one containing the APP personality and another containing the AM personality.

GUS nodes can be loaded manually from a fast-load cartridge mounted on a different US/GUS (not the local node) without requiring the system HM to be on-line as described in the following procedure.

5. TPN (TDC) procedures

5.8. Loading LCNP board with TPN personality

The following table lists the personalities that are loaded into the GUS node under various load options when the node is being loaded using fast-load cartridges that contain more than one personality.

Personalities on fast-load cartridge	Load option	NCF default option	Resulting GUS personality
OPR, &GUS	Manual Load/Operator Program	--	Operator
OPR, &GUS	Auto Load Local	Operator	Operator
OPR, &GUS	Manual Load/Universal Program	--	GUS
OPR, &GUS	Auto Load Local	Universal	GUS
UNP, &GUS	Manual Load/Operator Program	--	GUS
UNP, &GUS	Auto Load Local	Operator	GUS
UNP, &GUS	Manual Load/Universal Program	--	Universal
UNP, &GUS	Auto Load Local	Universal	Universal
OPR, UNP, &GUS	Manual Load/Operator Program	--	Operator
OPR, UNP, &GUS	Auto Load Local	Operator	Operator
OPR, UNP, &GUS	Manual Load/Universal Program	--	Universal
OPR, UNP, &GUS	Auto Load Local	Universal	Universal

Manual load of a GUS node Personality (non local)

The following table contains the procedure to load a GUS node with the GUS Personality from a US/GUS node on the LCN using the Fast-Load Zip Cartridge (Non Local) - Manual Load.

Step	Action
1	Mount a fast-load cartridge that includes GUS personality files (directory &UXS) on a different GUS/US. Ensure that the fast-load cartridge does not include OPR personality files (directory &OPR).
2	From a US/GUS node on the LCN, press the SYS STATS button to invoke the System Status Display.
3	Select the desired GUS node.
4	Select NODE STATUS .
5	Select the desired GUS node.
6	Select LOAD/DUMP target.
7	Select MANUAL LOAD .
8	Select UNIVERSAL .
9	For PROGRAM SOURCE , select ALTERNATE SOURCE .
10	Select the drive number for the fast-load cartridge.
11	Select EXECUTE COMMAND .
12	For DATA SOURCE , select ALTERNATE SOURCE .
13	Select the drive number that contains fast-load cartridge.
14	Select EXECUTE COMMAND .
15	Press ENTER to execute. When the load is complete and the node status is displayed as OK , the node type is displayed as GUS .

5. TPN (TDC) procedures

5.8. Loading LCNP board with TPN personality

Auto Load of a GUS node personality (local)

The following table contains the procedure to load a GUS node with the GUS Personality from a US/GUS node on the LCN using the Fast-Load Zip Cartridge (Non Local) – AutoLoad Local.

Step	Action
1	Mount a fast-load cartridge that includes GUS personality files (directory &UXS) on a different GUS/US. Ensure that the fast-load cartridge does not include OPR personality files (directory &OPR).
2	From a US/GUS node on the LCN, press the SYS STATS button to invoke the System Status Display.
3	Select the desired GUS node.
4	Select NODE STATUS .
5	Select the desired GUS node.
6	Select LOAD/DUMP target.
7	Select AUTOLOAD LOCAL .
8	Press ENTER to execute.
	Note: In some instances where the fast-load cartridge is mounted on the requesting node, it is not necessary to perform steps 9-15.
9	For PROGRAM SOURCE , select ALTERNATE SOURCE .
10	Select the drive number for the fast-load cartridge.
11	Select EXECUTE COMMAND .
12	For DATA SOURCE , select ALTERNATE SOURCE .
13	Select the drive number for the fast-load cartridge.
14	Select EXECUTE COMMAND .
15	Press ENTER to execute. When the loading is complete, the node status is displayed as OK ; the node type is displayed as GUS .

Manual load of an APP node personality using the fast-load cartridge

The following table contains the procedure to load an APP node with APP Personality from a US/GUS node on the LCN using the Fast-Load Zip Cartridge - Manual Load.

Prerequisite

The APP node should be in the PWR_ON, or QUALIF, or FAIL state before you can perform a manual load.

Step	Action
1	Mount the APP/AM fast-load cartridge that includes APP personality files (directory &AXM) on a GUS/US. Ensure that the fast-load cartridge does not include AM personality files (directory &AMO).
2	From a US/GUS node on the LCN, press the SYS STATS button to invoke the System Status display.
3	Select the desired APP node.
4	Select NODE STATUS .
5	Select the desired APP node.
6	Select LOAD/DUMP .
7	Select MANUAL LOAD .
8	Select desired mode from the options HOT LOAD , WARM LOAD , COLD LOAD , and NO POINT PROCESS .
9	For PROGRAM SOURCE , select ALTERNATE SOURCE .
10	Select the drive number for the fast-load cartridge.
11	Select EXECUTE COMMAND .
12	For DATA SOURCE , select ALTERNATE SOURCE .
13	Select the drive number for the fast-load cartridge.
14	Select EXECUTE COMMAND .
15	Press ENTER to execute. When the loading is complete, the node status is displayed as OK , and the node type changes from AM to APP .

5. TPN (TDC) procedures

5.8. Loading LCNP board with TPN personality

Auto load of an APP node personality (local)

The following table contains the procedure to load an APP node with the APP Personality from a US/GUS node on the LCN using the Fast-Load Zip Cartridge – AutoLoad Local.

Step	Action
1	Mount the APP/AM Fast-Load cartridge that includes APP personality files (directory &AXM) on a GUS/US. Ensure that the fast-load cartridge does not include AM personality files (directory &AMO).
2	From the US/GUS node on the LCN, press the SYS STATS button to invoke the System Status display.
3	Select the desired APP node.
4	Select NODE STATUS .
5	Select the desired APP node.
6	Select LOAD/DUMP .
7	Select AUTOLOAD LOCAL .
8	Press ENTER to execute.
	Note: In some instances where the fast-load cartridge is mounted on the requesting node, it is not necessary to perform steps 9-15.
9	For PROGRAM SOURCE , select ALTERNATE SOURCE .
10	Select the drive number for the fast-load cartridge.
11	Select EXECUTE COMMAND .
12	For DATA SOURCE , select ALTERNATE SOURCE .
13	Select the drive number for the fast-load cartridge.
14	Select EXECUTE COMMAND .
15	Press ENTER to execute. When the loading is complete, the node status is displayed as OK , and the node type changes from AM to APP .

Manual load of an APP node with AM personality using the fast-load zip cartridge

The following table contains the procedure to load an APP node with the AM Personality from a US/GUS node on the LCN using the Fast-Load Zip Cartridge - Manual Load.

Step	Action
1	Mount the APP/AM fast-load cartridge that includes AM personality files (directory &AMO) on a GUS/US.
2	From a US/GUS node on the LCN, press the SYS STATS button to invoke the System Status Display.
3	Select the desired APP node.
4	Select NODE STATUS .
5	Select the desired APP node.
6	Select LOAD/DUMP target
7	Select MANUAL LOAD .
8	Select the desired mode from the options COLD LOAD , WARM LOAD , HOT LOAD , and NO POINT PROCESS .
9	For PROGRAM SOURCE , select ALTERNATE SOURCE .
10	Select the drive number for the fast-load cartridge.
11	Select EXECUTE COMMAND .
12	For DATA SOURCE , select ALTERNATE SOURCE .
13	Select the drive number that contains fast-load cartridge.
14	Select EXECUTE COMMAND .
15	Press ENTER to execute. When the load is complete and the node status is displayed as OK , the node type is displayed as AM .

5. TPN (TDC) procedures

5.8. Loading LCNP board with TPN personality

Auto load of an APP node with AM personality using the fast-load zip cartridge

The following table contains the procedure to load an APP node with the AM Personality from a US/GUS node on the LCN using the Fast-Load Zip Cartridge – AutoLoad Local.

Step	Action
1	Mount the APP/AM Fast-Load cartridge that includes AM personality files (directory &AMO) on a GUS/US.
2	From a US/GUS node on the LCN, press the SYS STATS button to invoke the System Status Display.
3	Select the desired APP node.
4	Select NODE STATUS .
5	Select the desired APP node.
6	Select LOAD/DUMP target.
7	Select AUTOLOAD LOCAL .
8	Press ENTER to execute.
	Note: In some instances where the fast-load cartridge is mounted on the requesting node, it is not necessary to perform steps 9-15.
9	For PROGRAM SOURCE , select ALTERNATE SOURCE .
10	Select the drive number for the fast-load cartridge.
11	Select EXECUTE COMMAND .
12	For DATA SOURCE , select ALTERNATE SOURCE .
13	Select the drive number for the fast-load cartridge.
14	Select EXECUTE COMMAND .
15	Press ENTER to execute. When the loading is complete, the node status is displayed as OK , and the node type is displayed as AM .

6. Additional TPS implementation

6.1 Remote access procedures

Set Security folder permissions



ATTENTION

The procedures in this section are optional. It is only necessary to perform this procedure if you wish to allow other nodes to access the GUS node's LCNP, or if you wish to allow other nodes to remotely access the GUS node for a Native Window session.

Enable remote computer to connect to this computer (to LCNP)

Step	Action
1	Log on to the machine as Administrator.
2	Use Windows Explorer to locate the following folder: C:\Program Files\Honeywell\TPS\Emulators\Security
3	Right-click the folder and select Properties .
4	Click the Sharing tab.
5	Select Share This Folder .
6	Enter the following Share Name : GUS Security .
 ATTENTION	
There is space between "GUS" and "Security".	
7	Click Permissions to display the Permissions for GUS Security dialog box.
8	Select the Everyone group and click Remove .
9	Click Add .
10	Enter Authenticated Users and click OK . Make sure the Authenticated Users group has Read access.
11	Click OK to close the Permissions for GUS Security dialog box.
12	Click Apply .
13	Select the Security tab in the Security Properties window.

6. Additional TPS implementation

6.1. Remote access procedures

Step	Action										
14	Click Advanced. Uncheck the Inherit from parent the permission entries that apply to child objects checkbox. Click Remove. Click OK and Yes to confirm.										
15	To make the Access Control List (ACL), click Add, enter the group name, and then click OK: <u>Group Name</u> Authenticated Users SYSTEM Domain Admins DCS Administrators										
16	To complete the Access Control List (ACL), select each group name and then select the access permission, as follows: <table><tr><td><u>Group Name</u></td><td><u>Access Permission</u></td></tr><tr><td>Authenticated Users</td><td>Allow Read & Execute</td></tr><tr><td>SYSTEM</td><td>Allow Full Control</td></tr><tr><td>Domain Admins</td><td>Allow Full Control</td></tr><tr><td>DCS Administrators:</td><td>Allow Full Control</td></tr></table>	<u>Group Name</u>	<u>Access Permission</u>	Authenticated Users	Allow Read & Execute	SYSTEM	Allow Full Control	Domain Admins	Allow Full Control	DCS Administrators:	Allow Full Control
<u>Group Name</u>	<u>Access Permission</u>										
Authenticated Users	Allow Read & Execute										
SYSTEM	Allow Full Control										
Domain Admins	Allow Full Control										
DCS Administrators:	Allow Full Control										
17	Click OK to close the Security Properties window.										
18	Continue on to the following procedures, as appropriate for your node: <i>Enable Remote Native Window Sessions</i> <i>Limit Users Who Can Remotely Access the Native Window</i> <i>Limit Users Who Can Remotely Access the LCNP Status Display</i> If the following procedures are not required and you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.										

Enable remote Native Window sessions



ATTENTION

The procedure in this section is optional. It is only necessary to perform this procedure if you wish to allow other nodes to remotely access the GUS node for a Native Window session.

Step	Action
1	The LCNP Native Window "Enable Remote Connections" option has to be enabled using the TPS Configuration Utility: Start > Programs > Honeywell TPS > Configuration Utility
2	Select Configure > [Board 0 or your customized LCNP board name].
3	Select the Native Window tab.
4	Select Enable Remote Connections .
5	Select OK .
6	Close the Configuration Utility.
7	Continue to the following procedure "Limit Users Who Can Remotely Access the Native Window", if appropriate for your node. If the following procedure is not required and you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

Limit users who can remotely access the Native Window

Step	Action
1	Use Windows Explorer to locate the following folder: C:\Program Files\Honeywell\TPS\Emulators\Security\TdcBoard0
2	Right-click the Native Window file in the folder and select Properties .
3	Select the Security tab in the Native Window Properties window.

6. Additional TPS implementation

6.1. Remote access procedures

Step	Action
4	Click Advanced. Uncheck the Inherit from parent the permission entries that apply to child objects checkbox. Click Copy. Click OK to close the Advanced Security Settings dialog.
5	To make the following Access Control List (ACL), click Add, enter the group name, and then click OK. <u>Group Name</u> <u>Global (domain) Groups</u> Domain Admins SYSTEM DCS Administrators Engineers Operators Supervisors View Only Users <u>Local Groups</u> Administrators

Step	Action																
	To complete the Access Control List (ACL), select each group name and then select the access permission, as follows: <u>Group Name:</u> <u>Access Permission:</u> <u>Global (Domain) Groups</u> <table><tr><td>Domain Admins</td><td>Allow Full Control</td></tr><tr><td>SYSTEM</td><td>Allow Full Control</td></tr><tr><td>DCS Administrators</td><td>Allow Full Control</td></tr><tr><td>Engineers</td><td>Allow Read & Execute</td></tr><tr><td>Operators</td><td>Allow Read & Execute</td></tr><tr><td>Supervisors</td><td>Allow Read & Execute</td></tr><tr><td>View Only Users</td><td>Allow Read & Execute</td></tr></table> <u>Local Groups</u> <table><tr><td>Administrators</td><td>Allow Full Control</td></tr></table>	Domain Admins	Allow Full Control	SYSTEM	Allow Full Control	DCS Administrators	Allow Full Control	Engineers	Allow Read & Execute	Operators	Allow Read & Execute	Supervisors	Allow Read & Execute	View Only Users	Allow Read & Execute	Administrators	Allow Full Control
Domain Admins	Allow Full Control																
SYSTEM	Allow Full Control																
DCS Administrators	Allow Full Control																
Engineers	Allow Read & Execute																
Operators	Allow Read & Execute																
Supervisors	Allow Read & Execute																
View Only Users	Allow Read & Execute																
Administrators	Allow Full Control																
6	Add any other needed users and groups, and only grant them Read & Execute access to prevent security breaches. Example: To allow all authenticated users to connect a Native Window running remotely to the local LCNP, assign that group Read permissions.																
7	Click OK to close the Native Window Properties window. RESULT: The users and groups granted access above can now remotely access the GUS node for a Native Window session.																
8	Continue to the next procedure "Limit Users Who Can Remotely Access the LCNP Status Display", if appropriate for your node. If the procedure is not required and you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.																

6. Additional TPS implementation

6.1. Remote access procedures

Limit users who can remotely access the LCNP status display



ATTENTION

The procedure in this section is optional. It is only necessary to perform this procedure if you wish to allow other nodes to access the GUS node's LCNP status.

Step	Action
1	Use Windows Explorer to locate the following folder: C:\Program Files\Honeywell\TPS\Emulators\Security\TdcBoard0
2	Right-click the Status Applet file in the folder and select Properties .
3	Select the Security tab in the Status Applet Properties window.
4	Click Advanced . Uncheck the Inherit from parent the permission entries that apply to child objects checkbox. Click Copy . Click OK to close the Advanced Security Settings dialog.

Step	Action																						
5	To make the following Access Control List (ACL), click Add, enter the group name, and then click OK. <u>Group Name:</u> <u>Global (Domain) Groups:</u> Domain Admins SYSTEM DCS Administrators Engineers Operators Supervisors View Only Users <u>Local Groups:</u> Administrators																						
6	To complete the Access Control List (ACL), select each group name and then select the access permission, as follows: <table> <tr> <th><u>Group Name</u></th><th><u>Access Permission</u></th></tr> <tr> <td colspan="2"><u>Global (Domain) Groups</u></td></tr> <tr> <td>Domain Admins</td><td>Allow Full Control</td></tr> <tr> <td>SYSTEM</td><td>Allow Full Control</td></tr> <tr> <td>DCS Administrators</td><td>Allow Full Control</td></tr> <tr> <td>Engineers</td><td>Allow Read & Execute</td></tr> <tr> <td>Operators</td><td>Allow Read & Execute</td></tr> <tr> <td>Supervisors</td><td>Allow Read & Execute</td></tr> <tr> <td>View Only Users</td><td>Allow Read & Execute</td></tr> <tr> <td colspan="2"><u>Local Groups</u></td></tr> <tr> <td>Administrators</td><td>Allow Full Control</td></tr> </table>	<u>Group Name</u>	<u>Access Permission</u>	<u>Global (Domain) Groups</u>		Domain Admins	Allow Full Control	SYSTEM	Allow Full Control	DCS Administrators	Allow Full Control	Engineers	Allow Read & Execute	Operators	Allow Read & Execute	Supervisors	Allow Read & Execute	View Only Users	Allow Read & Execute	<u>Local Groups</u>		Administrators	Allow Full Control
<u>Group Name</u>	<u>Access Permission</u>																						
<u>Global (Domain) Groups</u>																							
Domain Admins	Allow Full Control																						
SYSTEM	Allow Full Control																						
DCS Administrators	Allow Full Control																						
Engineers	Allow Read & Execute																						
Operators	Allow Read & Execute																						
Supervisors	Allow Read & Execute																						
View Only Users	Allow Read & Execute																						
<u>Local Groups</u>																							
Administrators	Allow Full Control																						

6. Additional TPS implementation

6.1. Remote access procedures

Step	Action
7	Add any other needed users and groups, and only grant them Read & Execute access to prevent security breaches. Example: To allow all authenticated users to display this node's LCNP status display remotely, assign that group Read permissions.
8	Click OK to close the Status Applet Properties window.
9	Close Windows Explorer.
10	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

6.2 Configuring a local or network printer

Default printer installation permission

The system policies installed in a TPS node prevent the GUS local user and domain users in the groups Operators, Supervisors and View from configuring a Window's printer to use.

The following procedure, performed by the system administrator of a TPS node, configures a local printer as the default printer. The local printer is then mapped to a network printer when the user logs on to the machine, making the network printer available to users of that node.

For detailed instructions to configure the TPN (LCN) printer functionality, refer to "Configure TPN (LCN) printer settings".



ATTENTION

If the local printer is setup as the alarm printer, then do not perform these procedures.

6. Additional TPS implementation

6.2. Configuring a local or network printer

Local or network printer installation

Step	Action
1	Log on as administrator.
2	Select the following: Start > Settings > Printers and Faxes.
3	Select the following: Add Printer.
4	Select the following: Next.
5	Select the following: Local Printer>Next.
6	Select the following: LPT1: > Next.
7	Select the following: Printer Manufacturer > Printer.
	ATTENTION - If you do not find the printer in the list, you have to provide the printer driver on a disk. - If you are using a Signum Color Printer, a special driver is located on the TPS Software CD.
8	Type the name of the Printer. If it is a network printer, then enter the Network name of the printer.
9	Click Next.
10	Click Next.
11	If printer is not connected to, or managed by the node, select Do not share this printer.
12	Click Next.
13	Click No to not print a test page.
14	Click Finish.
15	Continue to the procedure <i>Redirecting LPT1 to a Network Printer.</i>

Redirecting LPT1 to a network printer

This procedure opens security to expand the node capabilities so it can print items other than GUS alarms, such as reports. The following procedure sets up a NET USE command in the run registry to connect a network printer to the LPT1 port configured in the procedure *Local or Network Printer Configuration*. The print server attempts to validate the user during the connection. If the user is valid, connection is made. If the print server does not recognize the user or password, then connection is rejected.

Use one of the following two methods to redirect LPT1 to a network printer:

- "High Security Method" - if security is an issue and printing is not required by the GUS user account.
- "Low Security Method" - if security is less of an issue and printing is required by the GUS user account.

High security method

Step	Action
1	Log on to the TPS node as administrator.
2	Create a file named printer.reg with the following contents using Notepad and save the file: REGEDIT4 [HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\Run] "NetworkPrinter"="NET USE LPT1 \\\PrntServer\HP4000N /PERSISTENT:NO" (Insert a blank line here.) Where: Replace PrntServer with the computer name of the Print Server. Replace HP4000N with the shared name of the network printer.
3	Use Windows Explorer to navigate to the folder that contains the printer.reg file created in the previous step.
4	Double-click the printer.reg file. RESULT: You should see an information box indicating printer.reg has been successfully entered into the registry.
5	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

6. Additional TPS implementation

6.2. Configuring a local or network printer

Low security method

Step	Action
1	Log on to the Printer Server as Administrator and create the following local user account: PrinterAccess
2	Assign this user a password of your choice. Record the password for later use.
3	Clear the following checkbox: Reset User must Change Password at Next logon
4	Check the following box: User Cannot Change Password
5	Check the following box: Password Never Expires
6	Depending on system configuration, it might be necessary to give the user the right to access the Printer Server from the network. If it is necessary, use the following to grant the right: User Manager > Policies > User Rights.
7	Log on to the TPS Node as Administrator.
8	Create a file named printer.reg with the following contents using Notepad and save the file: REGEDIT4 [HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\Run] "NetworkPrinter"="NET USE LPT1 \\\PrntServer\HP4000N /USER:PrntServer\PrinterAccess password /PERSISTENT:NO" (Insert a blank line here.) Where: Replace PrntServer with the computer name of the Print Server. Replace HP4000N with the shared name of the network printer. Replace password with the password you created in step 2.
9	Use Windows Explorer to navigate to the folder containing the printer.reg file created in the previous step.
10	Double-click the printer.reg file. RESULT: You should see an information box indicating printer.reg has been successfully entered into the registry.
11	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

Configuring page setup for a laser printer

Windows drivers for laser printers enforce margins at the page top and bottom. If you have the LCN file print setup set for letter paper size (8 ½ x 11) and the NCF configuration set to 66 lines, the printer uses two pages to print out the file. To prevent this, do one of the following:

- Set the paper size to 8 1/2 x 14 and the font size to 10, or
- Change the NCF configuration to a smaller number (60 or smaller).

6.3 Configuring Windows 2003 offline files and folders for replication



ATTENTION

Configuring Windows 2003 Offline Files is not required for the proper operation of a TPS system.

The information in this section can be adapted for use with third-party replication solutions.

Definition of offline files and folders

Offline Files is an IntelliMirror capability that allows for the synchronization of files and folders between the local hard drive and a network location. Files and folders marked for offline usage are available even after a network or server failure. To accomplish this, files and folders are cached to the local hard drive.

Windows 2003 Offline Files can be used to distribute GUS-related files, such as .pct files and SafeView workspace configuration files.

When does synchronization occur?

By default, files and folders marked for offline usage are synchronized with the file server **when a user logs on or logs off** the client computer. For example, when the user logs off, the files are synchronized (copied from the server to a reserved portion of the local hard drive called a cache).

Synchronization can be customized to occur automatically on a **predefined interval**.

Synchronization can occur **manually** by right-clicking the files or folders marked for offline usage and selecting the Synchronize item.

6. Additional TPS implementation

6.3. Configuring Windows 2003 offline files and folders for replication

Offline Files Caching Options

By default, the Offline Files feature is not enabled in the Windows Server 2003 family, because any Terminal Services mode that allows multiple, concurrent user logon sessions is not compatible with Offline Files.

When Offline Files is enabled in Windows Server 2003, the following three types of Offline Files caching are available:

- **Only the files that users specify will be available offline** (previously called Manual Caching for Documents). Provides offline access to only those files on a network share that have been manually selected. Manual caching for documents is the default option when you set up a shared folder to use offline.
- **All files and programs that users open from the share will be automatically available offline** (previously called Automatic Caching for Documents and Automatic Caching for Programs). Selecting this option makes every file that someone opens from your shared folder available offline. However, only those files that have been opened are available offline.

If you select the **Optimized for performance** check box, all programs will be automatically cached so that they can be run locally. This option is especially useful for file servers that host applications, because it reduces network traffic and improves server scalability.

- **Files or programs from the share will not be available offline.** Selecting this option prevents users from storing files offline.

Optimizing display callup times

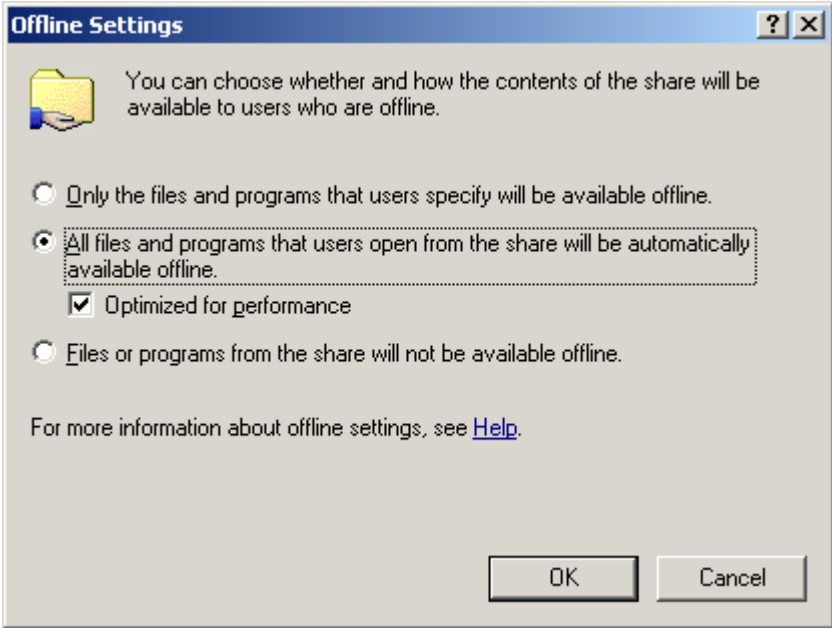
By default, offline files are configured with the option "**Only the files that users specify will be available offline**" (previously called Manual Caching for Documents). This configuration uses the network copy of a file when the network and server are both available. As a result, display file callup time can be adversely affected, due to the network access times.

To optimize display callup time when using offline files, Honeywell recommends selecting the option "**All files and programs that users open from the share will be automatically available offline**" and the option "**Optimized for Performance**".

This caching mode always uses the local cached version of the files for optimum display callup efficiency. The downside of this mode is that you may be using an out of date display file until synchronization occurs.

Checklist—implementation example

The following checklist summarizes the steps to implement offline files for GUS displays.

Step	Action
	At the Domain Controller, put the master files in C:\WINDOWS\Sysvol\domain\Scripts.
	At the Domain Controller, share the above folder and allow caching of files:
	Select All files and programs that users open from the share will be automatically available offline , and Optimized for performance .
	
	At the GUS, map the G drive to the 'Scripts' share.
	At the GUS, enable offline files (default for Windows XP).
	At the GUS, make G:\ available offline.

6. Additional TPS implementation

6.3. Configuring Windows 2003 offline files and folders for replication

Step	Action
	At the Domain Controller, modify the operator.bat files to map the GUS node's G: drive to the 'Scripts' share: NET USE G: /delete NET USE G: \\<domain>\Scripts At the GUS, modify the area database pathname catalog to point to the G:\drive.
	REFERENCE – EXTERNAL Refer to the Microsoft Windows 2003 Help for additional details and configuration guidelines for Offline Files.

Procedure to implement offline files

Before you begin

Before you perform this procedure, read the Microsoft online documentation for offline files.

File server configuration procedure for offline files

Step	Action
1	Log on to the Domain Controller as Administrator.
2	Navigate to the folder that is to be a master repository for files to be replicated. For example, \WINDOWS\system32\repl\exports or \WINDOWS\sysvol\domain\scripts. This is the location where you place the master copies of files, such as GUS displays and SafeView files.
	TIP • If you locate the files on a Windows 2003 Domain Controller, you can leverage the \WINDOWS\SYSTEMVOLUME\domain\scripts folder as the repository for shared files. The entire \WINDOWS\SYSTEMVOLUME folder is replicated automatically to every Domain Controller node. • You can continue to use the legacy Repl\$ share folder if it is established.
3	Establish the directory as a Windows share: Right-click the folder, select Sharing , and then select Share this folder .
4	Apply appropriate permissions to the share folder. The default permission is to allow Everyone Full access to the share. To restrict access, click Permissions and apply the permissions as appropriate for your site.
5	Click Caching and apply the appropriate caching method. (Honeywell recommends that you select All files and programs that users open from the share will be automatically available offline and Optimized for performance .)



REFERENCE

For more information, see ["Optimizing display callup times"](#).

6. Additional TPS implementation

6.3. Configuring Windows 2003 offline files and folders for replication

Step	Action
6	Click OK .
7	Continue to the next procedure, "Client configuration procedure for offline files".

Client configuration procedure for offline files

Perform the following procedure for each of the nodes that are to have Offline Files, such as GUS nodes that are to access GUS displays using Offline Files.

Step	Action
1	Log on to the node as Administrator.
2	In Windows Explorer, map a drive to the network share: Tools > Map Network Drive .
3	In Windows Explorer, enable offline files as follows: a) Select Tools > Folder Options. b) Select the Offline Files tab. c) Make sure the Enable Offline Files checkbox is checked. d) Click OK.



ATTENTION

- By default, the Offline Files option is enabled on any client running Windows XP. The default on Windows 2003 Server is disabled (not checked).
 - If the option is controlled by a Group Policy setting, it cannot be changed on the Offline Files tab of the local computer.
-

4	In Windows Explorer, open the newly mapped drive.
5	Select all the files and/or folders that should be available offline (synchronized with the file server), then right-click and select Make Available Offline .



ATTENTION

If you want to make ALL files on the mapped drive available offline (rather than specific files and folders), right-click the mapped drive and select **Make Available Offline**.

Step	Action
6	In the 'Offline Files Wizard', click Next .
7	Make sure that the Automatically synchronize ... log on and log off ... option is checked, and then click Next .
	ATTENTION By default, offline files on the client are synchronized with the file server when a user logs on or logs off the client computer.
8	a) Leave the Enable reminders option selected. b) Leave the Create a shortcut ... option de-selected. c) Click Finish. Result: You should see a dialog box indicating that synchronization is occurring. When synchronization is complete, the icons for the files/folders change appearance, indicating that they have been made available offline.
9	To view all of the offline files, in Windows Explorer go to Tools > Folder Options . Select the Offline Files tab. Click View Files .
10	For GUS nodes that are to access GUS displays using Offline Files, modify the appropriate Area Database Pathname Catalog to point to the mapped drive (for example, G:\).
11	a) At the DC, edit the Operator.bat file for the GUS stations to map the network drive (Ex: G:\) to the 'Scripts' share (include the NET USE command in the operator.bat file). b) Distribute the modified operator.bat file to the GUS stations. Refer to Section 3.8 "Editing OPERATOR.BAT file" for more information.
12	If you arrived at this procedure from a checklist for node implementation, migration or upgrade, return to the checklist.

6.4 Starting a GUS node for operations

Purpose

When a GUS node is started, the Native Window must be loaded and a SafeView workspace environment started. With these tasks completed, the desired result is that the operator be able to navigate to the displays of interest and manipulate the process.

This discussion outlines two options for starting a SafeView workspace environment that manages a Native Window and displays a “task panel” display. The “task panel” display is a user-defined display providing navigation to other displays of interest.



REFERENCE – INTERNAL

- For additional details and implementation scenarios, see “Enable LCNP Auto Restart” in the *TPS System Configuration Utility User's Guide*.
 - For additional information on configuring the SafeView workspace definition file, refer to the *SafeView User's Guide*.
-

For the detailed procedure to enable a node's LCNP Auto Restart function, refer to section 4.6, "Configure LCNP (Board0) Settings".

For additional information on creating a logon script (Operator.bat) file, refer to section 3.8 "Editing OPERATOR.BAT file" for more information.

GUS startup option A

Step	Desired operator action upon GUS startup	Engineering work to enable desired operator action
1	No operator action - The operator sees a Native Window invoked automatically during startup and displayed on the desktop.	Put the statement to run the Native Window in the operator.bat file. <u>Example statement:</u> (The following example shows a single statement displayed with word wrapping.) START /max "C:\PROGRAM FILES\HONEYWELL\TPS\NATIVEWINDOW\lcnwindow.exe"
2	No operator action - The operator waits for the Native Window to complete automatic loading. Or The operator manually loads the Native Window.	(Optional) To cause the Native Window to load automatically from the PWR_ON state, enable the node's "LCNP Auto Restart" function through the TPS Configuration Utility.
3	The operator selects a shortcut in the Start menu that loads the SafeView operations workspace. (The SafeView workspace manages the Native Window, manages previously running windows, and invokes the user's "task panel" display.)	1.) Create a shortcut in the Start menu to run SafeView, load the operations workspace, and manage previously running windows. <u>Example shortcut command:</u> "C:\Program Files\Honeywell\TPS\SafeView\safeview.exe" /fc:\picture\operator.wdl /cp /mp" Where: operator.wdl is the name of the user's SafeView workspace definition file. 2.) Include a statement in the startup command of the SafeView workspace definition file (operator.wdl) to invoke a "task panel" at workspace invocation. <u>Example workspace startup command:</u> "c:\program files\honeywell\tps\gus\runpic.exe" c:\picture\taskpanel.pct Where: taskpanel.pct is the name of a user-defined "task panel" display.

6. Additional TPS implementation

6.4. Starting a GUS node for operations

GUS startup option B

Step	Desired operator action upon GUS startup	Engineering work to enable desired operator action
1	No operator action – The operator sees the SafeView operations workspace invoked with a Native Window.	1.) Put the statement to run the operations workspace in the operator.bat file. <u>Example statement:</u> (The following example shows a single statement displayed with word wrapping.) Start C:\Program Files\Honeywell\TPS\SafeView\safeview.exe" /fc:\picture\operator.wdl /cp /mp Where: operator.wdl is the name of the user's SafeView workspace definition file. 2.) Include a statement in the startup command of the SafeView workspace definition file (operator.wdl) to run the Native Window at workspace invocation. <u>Example workspace startup command:</u> "c:\program files\honeywell\tps\nativewindow\lcnwindow.exe"
2	No operator action - The operator waits for the Native Window to complete automatic loading. Or The operator manually loads the Native Window.	(Optional) To cause the Native Window to load automatically from the PWR_ON state, enable the node's "LCNP Auto Restart" function through the TPS Configuration Utility.
3	The operator invokes the user's "task panel" display using a dedicated key or the SCHEM key.	(Optional) Configure a dedicated key to invoke the "task panel" display.



ATTENTION

The operator should wait 10 seconds after the Native Window has completed loading before invoking the "task panel".

6.5 Setting Dr. Watson crash dump path

Crash log file and dump file

If an application encounters a fatal error and crashes, the Windows Operating System may attempt to capture crash information by generating a Dr. Watson log (Drwtsn32.log) and dump file. Dr. Watson dump files are useful for troubleshooting at the site, and also for providing information to the Honeywell Technical Assistance Center.

Reason for changing the default crash dump path

Honeywell recommends changing the default path used by Dr. Watson because its default crash dump directory may be inaccessible to these user groups: Engineer, Operator, and Supervisor.

Procedure to set crash dump path



ATTENTION

If a crash occurs while a user is logged on to one of the above user groups and the default path is inaccessible, a Dr. Watson message prompts the user to browse to a new dump directory. The user should save the dump file for troubleshooting to the directory of their choice.

The paths are set the first time Dr. Watson runs, which can be either when a program error occurs, as described above, or when you manually start Dr. Watson yourself.

1. The paths may be set by using the Administrator account and starting the Dr. Watson program through the Run command: **Start > Run**, enter **drwtsn32**.
2. In the resulting Dr. Watson dialog box, enter the paths of your choice (Honeywell suggests using the C:\TEMP directory:

Log File Path: C:\TEMP\

Crash Dump: C:\TEMP\user.dmp

6.6 Working with fonts

Considerations for installing fonts on a GUS node



ATTENTION

When installing fonts on a GUS node, convert Type 1 Fonts to True Type Fonts. If this conversion is not done, you cannot see the newly installed fonts in the Font dialog invoked in a text object's property sheet.

Increasing the font size for the Native Window

In Windows XP, maximizing the Native Window does not properly increase the font size. The LCN fonts are not true scalable fonts—they are pixel-mapped. Use one of the following methods to properly increase the font size of the Native Window.

Method 1

Do the following to double the size of the Native Window:

1. Set the node's Windows XP Display settings at 1280x1024 pixels or higher.
2. Select **View > Double Size** on the Native Window menu.

RESULT: The LCN font displays properly in a maximized window, if there are 1280 pixels available. (At display settings lower than 1280x1024, the View > Double Size item on the Native Window menu is grayed out. If 1280x1024 pixels are not possible, use Method 2.)

Method 2

If Method 1 is not possible, then try using other fonts. Do the following to change the font being used by the Native Window:

1. Go to the Honeywell TPS **Configuration Utility** and select **LCNI18N**.
2. Change the Full Height Font and the Half Height Font to the required fonts.
3. Select **OK** and then re-invoke the Native Window.



REFERENCE - INTERNAL

To select an appropriate font for the Native Window, refer to the "Selecting Fonts" topic in the *Configuration Utility User's Guide*.

7. Accessing an HCI Server from a Remote Client

7.1 Configuration Considerations for Remote Clients

Definition of remote client

For the purpose of this discussion, a Remote Client is a node that resides outside of a TPS Domain. A Remote Client may exist on the Windows domain that contains the TPS Domain or on a separate Windows domain. The procedures in this section allow HCI or OPC client applications to access remote HCI servers residing within a TPS Domain.

Also, for the purpose of this discussion, an HCI client is an OPC client built with the HCI Client Developer Toolkit. An HCI client may or may not make use of the added value HCI features, such as timeouts or server location facilities. Use of the server location facilities requires additional client node configuration when the client node is not in the same multicast scope or name service scope as the target HCI server.

Two-way trust relationships

Unique configuration challenges emerge when clients in one domain attempt to communicate with servers in another domain. Clients using DCOM (including OPC clients) as a communication mechanism are particularly impacted.

In the Windows NT 4.0 environment, users are required to establish, at a minimum, a one-way trust relationship between domains. This implementation presents challenges as a one-way trust, typically, only enables synchronous communication (polling). As a result, clients using the OPC OnDataChange functionality (DCOM callback) fail to receive update notifications. To enable DCOM callbacks including this OPC OnDataChange functionality, customers are required to establish two-way trust relationships.

Windows 2000/2003 simplify configuration problems by automatically establishing transitive two-way trust relationships for all domains within a forest. If the domains are not members of a common forest, users can still establish one or two-way legacy Windows NT 4.0 trust relationships between domains (Windows 2000/2003 to Windows NT 4.0 or Windows 2000/2003 to Windows 2000/2003 in another forest) to achieve the desired functionality.

Security through local groups

After appropriate trust relationships are in place, additional security configuration is recommended for non-TPS nodes. TPS applies security to numerous files, registry keys, and user rights to provide a highly secure computing environment. All security is applied using local Windows 2003/XP groups. To operate correctly in a domain environment, global groups are added to local groups on each node. The default TPS security installation performs these steps automatically. If, however, you want users in other domains to be able to access local resources (servers or the registry, for example), you must add the appropriate global groups or users from the remote domain into the local groups on each machine.

Example—local group security

Local Group Configuration:

The **Local Engineers** local group is applied to the User Right **Access this computer from the network**.

The **Engineers** global group in ProcessControlDomain1 is automatically added to the **Local Engineers** group on each node.

Resulting User Access:

Users in the Engineers global group in ProcessControlDomain1 can access each TPS node from the network.

Users from BusinessDomain1 are unable to access each TPS node.



TIP

In the above example, to give access to users from BusinessDomain1, create an Engineers global group on BusinessDomain1 and add this new group to each of the local Local Engineers groups.



REFERENCE – INTERNAL

Refer to the *TPS System Administration Guide* for additional information on TPS Security.

7.2 Implementing an HCI/OPC Client Node

Prerequisites

Prior to performing the procedures to implement an HCI/OPC client node, ensure the following prerequisites have been met:

- If the client node is in a different domain than the HCI server domain, the client domain must have a trust relationship with the server domain.
- For asynchronous communications to occur between domains, a two-way trust must be established.
- The client node and its Domain Controller and the HCI Server node and its Domain Controller must all be visible to each other over the network.

Overview of procedures

Perform the steps in the following sections to allow HCI and OPC client applications communicate with an HCI server residing within a TPS Domain:

- Configure HCI/OPC client node.
- Install System Management Runtime or HCI Runtime.
- Install HCI Client Connectivity software.
- Enable HCI Server to accept communications from HCI/OPC Client.
- Generate HCI Server Component Name file.

Configure HCI/OPC client node

Step	Action
1	Verify that the client node is a member of a Windows domain.
2	Verify that the client node has been upgraded to Windows 2003/XP or Windows 2000 (with the latest hot fixes and Microsoft service packs supplied by Honeywell, as defined in the TPS R4xx Software Change Notice). If you need to install the System Management Runtime software package, then the client node must be running Windows 2000/2003/XP.

7. Accessing an HCI Server from a Remote Client

7.2. Implementing an HCI/OPC Client Node

Step	Action
	ATTENTION The following step is necessary only if the HCI/OPC client application is to be run as a user account in the node's Windows domain.
3	Create a Windows Global Group and add users who need to run HCI/OPC client applications. For example, create a global group called Engineers, and add user account joesmith to the newly created group.
4	At the HCI Server node, configure security as follows: a) Configure or verify the following DCOM settings (DCOMCNFG): – Default Authentication Level: Connect – Default Impersonation Level: Impersonate – Default Access: Everyone/Allow – Default Launch: Everyone/Allow b) Configure or verify that the Everyone group is applied to the user right setting called Access This Computer From the Network.
5	Continue to the next procedure, "Install System Management Runtime or HCI Runtime".

Install System Management Runtime or HCI Runtime

Install the **HCI Runtime** software component on a Windows 2003/XP or Windows 2000 node that is to access a Honeywell HCI Server.

Install the **System Management Runtime** package on a Windows 2003/XP or Windows 2000 client node if *both* of the following are true:

- The client node runs HCI client applications that call the HCI runtime routine GetComponentInfo() to retrieve the hostname and ProgID of the target server.
- The client node is NOT configured in the same multicast scope or name service scope as the target HCI server(s).



ATTENTION

- If any of the following licensed packages from the **TPS Software CD** have been installed on the client node, then the System Management Runtime and HCI Runtime packages are already installed on the node, and you do not need to perform this procedure:
 - Fault Tolerant Ethernet Driver
 - Honeywell System Management Display
 - Redirection Manager
- System Management Runtime uses Multicast to process information between nodes. Multicast must be enabled on all network switches and routers that interconnect participating nodes.
- System Management Runtime depends on services provided by the Windows 2003/XP/2000 operating system. System Management Runtime can be installed only on nodes running Windows 2003/XP/2000. This requirement also applies to remote clients.

Step	Action
1	Insert the TPS Software CD .
2	Install the Honeywell Repackaged Redistributables by running Setup.exe from \I386\Packages\hwredist and then restart the machine.
3	Reinsert the TPS Software CD , and then install System Management and HCI Runtime .
4	Continue to the next procedure, "Install HCI Client Connectivity software".

7. Accessing an HCI Server from a Remote Client

7.2. Implementing an HCI/OPC Client Node

Install HCI Client Connectivity software



ATTENTION

You must ensure to install the following:

- Install System Management Runtime and .NET framework prior to the HCI Client Connectivity package installation.
- Install both HCI runtime and System Management runtime from the Common Components CD before installing the TPNServer.

Step	Action
1	Insert the TPS Software CD .
2	Click Install Packages .
3	When the <i>Welcome</i> dialog displays, read the details and if you agree, click Next .
4	Accept the software license agreement terms by clicking Next .
5	When the <i>Third party Software Compatibility</i> dialog is displayed, read the details and then click Next .
RESULT: The <i>User Information</i> dialog is displayed.	



REFERENCE – INTERNAL

For more information on the compatibility of third party software with TPS software components, refer to the TPS SCN and the *TPS System Planning Guide*.

6	If this dialog has been used before, the information fills in. If not, enter the Name , Company , License No. , and Authorization No. , and then click Next .
RESULT: After the License No. and Authorization No. have been validated, the <i>Package Selection</i> dialog displays a list of the available licensed TPS packages.	

7 Accessing an HCI Server from a Remote Client

7.2 Implementing an HCI/OPC Client Node

Step	Action
7	Select <i>only one</i> of the following, depending on your implementation requirements: • GUS Display HCI Client Add-in • HCI Client Connectivity • TPS Open Layer Software For information on licensed TPS packages, refer to section 4.11, "TPS Software package usage by node type".
8	Click Install Package. RESULT: The Installable Components dialog displays.
9	If connecting to a TPN Server, deselect all checkboxes except TPN Server Connectivity Client. If connecting to a CL Server, clear all checkboxes except CL/NT Application Server Connectivity Client.
10	Click OK.
11	Click Finish.

7. Accessing an HCI Server from a Remote Client

7.2. Implementing an HCI/OPC Client Node

Step	Action
	ATTENTION HCI Runtime Location For GUS R400 and greater and APP R400 and greater (TPS R400 and greater), the software component HCI Runtime is on the TPS Software CD. For earlier releases, it is on the Common Components CD. When using the TPS R130 and greater software CD, selecting one or more TPS software packages that depend on the HCI Runtime component always displays the following message: “Please ensure that the correct HCI Runtime package has been installed before performing this install”. If the installer detects that HCI Runtime is not registered in the registry, the install does not continue and the following message displays: “The HCI Runtime package MUST be installed before performing this install”. If these messages appear, insert the TPS Software CD, install HCI Runtime, then return to the beginning (step 1) of this procedure to install the TPS software packages.

Step	Action
	ATTENTION <u>Uninstallation Errors</u> If a serious problem such as a power failure or a node failure occurs while you are installing TPS software, then the install log file generated by Install Shield can get corrupted when you reboot the node. If this occurs, you get the error message "Uninstaller Setup failed to initialize. You will not be able to uninstall this product". and you cannot uninstall the software package. If this occurs, you must manually delete the package's uninstall log file and then repeat the installation procedure. The uninstall log files have an ".isu" extension and are located in the \Program Files\Honeywell\TPS directory. Deleting the wrong files from this directory can cause package uninstallation problems. If you are unsure of what file to delete, call the Honeywell Technical Assistance Center for assistance. <u>Read-Only Files</u> If a read-only file is encountered during TPS software packages installation, a message displays asking if it is OK to overwrite the read-only file. Check the box to indicate that no further messages should be displayed and then click Yes to continue.
12	Install other optional packages as needed (such as HCI Client Developer Toolkit or HCI Server Developer Toolkit).
13	Continue to the next procedure, "Enable HCI Server to accept communications from HCI/OPC Client".
	REFERENCE – EXTERNAL For client installation requirements of servers other than HCI Servers, refer to the documentation of the specific servers.

7. Accessing an HCI Server from a Remote Client

7.2. Implementing an HCI/OPC Client Node

Enable HCI Server to accept communications from HCI/OPC Client

Step	Action
1	Log onto the HCI Server node as Administrator.
2	Add Domain Global Groups to Local Groups on the HCI Server node. For example, if an Engineers group was created in the enterprise domain (domain1), add the Domain1\Engineers group to the local group on the TPS Node called "Local Engineers".
3	Set the HCI Server's method security to permit access to the server's secure methods (Read, Write, and Shutdown).
	REFERENCE – INTERNAL For additional information on the previous step (step 3), refer to the "TPS Security. HCI/OPC server Capability File Customization" topic in the <i>TPS System Administration Guide</i> .
4	On the HCI Server node, update each TPS proxy file in the [install drive]\HWIAC\Security directory structure to include the new global group(s) from the enterprise domain. RESULT: Specific users or groups in the enterprise domain are permitted to access an HCI server on an alternate domain.
5	Continue to the next procedure, "Generate HCI Server Component Name file".

Generate HCI Server Component Name file

This procedure requires that System Management Runtime be installed on the node (to provide the Alias Generator tool).

Perform this procedure to add the ProgID and Hostname of the target HCI server(s) to the file \HWIAC\ComponentAlias\Componentalias.xml on the client node.



ATTENTION

This procedure is required only if *both* of the following are true:

- An HCI client application uses GetComponentInfo () to retrieve the target HCI server ProgID and Hostname.
- The client node is NOT in the same multicast scope as the HCI server node.



REFERENCE – INTERNAL

For detailed instructions, refer to the “Generating a File using Alias Generator” topic in the *System Management Configuration Guide*.

Step	Action
1	On the client node, select Start > Programs > Honeywell TPS > Alias Generator .
2	Open the componentalias.xml file.
3	Select Component > Add and add to the file the names of the HCI server(s) that the client node needs to access.
4	Save the file.

7. Accessing an HCI Server from a Remote Client

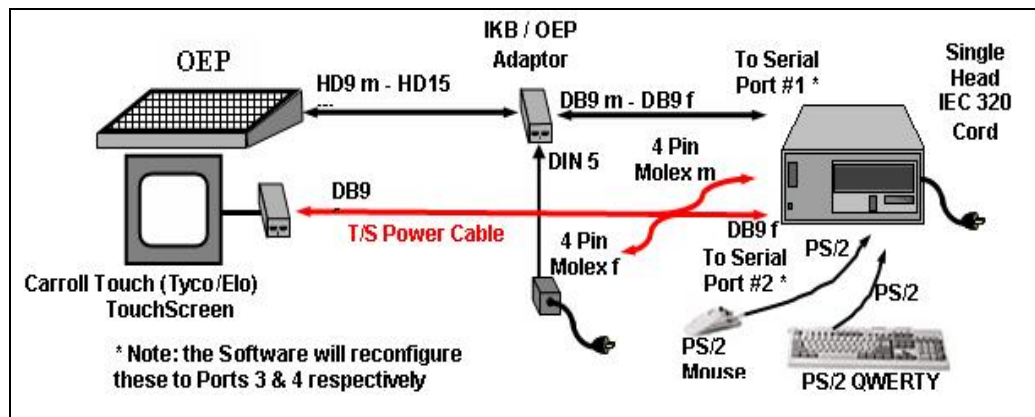
7.2. Implementing an HCI/OPC Client Node

8. Touchscreen, IKB/OEP, monitor configuration

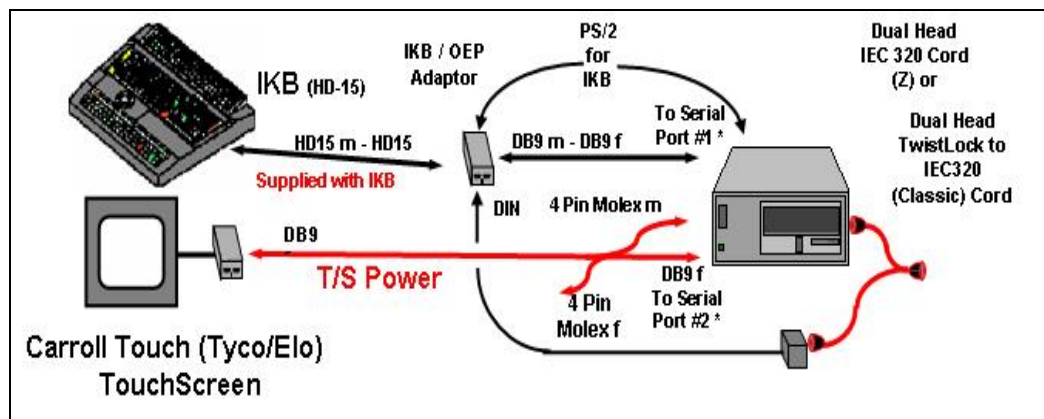
8.1 Connecting IKB/OEP/Touchscreen

This section contains the diagrams and details for connecting IKB/OEP and Touchscreen to a GUS node.

Connecting Touchscreen and OEP—Serial Port Solution



Connecting Touchscreen and IKB—Serial Port Solution



8 Touchscreen, IKB/OEP, monitor configuration

8.1 Connecting IKB/OEP/Touchscreen

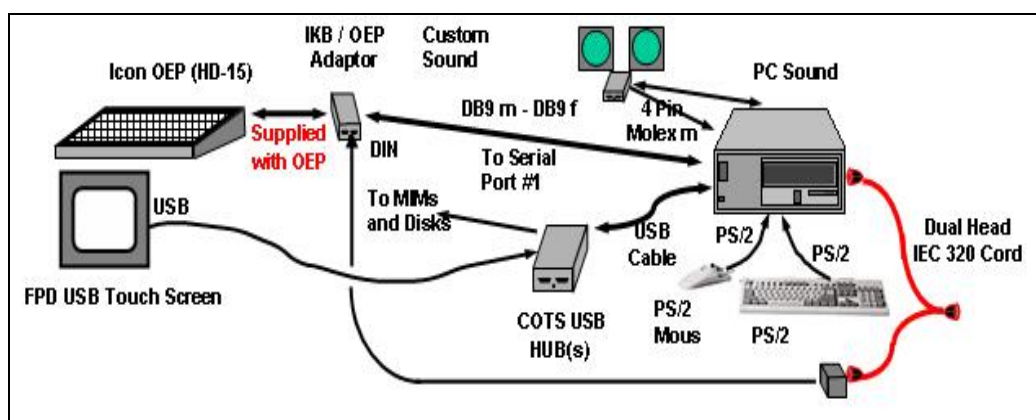
Connecting FPD Touchscreen (SAW Touchscreen)

Connect the USB connector of the FPD Touchscreen to the USB port of the GUS node.

Connecting USB-IKB

Connect the USB connector of the USB IKB to the USB port of the GUS node.

Connecting Icon OEP—USB and Serial Port Solution



8.2 Managing COM Port Settings

This section contains information on using the Windows XP Device Manager and Hardware Wizard to manage the communication port settings.

Adjust communication port number

The following table provides information on using the Windows XP Device Manager to adjust the Port Number for one of its communication ports.

Step	Action	Result
1	On the System Properties Hardware page, click Device Manager .	The Device Manager window is displayed a list of all the device types.
2	Expand the Ports (COM & LPT) entry by clicking the "+" box on the left.	The entry expands and shows all the currently defined Ports.
3	Double-click the entry for the port to be examined or changed.	The Communication Port Properties window is displayed, with the General page on display.
4	Click the Port Settings tab.	The Port Settings page is displayed. This page displays the major parameters for the port and its name.
5	The Port Number can be adjusted. To change the port number, click Advanced .	The Advanced Settings page displays. The number can be adjusted by selecting a new entry in the COM Port Number box.
6	When you are finished with this page, press OK .	Properties window is displayed.
7	Click OK on the Properties window.	The System Settings Change window is displayed, stating that the computer must be restarted for the changes to take effect.
8	Click NO to skip the restart.	

8.3 Configuring a CarrollTouch touch screen

Introduction

The CarrollTouch touchscreen drivers for Windows XP are called the "Universal Pointing Device Drivers" (UPDD). You do all touch screen configuration and calibration through the UPDD program group. The following table shows the driver information for the interface.

Driver Package	Driver Distribution Locations
Touch Screen Drivers Carroll Touch	EXPPlus Initialization DVD: \\UTILITIES\\TBUPDDS



TIP

Rename the COM Port number to support the Carroll Touchscreen as described in the section 8.2. For example, if the Carroll Touchscreen is connected to the COM1 port, COM1 needs to be renamed to COM4 in Windows Device Manager.

The Device Manager is found by starting the Control Panel, selecting the System icon, and selecting the hardware tab.



REFERENCE – EXTERNAL

Refer to the Microsoft documentation for additional Device Manager information.

Running the UPDD configuration and calibration tools

When the CarrollTouch drivers are installed, a UPDD folder is placed in the Start menu, providing access to the following applications:

- Calibrate
- Event Selector
- Help
- Settings
- Test



TIP

To start applications, click **Start** and then navigate to the desired application:
Start > Programs > UPDD > [application].

The following table describes how each application is used:

Application	Usage
Calibrate	The Calibrate application draws a white background in the area configured for the touch events and then places an "X" in one of the corners. When the user touches the "X" it is redrawn in the opposite corner. Touching the second "X" completes the calibration sequence and closes the window. The calibration window closes automatically at the end of the calibration sequence or if 15 seconds pass without receiving any touch events.
Event Selector	The Event Selector is used to indicate the button modes that perform when a touch event occurs.
Help	This application provides help information on the touchscreen and should be accessed for configuration and operational information.
Settings	The settings application brings up a set of property pages used to configure various properties and for viewing the driver state. It also provides access to the test and calibrates functions as settings are adjusted.
Test	The test application draws a window in the area configured for the touch screen and responds to touches by reporting the touch down and up events. In addition, if the touch point is moved between down and up events, a line is drawn as the touch point is moved. The test screen can be closed by pressing escape on the keyboard or by using the close box on the display.

8 Touchscreen, IKB/OEP, monitor configuration

8.3 Configuring a CarrollTouch touch screen

Using the touchscreen on multi-monitor systems

TPS Systems support multiple monitor configurations with dual screens arranged either side by side (horizontally) or one-over (vertically).

The desktop segment setting identifies the portion of the virtual desktop that the touch screen monitors. The parameter can be adjusted, if necessary, by using the "Settings" application and going to the Advanced tab on the Devices page.

The following table shows the configuration settings. The "Standard" TPS settings are indicated as underlined entries.

Display hardware	Orientation	Screen location	Desktop segment
ATI FireGL 3100 (WS470-based Honeywell workstation) ATI FireGL 3400 (WS490-based Honeywell workstation) nVidia Quadro (WS360-based Honeywell workstation)	Side By Side	<u>Left</u>	<u>Monitor 1</u>
		Right	Monitor 2
	One Over	<u>Bottom</u>	<u>Monitor 1</u>
		Top	Monitor 2

8.4 Configuring a MicroTouch screen

Overview

The MicroTouch touch screen can be used in a single or dual screen configuration. In the dual screen configuration, one of the two monitors is the touch screen. The monitors can be oriented either horizontally or vertically.

If a MicroTouch touch screen exists on the GUS, a small touch screen controller box attaches to the back of the monitor case with the name “MicroTouch” printed on the surface.

This section describes two procedures: configuration and calibration.

MicroTouch dual screen configuration

Execute the following configuration procedure only if the following is true:

- You have dual screens.
- The MicroTouch touch screens were not delivered with your system.
- The MicroTouch touch screens were delivered preinstalled with your system, but you wish to reconfigure.

Note: that the MicroTouch drivers are located on the EXPPlus Initialization DVD.

Step	Action
1	Log on to the machine as Administrator.
2	Insert the EXPPlus Initialization DVD into the DVD drive.
3	Double-click the My Computer icon.
4	Click Folders in the Task Bar.
5	Navigate to the following directory on the DVD: <EXPPlusDVD>\UTILITIES\MTSCREEN\HWIAC\MTRX
6	For side-by-side configuration, double-click hleft.reg . OR For one-over-one configuration, double-click vbottm.reg . Result: The Registry Editor window is displayed and the configuration procedure is initiated.

8 Touchscreen, IKB/OEP, monitor configuration

8.4 Configuring a MicroTouch screen

Step	Action
7	If the configuration procedure was successful, click OK .
8	Restart the machine.
9	Continue to the procedure to calibrate the touchscreen.

MicroTouch screen calibration

Step	Action
1	Log on to the machine as Administrator.
2	Call up the MicroTouch Touch Screen window.
3	Select Calibrate .
4	Touch the spots on the Display when displayed.
5	Verify the Calibration Complete prompt.
6	Select OK .
7	Select OK again.

8.5 Configuring a dual/triple/quad screen

Enabling dual/triple/quad screen operation

If only a single screen appears, perform the following procedure to reconfigure the display properties.

Step	Action
1	Open the video display properties: Start > Settings > Control Panel > Display > Settings tab.
2	To enable a display, select a grayed out monitor box, right-click and select Attached in the menu.
3	One of the monitors must be assigned as a primary: a) Select a monitor. b) Select the checkbox labeled “use this device as the primary monitor”.



TIP

To view the monitor numbers, click **Identify**. The monitor number shows on each monitor.

8 Touchscreen, IKB/OEP, monitor configuration

8.5 Configuring a dual/triple/quad screen

Honeywell

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